Type Selection Report

FOR

ALTA MESA RD AT LAGUNA CREEK

August 2013

Submitted To: Sacramento County

Prepared By: MARK THOMAS & COMPANY, INC.
Providing Engineering, Surveying and Planning Services

STATE OF CALIFORNIA DEPARTMENT OF TRANSPORTATION

STRUCTURE TYPE SELECTION MEMO

PROJECT IDENTIFICATION					DATE	
Alta Mesa Road Bridge Replacement					August 2013	
DIST	CO	RTE	PM	EA	CONSULTANT	
3	Sac				**Mark Thomas & Co. Inc.**	
BRIDGES NAME(S)					**BR NO**	**BRIDGE CONSTRUCTION COST**
Alta Mesa Rd Bridge - Alt. 1 (CIP Reinforced Concrete Slab)					24C0306	$2,650,000
Alta Mesa Rd Bridge - Alt. 2 (CIP P/S Concrete Box Girder)					24C0306	$2,950,000
Alta Mesa Rd Bridge - Alt. 3 (PC/PS Concrete Bulb-Tee)					24C0306	$2,940,000

Brief Project Description:
Alta Mesa Road Bridge over Laguna Creek (Br. No. 24C-0306) is located just north off of State Route (SR) 104, approximately two miles northeast of the community of Herald. The roadway is approximately 21 feet wide, consisting of two lanes with minimal shoulders. The existing bridge currently serves agricultural and residential land uses between SR 104 and Valensin Road. The Caltrans Inspection Report (August 2010) classifies this bridge as Functionally Obsolete with a Sufficiency Rating of 35.2. Since the County would like to replace the existing structure, the three alternatives for replacement of Alta Mesa Road Bridge are presented below.

Foundations:
We discussed the foundations with our subconsultant, Taber Consultants. Based on the attached preliminary foundation memo, we decided to use Caltrans Standard Plans Class 140 pipe piles at the abutments for all three alternatives and at the piers for Alternative 2 and 3. For the piers of Alternative 1, we assumed using 15” Pile Extensions.

Aesthetics:
California ST-70 Bridge Rails are being used to allow views of the creek. No additional aesthetics are being considered at this time.

Hydrology & Hydraulics:
The preliminary hydrology and hydraulics were completed for the Alta Mesa Road over Laguna Creek by WRECO in April 2013. The USGS regional flood-frequency equations and the “estimation for an ungagged site near a streamgage” approach published by the USGS were used to estimate peak discharged at the Project site. Below is the discharge summary.

Discharge Summary

Method	Location	Peak Flow (cfs)	
		50-Year	100-Year
USGS Regional Flood-Frequency Equation	Alta Mesa Road at Laguna Creek	13,322	14,810
USGS Estimation for an ungagged site near a streamgage		14,936	16,491

Based on the discharge information, HEC-RAS models were used for the analysis of the Water Surface Elevations shown below.

Location	Water Surface Elevation (ft, NAVD 88)	
	50-Year	100-Year
Laguna Creek - Just Upstream of Alta Mesa Road	71.3	71.8

The proposed bridge minimum soffit elevation is determined to be at elevation 73.36 feet. This elevation passes the governing 50 year storm event plus 2 feet of freeboard (73.30 feet). The Central Valley Flood Protection Board has indicated that this site is not within their jurisdiction.

Scour:
No scour analysis has been done but rock slope protection may be required at the bridge site.

Barrier Rails:
California ST-70 Bridge Rails will be used. This 46.5 inches tall rail satisfies the minimum height (42 inches) for pedestrians and bicyclists.

Utilities:
Overhead utilities and underground fiber optic lines running along the existing bridge will be relocated.

Detour:
The existing bridge will be closed during construction and traffic will be detoured. Staged construction will not be required.

Bridge Construction:
Temporary shoring will be required for the construction of the pier footings since water is always present in the creek.

Right-of-Way:
Right-of-Way acquisitions will be required for the construction of bridge.

Approach Structure:
Structure Approach Type EQ (10) will be used for a smooth transition from the roadway and the bridge.

Alternatives Relative Ranking:

	Relative Ranking					
Alternatives	Cost	Roadway and ROW Cost	Maintenance	Construction Duration / Impact to Residents	Environmental Impacts	Preferred Alternative
#1 Eight Span CIP Slab	1- $2.65M	1- $0.93M	2	2	3	
#2 Three Span CIP/PS Box Girder	2- $2.95M	2- $1.27M	1	2	2	
#3 Three Span Precast PS Girder	2 -$2.94M	2- $1.27M	1	1	1	X

Alternatives Considered:

Alternative 1: The cast-in-place reinforced concrete slab bridge is an eight-span structure of approximately 324 feet long by 40 feet wide. The required structure depth is 22.5 inches. Alternative 1 is the least expensive structure type with a total construction cost of approximately $3.58 million. Most of the cost savings come from the use of diaphragm abutments which require fewer piles and less earthwork compared to seat type abutments. The Roadway and ROW cost would be lower compare to the other alternatives. This alternative will require the most amount of maintenance due to the largest number of supports that are physically in the creek. This cast-in-place structure will require falsework to temporarily support the structure until the required concrete strength is reached. Not only will the construction of this alternative be longer but also causing the most environmental impact to the surrounding (see General Plan - Alt 1).

Alternative 2: The cast-in-place prestressed concrete box girder is a three-span structure of similar plan dimensions to Alternative 1. The required structure depth is 5’-0”. This structure type will cost approximately $4.22 million for the total construction of the project. The higher cost is associated with the seat type abutments and the required earthwork. The Roadway and ROW cost will also be higher compare to Alternative 1. This cast-in-place structure will also require falsework for construction but the level of future maintenance is minimal compare to Alternative 1. Similar to Alternative 1, the construction and environmental impact to the surrounding will be more compare to Alternative 3 (see General Plan - Alt 2 & 3).

Alternative 3: The precast prestressed concrete bulb-tee is also a three-span structure of similar dimensions and depth to Alternative 2. The total construction cost for this structure type will be approximately $4.21 million. One of the advantages of the precast option is shorter construction time. Since falsework will not be required for the construction of the superstructure, Alternative 3 will allow the road to open approximately 6 to 8 weeks sooner than the cast-in-place options. Similar to Alternative 2, this alternative will require lower future maintenance when compare to Alternative 1 since only two piers are in the creek as opposed to seven. This alternative will also have the least environmental impact to surrounding. (See General Plan - Alt 2 & 3).

Recommendation:

Based on the total construction cost alone Alternative 1 appears to be the most economical. However, when considering the construction time and future maintenance and environmental impact, Alternative 3 is the recommended option. Although the construction cost for Alternative 3 is higher than Alternative 1, however having the benefit of opening the road sooner and the cost saving due to future maintenance will offset the higher construction cost. Taking into consideration the construction cost, construction duration, lower future maintenance and environmental impact, we recommend Alternative 3 using precast prestressed concrete bulb-tee for the replacement of Alta Mesa Road over Laguna Creek.

ALTERNATIVE 1
CAST-IN-PLACE REINFORCED CONCRETE SLAB

DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				

REGISTERED CIVIL ENGINEER DATE

PLANS APPROVAL DATE

REGISTERED PROFESSIONAL ENGINEER PO-KANG CHEN No. S3112 Exp. 9/30/13 STRUCTURAL STATE OF CALIFORNIA

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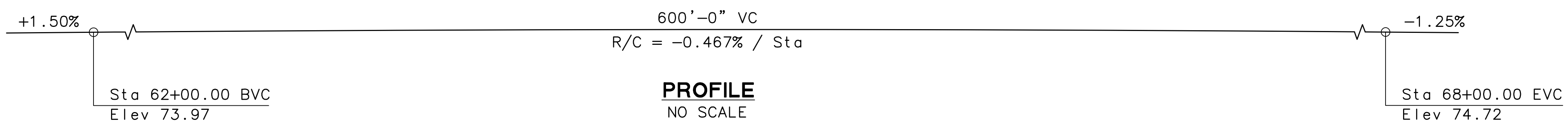
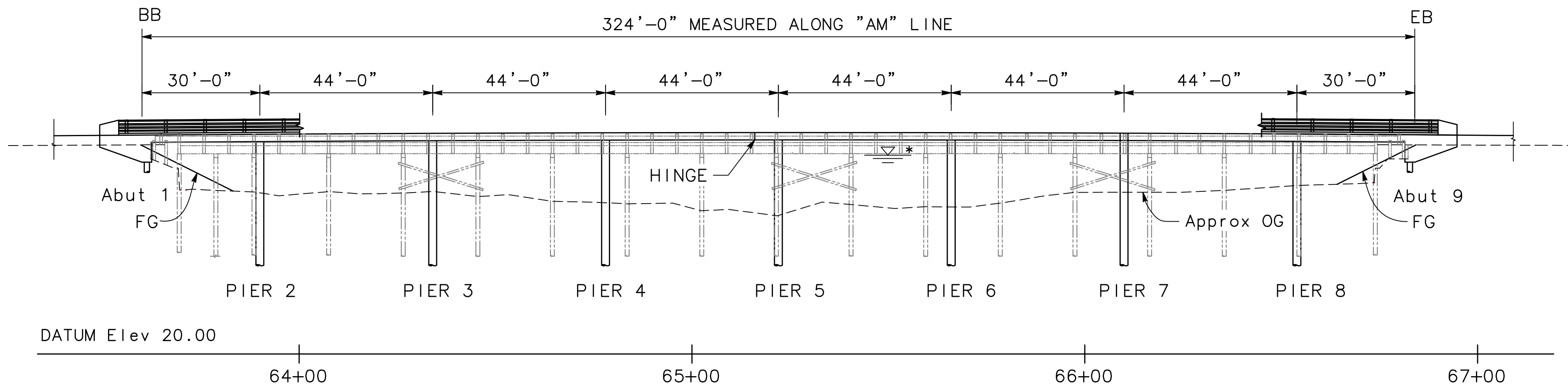
SACRAMENTO COUNTY
906 G STREET, SUITE 510
SACRAMENTO, CA 95814

MARK THOMAS & COMPANY, INC.
7300 FOLSOM BOULEVARD, SUITE 203
SACRAMENTO, CA 95826

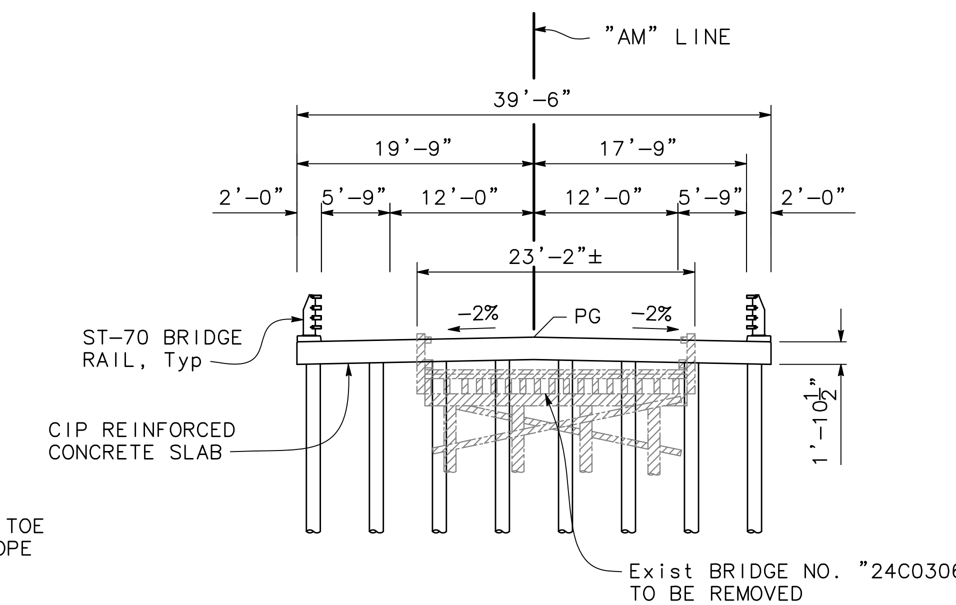


PROFILE
NO SCALE

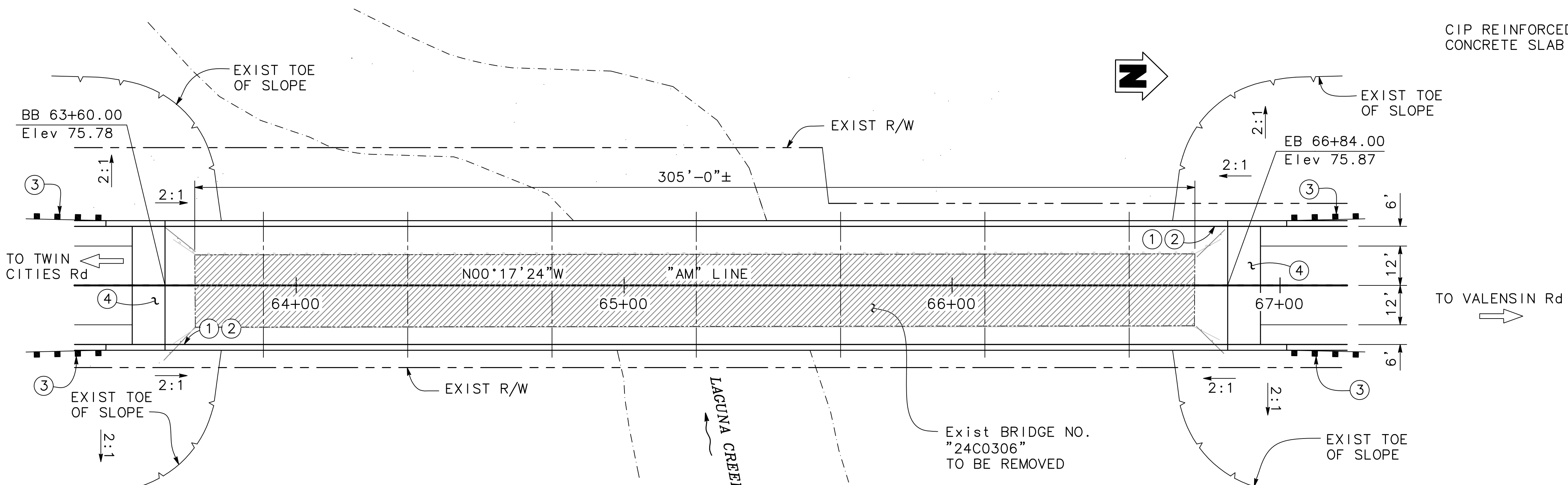


ELEVATION
1" = 20'

* Soffit elevation designed to meet the higher of 50-year + 2 ft of freeboard or 100-year



TYPICAL SECTION
1/8" = 1'-0"



PLAN
1" = 20'

NOTES:

1. Paint "ALTA MESA ROAD BRIDGE"
2. Paint "BR NO. XX-XXXX"
3. MBGR, SEE "ROAD PLANS"
4. Structure Approach Type EQ(10)

LEGEND:

- Indicates New Structure
- Indicates Existing Structure
- Indicates Bridge Removal

Alt 1

DESIGN OVERSIGHT	DESIGN	BY T. PHAM	CHECKED	LOAD & RESISTANCE FACTOR DESIGN	LIVE LOADING: HL93 W/"LOW-BOY"; PERMIT DESIGN VIHICLE	
	DETAILS	BY G. BOYKO	CHECKED	LAYOUT	BY J. WEIR	CHECKED
SIGN OFF DATE	QUANTITIES	BY	CHECKED	SPECIFICATIONS	BY	PLANS AND SPECS COMPARED

DESIGN GENERAL PLAN SHEET (ENGLISH) (REV.7/16/10)

ORIGINAL SCALE IN INCHES FOR REDUCED PLANS 0 1 2 3

PREPARED FOR THE **SACRAMENTO COUNTY** **DEPARTMENT OF TRANSPORTATION**

JASON HICKEY
PROJECT ENGINEER

BRIDGE NO. XX-XXXX
POST MILES

ALTA MESA ROAD BRIDGE (REPLACE)
GENERAL PLAN

UNIT: X
PROJECT NUMBER & PHASE: X

DISREGARD PRINTS BEARING EARLIER REVISION DATES

REVISION DATES (PRELIMINARY STAGE ONLY) 05/14/13

SHEET 1 OF X

FILE => $REQUEST

CONTRACT NO.: X

PROJECT ID:

USERNAME => $USER DATE PLOTTED => $DATE TIME PLOTTED => $TIME

ALTERNATIVE 2
CIP PRESTRESSED CONCRETE BOX GIRDER

AND

ALTERNATIVE 3
PRECAST P/S CONCRETE BULB-TEE

DIST	COUNTY	ROUTE	POST MILES TOTAL PROJECT	SHEET No	TOTAL SHEETS
03	SAC				

REGISTERED CIVIL ENGINEER DATE

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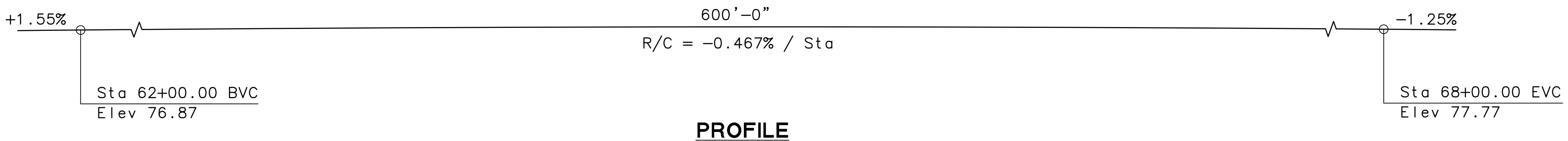
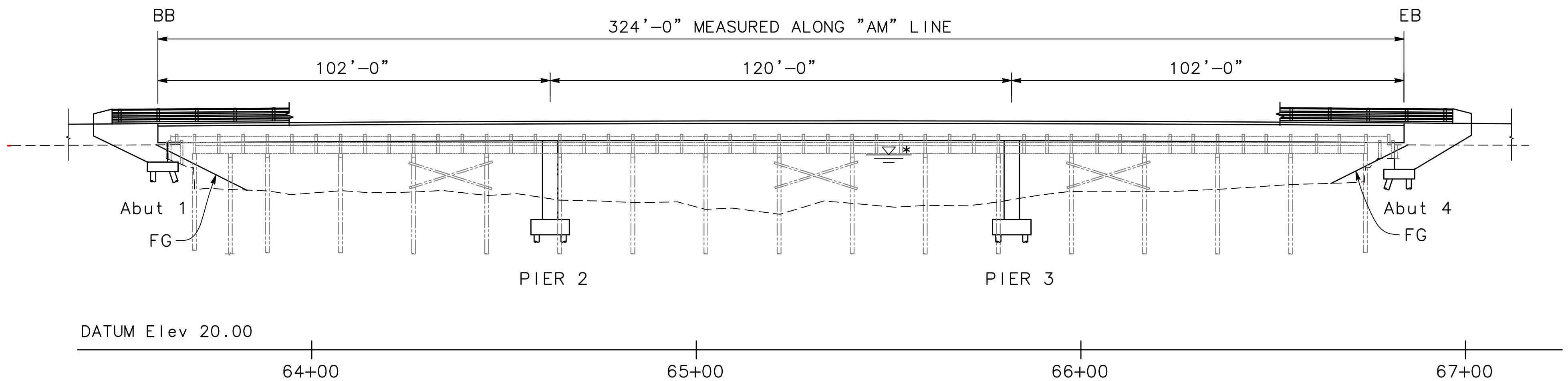
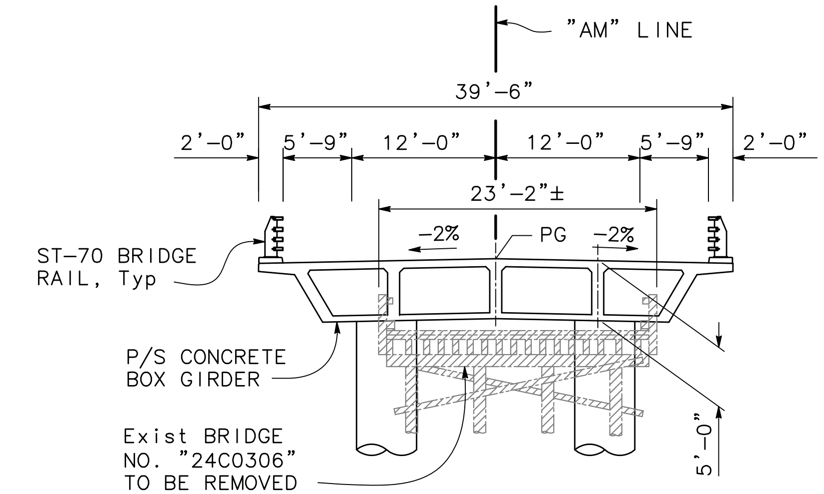


PROFILE
NO SCALE

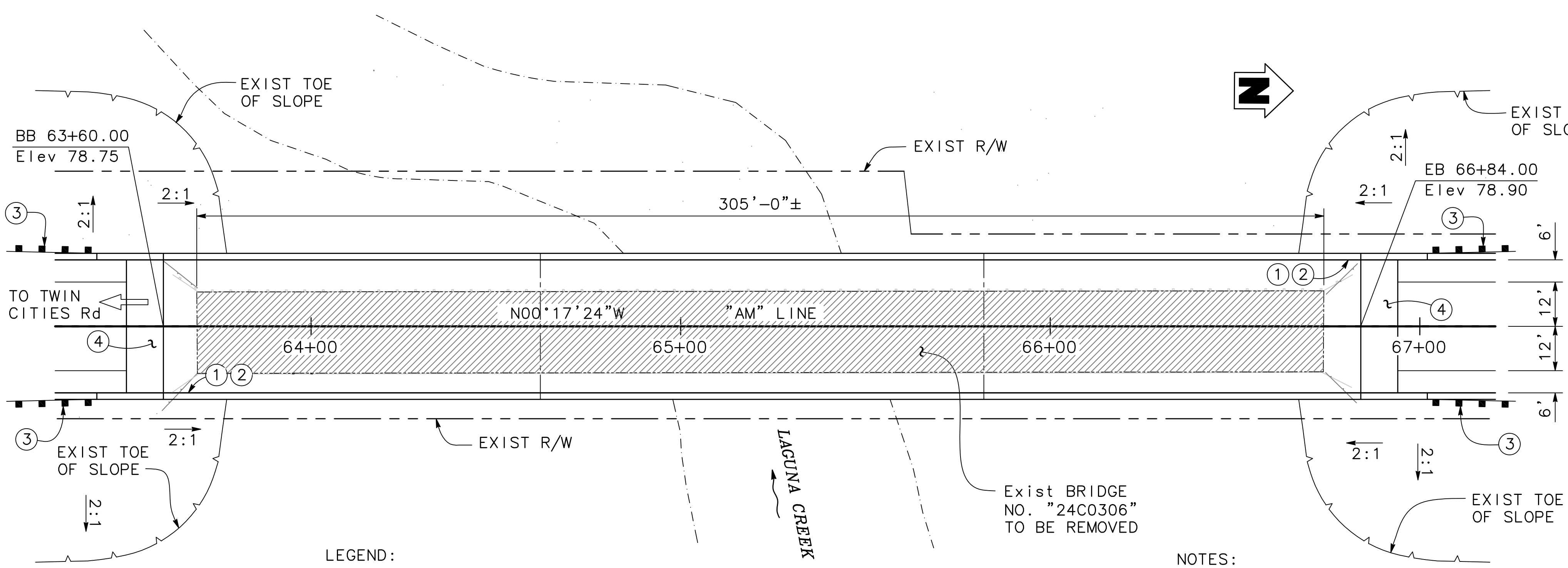


ELEVATION
1" = 20'

* Soffit elevation designed to meet the higher of 50-year + 2 ft of freeboard or 100-year



TYPICAL SECTION - Alt 2
1/8" = 1'-0"



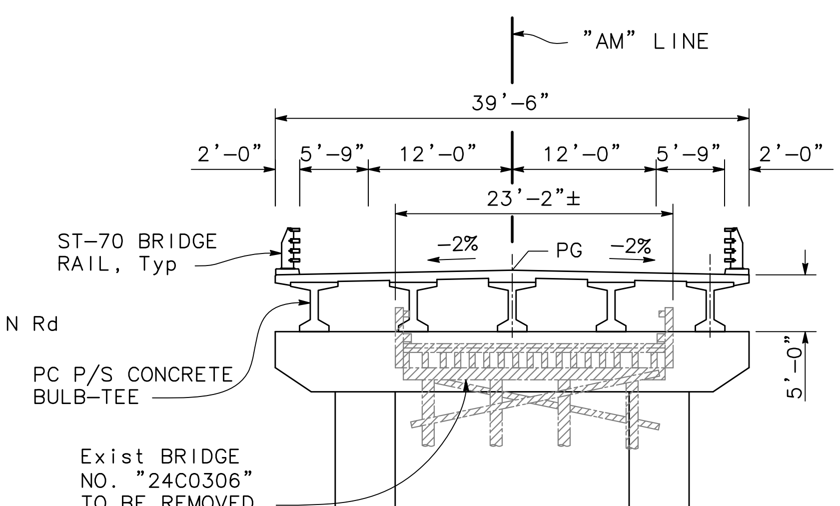
LEGEND:

- Indicates New Structure
- Indicates Existing Structure
- Indicates Bridge Removal

PLAN
1" = 20'

NOTES:

1. Paint "ALTA MESA ROAD BRIDGE"
2. Paint "BR NO. XX-XXXX"
3. MBGR, SEE "ROAD PLANS"
4. Structure Approach Type EQ(10)



TYPICAL SECTION - Alt 3
1/8" = 1'-0"

Alt 2 & Alt 3

DESIGN OVERSIGHT	DESIGN	BY T. PHAM	CHECKED	LOAD & RESISTANCE FACTOR DESIGN	LIVE LOADING: HL93 W/"LOW-BOY"; PERMIT DESIGN VIHICLE	
	DETAILS	BY G. BOYKO	CHECKED	LAYOUT	BY J. WEIR	CHECKED
SIGN OFF DATE	QUANTITIES	BY	CHECKED	SPECIFICATIONS	BY	PLANS AND SPECS COMPARED

PREPARED FOR THE **SACRAMENTO COUNTY DEPARTMENT OF TRANSPORTATION**

JASON HICKEY
PROJECT ENGINEER

BRIDGE NO. XX-XXXX
POST MILES

ALTA MESA ROAD BRIDGE (REPLACE)
GENERAL PLAN

DESIGN GENERAL PLAN SHEET (ENGLISH) (REV.7/16/10)

ORIGINAL SCALE IN INCHES FOR REDUCED PLANS 0 1 2 3

UNIT: X
PROJECT NUMBER & PHASE: X

DISREGARD PRINTS BEARING EARLIER REVISION DATES
REVISION DATES (PRELIMINARY STAGE ONLY) 05/14/13
SHEET 1 OF X

CONTRACT NO.: X   PROJECT ID:

FILE => $REQUEST

USERNAME => $USER   DATE PLOTTED => $DATE   TIME PLOTTED => $TIME

ROAD WAY EXHIBIT

LEGEND

PARCEL BOUNDARIES
AREA OF POTENTIAL EFFECTS
LIMITS OF WORK
PROJECT DESIGN

LOCATION DESCRIPTION

THE ALTA MESA ROAD OVER LAGUNA CREEK BRIDGE CROSSES LAGUNA CREEK 0.4 MILES NORTH OF SR104, HERALD, CALIFORNIA.

AREA OF POTENTIAL EFFECTS MAP ALTA MESA ROAD OVER LAGUNA CREEK HERALD, CALIFORNIA

ERIN DWYER
PQS/PI – PREHISTORIC ARCHAEOLOGY
CALTRANS DISTRICT 3

DATE

MICHAEL MCCOLLUM
LOCAL ASSISTANCE ENGINEER
CALTRANS DISTRICT 3

DATE

0 150 300
FEET

ALTA MESA ROAD BRIDGE REPLACEMENT APE EXHIBIT

PRELIMINARY FOUNDATION REPORT

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June 7, 2013

Mark Thomas & Company, Inc.
7300 Folsom Boulevard, Suite 203
Sacramento, CA 95826

Attention: Mr. Matt Brogan, P.E.

Subject: **Preliminary Foundation Report** 2012-0128-1
Alta Mesa Road Bridge at Laguna Creek 38121-C2:220N;290W
Existing Bridge No. 24C0306
Sacramento County, California

Taber Consultants is pleased to submit this Preliminary Foundation Report for the Alta Mesa Road Bridge at Laguna Creek Project in Sacramento County, California. We prepared this report in accordance with the agreement between Mark Thomas & Company and Taber Consultants dated April 19, 2013. Subsurface exploration and laboratory testing will be completed as part of final design.

We appreciate this opportunity to be of service. Please call if you have questions or require additional information.

Very truly yours,

TABER CONSULTANTS

Ronald E. Loutzenhiser
Senior Geotechnical Engineer

REL/WEN
Accompanying: Reports (4)

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PRELIMINARY FOUNDATION REPORT

Alta Mesa Road Bridge at Laguna Creek
Existing Bridge No. 24C0306
Sacramento County, California

Sacramento County Department of Transportation
Owner
906 G Street
Sacramento, CA 95814

Mark Thomas & Company, Inc.
Design Engineer
7300 Folsom Boulevard, Suite 203
Sacramento, CA 95826

2012-0128-1
38121-C2:220N;290W

June 7, 2013

Taber Consultants
Engineers and Geologists

PRELIMINARY FOUNDATION REPORT
Alta Mesa Road Bridge at Laguna Creek
Existing Bridge No. 24C0306
Sacramento County, California

Taber
Since 1954

2012-0128-1

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PRELIMINARY FOUNDATION REPORT
Alta Mesa Road Bridge at Laguna Creek
Existing Bridge No. 24C0306
Sacramento County, California

Taber
Since 1954

2012-0128-1

1 INTRODUCTION

1.1 Purpose

A limited study of preliminary foundation conditions has been completed at the above site in accordance with the agreement between Mark Thomas & Company, Inc. (MTCo) and Taber Consultants. This report provides preliminary geologic, seismic, and foundation information for use in preliminary bridge design. Taber Consultants will complete subsurface exploration and laboratory testing as part of final design.

Limitations of this study are discussed in the attached "General Conditions."

1.2 Geotechnical Services

To prepare this report, Taber Consultants:

- discussed the project with Mr. Matt Brogan and Mr. Jason Hickey at MTCo,
- reviewed published geologic maps and literature pertaining to the site,
- attended a project kick-off meeting on March 7, 2013,
- reviewed a topographic survey of the project site dated March 13, 2013,
- reviewed a draft "Type Selection Report" by MTCo for the project including a General Plan sheet titled "Alt 2 & Alt 3, Alta Mesa Road Bridge (Replacement)" dated May 14, 2013,
- reviewed an undated "Alt3 – Precast" drawing prepared by MTCo, received May 21, 2013,
- completed a site review and performed a seismic refraction profile on May 29, 2013, and
- completed preliminary engineering, evaluation, and analysis.

2 PROJECT LOCATION

The project site is located in southern Sacramento County northeast of the town of Herald, California (Figure-1). The bridge location is on Alta Mesa Road approximately 0.35 miles north of the intersection with State Route 104 (Twin Cities Road). Site coordinates are approximately latitude 38.31091° N and longitude 121.22637° W.

3 SITE DESCRIPTION

At the project location, Alta Mesa Road is a two lane road aligned north-south. The existing road grade is elevated up to 6± feet above the adjacent existing ground surface in the general vicinity of the bridge, but up to 8± feet is indicated at the southeast bridge corner.

The existing bridge (built in 1940) is 306.1 feet long by 22.0 feet wide, and consists of 16 timber-stringer spans with asphalt-paved concrete decking (Photo 1). The substructure consists of timber-bulkhead abutments and multi-timber pile (4) bents. A supplemental/temporary bent support has been erected adjacent to the south abutment, consisting of 6 timber columns and what appears to be a shallow timber "footing" (Photo 2). Riprap comprised of imported rock and broken concrete slabs is present at the bents near the south abutment of the bridge.

Site topography in the project vicinity is shown on the attached "Vicinity Map" (Figure-1). The existing ground surface in the vicinity of the bridge is relatively level to slightly undulating with surface drainage generally toward the west. The bridge site is located in a rural area of the county and land use near the bridge site is agricultural/open land with a few nearby structures (e.g., single family residences and small agricultural structures).

Laguna Creek flows approximately west at the bridge location, mirroring the general creek flow in the project vicinity. The creek is incised into the alluvial materials with channel banks steeper than 1:1 (horizontal:vertical) and locally near-vertical. At the bridge location the channel branches, with flow to the west and southwest (main channel). Vegetation within the channel and adjacent to the project site consists of tall grasses and weeds with scattered mature trees to the east and thick vegetation (numerous trees, brush, etc.) to the west. Channel thalweg is about 16 feet below bridge deck.

Photo 1: Alta Mesa Road Bridge over Laguna Creek looking northeast, upstream (October 25, 2012).

Photo 2: Supplemental/temporary bent support at south abutment looking east (October 25, 2012).

Details regarding the existing bridge structure are listed in the following table.

Table 1: Existing Bridge Structure Details

Bridge Element	Description
Bridge Number	24C0306
Date Constructed	1940
Number of Spans	16 (not including the temporary support near the south abutment)
Total Bridge Length	306.1 feet
Bridge Width	22.0 feet
Abutment Skew From Roadway	0°±
Abutment 1 Location	BB STA 63+69±
Abutment 2 Location	EB STA 66+74±
Abutment 1 Roadway Elevation	73.3 feet
Abutment 2 Roadway Elevation	73.3 feet
Channel Bottom (Thalweg) Elevation	57± feet
Abutment Foundation Type	Unknown

Note: Existing abutment locations/elevations referenced to current project stationing/datum.

Overhead utilities are present on the west side of the existing bridge. An underground utility marker is also located on the roadway shoulder west of the bridge. Locations of any other utilities, if present, are unknown.

4 AS-BUILT FOUNDATION DATA

At the time of this report, no as-built foundation data was available for review.

5 PROJECT DESCRIPTION

It is proposed to replace the existing bridge on a similar alignment to the existing structure with the new deck grade at about 5.5 feet higher than existing. The proposed bridge will consist of a three-span concrete slab structure supported by precast, prestressed concrete bulb-tees on two-column concrete bents and concrete wall, seat-type abutments (Alternative 3 in the "Type Selection Report"). The abutments are shown to have mechanically stabilized earth (MSE) wingwalls about 35 feet in length that parallel the road.

New bridge support is indicated to be driven Caltrans Standard Plans Class 140 (Alternative V) piles at all locations in the "Type Selection Report." The new abutments will be located on the order of 10 feet behind existing bridge abutments.

Details regarding the proposed replacement bridge structure are listed in the following table.

Table 2: Proposed Replacement Bridge Structure Details

Bridge Element	Description
Number of Spans	3
Total Bridge Length	324.0 feet (102 feet – 120 feet – 102 feet)
Bridge Width	39.5 feet
Abutment Skew From Roadway	0°
Abutment 1 Location	BB STA 63+60.00
Abutment 2 Location	EB STA 66+84.00
Abutment 1 Roadway Elevation	78.75 feet
Abutment 2 Roadway Elevation	78.90 feet
Abutment 1 Bottom of Pile Cap Elevation	66± feet
Piers 2 and 3 Bottom of Pile Cap Elevation	50± feet
Abutment 2 Bottom of Pile Cap Elevation	64± feet

Note: Proposed abutment locations/elevations referenced to current project stationing/datum.

Detailed foundation loads and pile layout have not been provided at this time.

Approach roadway improvements are anticipated to be limited to within about 400 feet of each end of the proposed bridge. No rock slope protection is shown at support locations at this time. No other channel improvements are indicated for this project. New/widened roadway embankments are indicated to have 2:1 (Horizontal:Vertical distance) side slopes. No separate retaining walls are proposed as part of this project.

6 FIELD EXPLORATION AND TESTING

Field work for this study was made on May 28, 2013 and consisted of site review/reconnaissance at the bridge location and a seismic survey. Glen Wade, PG, and Ron Loutzenhiser, GE, were the field personnel for this site visit.

A Multichannel Analysis of Surface Wave (MASW) seismic profile was completed along the west shoulder of Alta Mesa Road north of the existing bridge to determine the shear wave velocity profile at the bridge site. The data was recorded with a 24 channel ES-3000 seismometer with geophones arranged in a line running approximately north from 57 feet north of Abutment-17 of the existing bridge. Twenty-four geophones spaced at 11 foot intervals were used for this sounding. The seismic line was 253 feet long and the refraction sounding extents were located at approximately STA. 67+37 and STA. 69+90 (Figure-2).

The energy source for this testing was a 40 lbs falling weight, with a 24 inch drop, striking a steel plate. The source was located 50± feet east of the north end of the geophone string.

The data was analyzed using the SeisImager/SW software package by Geometrics to determine the shear wave profile. The results of this testing gives a generalized 1D profile of the soils centered on the middle of the geophone spread showing the shear wave velocity of the soil versus depth (Figure-3).

7 SITE GEOLOGY AND SUBSURFACE CONDITIONS

7.1 Site Geology

According to published geologic mapping[1], the site is underlain by the Modesto and Riverbank Formations, consisting of alluvial deposits. Shown under the northern abutment of the bridge (undivided) and south of the south abutment (middle unit), the Riverbank Formation consists of silt and sand, forming alluvial terraces. Shown in the channel and south abutment area, the upper member of the Modesto Formation consists of undivided alluvium (sand with minor gravel and silt).

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[1] Dawson, T.E., 2009, "Preliminary Geologic Map of the Lodi 30' X 60' Quadrangle, California: California Geological Survey, Preliminary Geologic Maps," scale 1:100,000.

No faults, landslides, or other geologic hazards are mapped at the site. Nearby faults and associated potential ground motion at the site are discussed in the site seismic conditions section of this report.

7.2 Subsurface Conditions

Based upon our site visit, geologic mapping and previous experience in the general project area, we anticipate subsurface conditions at the project site to generally consist of a near-surface layer of loose to compact silty sand and gravel (including roadway fill materials) underlain by compact to dense silty and sandy soils.

The interpreted seismic profile data shows shear wave velocities that are generally consistent with a minimum of about 1,200 feet/second (fps) near the surface and a maximum of approximately 1,400 fps at about 75 foot depth. The shear wave velocity is about 1,300 fps at 100 foot depth. The shear wave velocities appear consistent with partly cemented/consolidated alluvial materials, similar to what would be expected based on the referenced geologic mapping as discussed above.

The average shear wave velocity in the upper 100 feet (V_{S30}) of the profile is approximately 1,330 fps. Until we can verify soil conditions with geotechnical exploration, a V_{S30} velocity of 1,200 fps (360 meters/second), which corresponds to a Caltrans Soil Profile Type D (borderline Type C) per Caltrans SDC 1.6 (shear wave velocity $600 < V_{S30} < 1{,}200$ fps), is considered appropriate for preliminary bridge design.

A generalized one-dimensional soil profile with shear wave velocity of the soil versus depth is shown on Figure-3.

7.3 Groundwater

At the time of our initial site visit (October 25, 2012) there was approximately 3 feet of water in Laguna Creek (water surface about 14 feet below bridge deck, approximate elev. 60).

According to information available from the California Department of Water Resources Water Data Library website[2], the following table shows the groundwater elevation measured in three wells located near the site. The groundwater information provided below is for the last few years immediately prior to the last reliable reading.

Table 3: Groundwater Data from Nearby Wells

Well Number	Date of Last Reliable Reading	Groundwater Elev. (feet)	Well Location From Site
06N07E32P001M	November 18, 2008	-44.4 to -59.8	0.8± miles northwest
05N07E06A001M	March 18, 1986	-32.2 to -49.3	1± mile west
05N07E10D001M	October 25, 2004	-41.7 to -53.1	1.2± miles southeast

The general trend in the groundwater data indicates a lowering of the groundwater elevation. Based on the above, groundwater is not anticipated to be present for abutment construction excavations. Seepage from Laguna Creek surface water may be present in excavations extending below water surface elevation (pier locations), especially if granular soils are encountered. More detailed information on subsurface materials will be forthcoming from explorations during our design studies. Groundwater elevations are expected to vary with rainfall, creek level, local irrigation, and possibly other factors.

8 SCOUR EVALUATION

No scour study has been provided for our review. A scour analysis will be completed for this project by others. Surficial soils are considered subject to erosion and scour.

9 CORROSION EVALUATION

Corrosion evaluation of soil will be completed after the field exploration phase and included in the foundation report.

[2] http://www.water.ca.gov/waterdatalibrary/, accessed June 6, 2013.

10 SITE SEISMIC CONDITIONS

10.1 Ground Motion

The following seismic analysis and data is intended for use in preliminary design of the proposed Alta Mesa Road Bridge at Laguna Creek. The preliminary site shear wave velocity is based on the results of seismic line S-1, conservatively reduced to 360 m/s to reflect a velocity at the Site Class C / D boundary (the seismic line result was 403 m/s). The controlling spectrum for structure design at the site is a combination of the Minimum Deterministic Spectrum for California and Caltrans ARS Online Probabilistic Spectrum [2008 USGS seismic hazard map with 5% in 50 years (975-year) return period]. Figure-4 shows our recommended Preliminary ARS Curve per Caltrans "Seismic Design Criteria" (SDC 1.6) Rev. 10/2010, using the Caltrans ARS Online (V2.2.06)[3] website. A summary of our analysis follows.

The Caltrans ARS Online tool deterministic spectrum for the site is based on the minimum deterministic spectrum for California. The controlling deterministic fault is the Great Valley 06 (Midland Fault) alt2. The Foothills Fault System – south central reach section (Ione Fault) is closer to the site and has a similar spectrum.

These deterministic faults are summarized in the following table.

[3] http://dap3.dot.ca.gov/ARS_Online/, accessed May 30, 2013.

Table 4: Fault Data

Fault Parameters	Foothills Fault System-South Central Reach Section (Ione Fault)	Great Valley 06 (Midland Fault) alt2
Fault Identification Number (FID)	420	116
Maximum Moment Magnitude (M_{max})	6.1	6.8
Fault Type	Normal	Reverse
Fault Dip (degrees)	50°	30°
Dip Direction	West	West
Site-to-Fault Distance, R_{RUP}	22 km / 14 miles	41 km / 25 miles

Prior to completion of geotechnical investigation, we tentatively assign the site a shear wave velocity (V_{S30}) equal to 360 meters per second (1,200 feet per second), corresponding to the upper boundary of Soil Profile Type D, for the upper 100 feet of the soil profile. This V_{S30} is based on the shear wave velocity obtained from our seismic line S-1 conservatively reduced to the Soil Profile Type D maximum (the soil profile type typical for sites in the general project vicinity).

Caltrans structure design practice also requires an increase to spectra due to fault proximity (near-fault factor) and when the site is located over a deep sedimentary basin (basin factor). The near-fault adjustment factor is applied for locations with a site-to-rupture plane distance (R_{rup}) of 25 km (15.5 miles) or less to the causative fault. The near-fault factor increase applies to the site; the basin factor increase does not.

Based on the above information, we recommend that preliminary structure design be based on the following Caltrans SDC parameters:

- Shear Wave Velocity, V_{S30}: 360 meters per second / Soil Profile Type D;
- Maximum Moment Magnitude (M_{max}): 6.8;
- Peak Ground Acceleration (PGA): 0.23 g; and
- Controlling Spectra: Combination of the Minimum Deterministic Spectrum for California and the Caltrans ARS Online Probabilistic Spectrum (2008 USGS seismic hazard map with 975-year return period).

It should be noted that per Caltrans SDC v.1.6 Section 2.1.5, if the bridge meets the criteria of an Ordinary Standard Bridge, a reduction factor could be applied to the 5% damped response spectrum used to calculate displacement demand.

We include the Preliminary ARS Curve as Figure-4. The Preliminary ARS Curve may be updated based on data obtained from our subsurface exploration.

10.2 Liquefaction Potential

Liquefaction is a secondary effect associated with seismic loading. It can occur when saturated, loose to semicompact, granular soils (generally within 50 feet of the surface), or specifically-defined cohesive soils, are subjected to ground shaking. Tentatively, the potential for liquefaction is considered low based on the geologic mapping and our seismic line results.

Liquefaction potential of soils will be evaluated and analysis performed, if necessary, after subsurface exploration and laboratory testing of soil samples.

10.3 Surface Fault Rupture

No active fault traces are shown on published mapping to cross the site and the site is not within or adjacent to an Alquist–Priolo Earthquake Fault Zone for fault rupture hazard. No evidence of surface fault rupture was observed from our geologic reconnaissance at the bridge site. On these bases, the potential for surface fault rupture within the project limits is considered low.

10.4 Seismically Induced Settlement

During a seismic event, ground shaking can cause densification of granular soil above the water table that can result in settlement of the ground surface. For similar reasons as indicated above, we would not expect significant densification of site soils. We will further address the potential for seismically induced settlement based on data from our proposed borings.

10.5 Seismic Slope Instability

No indications of slope instability were observed at the site. We consider the potential for seismically induced slope instability along the channel banks to be very low, limited to some minor distortion.

11 DISCUSSION AND CONCLUSIONS

Soil conditions at the project site are anticipated to be adequately stable and suitable for construction of the proposed replacement bridge. The risk of fault rupture hazard appears to be remote. No over-riding geologic hazards (e.g., faulting, landslides, subsidence, etc.) were identified by either published geologic mapping or site reconnaissance performed for this study. Geologic mapping indicates the site is underlain by alluvial sediments of Pleistocene age. The bedload/alluvium materials are considered susceptible to erosion, and within the channel, to scour.

Based on the limited data presented above, structure support achieved by means of driven piles or cast-in-drilled-hole (CIDH) piles can be considered at this site. We would not expect spread footings to be applicable for the proposed bridge due to potentially variable soil conditions at footing level and possible scour concerns. The applicability of any particular foundation type at this site will need to be determined / confirmed after completion of subsurface exploration based on actual subsurface soil conditions and defined loading (compressive, tensile and lateral) requirements. A general discussion of both foundation alternatives is provided below.

Based on geologic mapping and the shear wave velocity profile, the possibility of encountering loose or soft materials that are susceptible to secondary seismic affects appears limited to near-surface soils (existing fill, channel bedload, and upper alluvial materials). If present, these materials will likely be removed and replaced near foundation and abutment locations during replacement bridge construction.

Existing foundations are of unknown type. Their removal could result in disturbed soils at / near proposed bearing elevations for new bridge foundation elements. Existing utilities will need to be protected or moved / replaced during construction.

12 PRELIMINARY FOUNDATION RECOMMENDATIONS

12.1 Driven Piles

Tentatively, the use of Caltrans Standard Class 90 or Class 140 concrete and steel pipe displacement-type piles appears feasible at this site. The presence of dense to very dense consistency soils and/or cemented soils at shallow depth may preclude the use of such piles or require undersize drilling to assist driving and obtain adequate embedment for lateral stability. Driven steel H-piles may also be feasible at this site and possibly more suitable than displacement-piles depending on subsurface conditions encountered, controlled by depth to dense/cemented soil. We would expect that driven steel H-pile sections would be longer than concrete piles of the same capacity

Without consideration of future scour, presence of unsuitable soil layers, or other adverse factors, the required embedment length for a 12-inch square-section concrete pile driven in compact/dense soils is generally estimated to range from 30 to 35 feet for 90 kip (service) loads and 40 to 45 feet for 140 kip (service) loads.

12.2 Cast-In-Drilled-Hole (CIDH) Piles

The use of CIDH piles is also considered feasible at this site. Such piles may require special installation measures, including temporary casing, slurry drilling methods (with inspection tubes) and the use of minimum 24-inch diameter CIDH piles for tremie concrete placement if granular soils and significant seepage of surface / channel water are encountered. Caltrans Standard 16-inch diameter CIDH piles are not considered suitable for wet installation methods and should not be considered for foundation support unless subsurface exploration for the design level foundation report indicates wet installation methods will not be required.

Standard (Caltrans) 24-inch diameter CIDH piles can be assigned up to 200 kip design (service) load per pile in compression. The tip elevations for CIDH pile foundations will depend on the subsurface soil profile, depth to scour, pile diameter and defined compressive, tensile and lateral loading requirements.

13 APPROACH FILL EARTHWORK

It is expected that embankment construction will be generally feasible without special fill foundation preparation or settlement concerns. However, after completing our field explorations, emergent conditions may be realized that would require special considerations for settlement or overexcavation and compaction to prepare fill foundations. The existing fill materials might also be considered for re-use depending on the make-up and suitability of the existing fill. Site Grading and Earthwork should be performed in accordance with Section 16 and Section 19 of Caltrans "Standard Specifications", respectively.

14 CONSTRUCTION CONSIDERATIONS

14.1 Existing Foundations

The existing structure is to be removed prior to construction and disruption from existing bridge removal might require increased foundation depth or other considerations. Based on the preliminary project data, there is apparent conflict between the proposed pier locations and existing bridge piers. If possible, locate new bridge foundations clear of existing foundations. If / where existing foundation elements remain, we recommend that they be cut off one foot below ground line and left in place to minimize disturbance of bearing materials for new foundations.

14.2 Excavation and Shoring

Based on our review of existing site data, surface observations, and interpretation of the refraction seismic profile, we expect excavation can be achieved with typical heavy construction equipment. Tentatively, we anticipate that granular soils associated with existing fill, bedload, and alluvium materials will be consistent with Type C soils in accordance with CalOSHA.

14.3 Dewatering

Groundwater can be expected to be encountered below the creek surface water elevation (pier excavations). We expect seasonal fluctuation of groundwater level and possibly perched groundwater. We anticipate that groundwater would likely not be encountered in abutment excavations during dry season construction (June through October).

15 ADDITIONAL GEOTECHNICAL SERVICES

This report is preliminary and not to be used for final design. Taber Consultants current scope of services includes subsurface exploration, associated laboratory testing of soil samples, and engineering analysis to complete a foundation report for final bridge design. The subsurface exploration will further define the quality and characteristics of the subsurface materials with respect to bearing capacity, excavation characteristics, and presence of seepage / groundwater.

We expect to complete the borings and laboratory tests within the next few months. Taber will obtain earth material samples for evaluation and laboratory testing that is expected to include: moisture content and unit weight, unconfined compression tests, soil classification, and corrosivity analysis (resistivity, pH, Sulfate Content and Chloride Content). Laboratory testing will also include R-value tests for pavement design of approach roadways.

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2012-0128-1

TABER CONSULTANTS

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W. Eric Nichols
C.E.G. 2229

June 7, 2013
REL/WEN

GENERAL CONDITIONS

The conclusions and recommendations of this study are professional opinion based upon the indicated project criteria and the limited data described herein. It is recognized there is potential for variation in subsurface conditions and modification of conclusions and recommendations might emerge from further, more detailed study.

This report is intended only for the purpose, site location and project description indicated and construction in accordance with Caltrans practice. This report is preliminary and not intended for final design. Additional study, including subsurface exploration, laboratory testing, and engineering analysis is required for final design.

As changes in appropriate standards, site conditions and technical knowledge cannot be adequately predicted, review of recommendations by this office for use after a period of two years is a condition of this report.

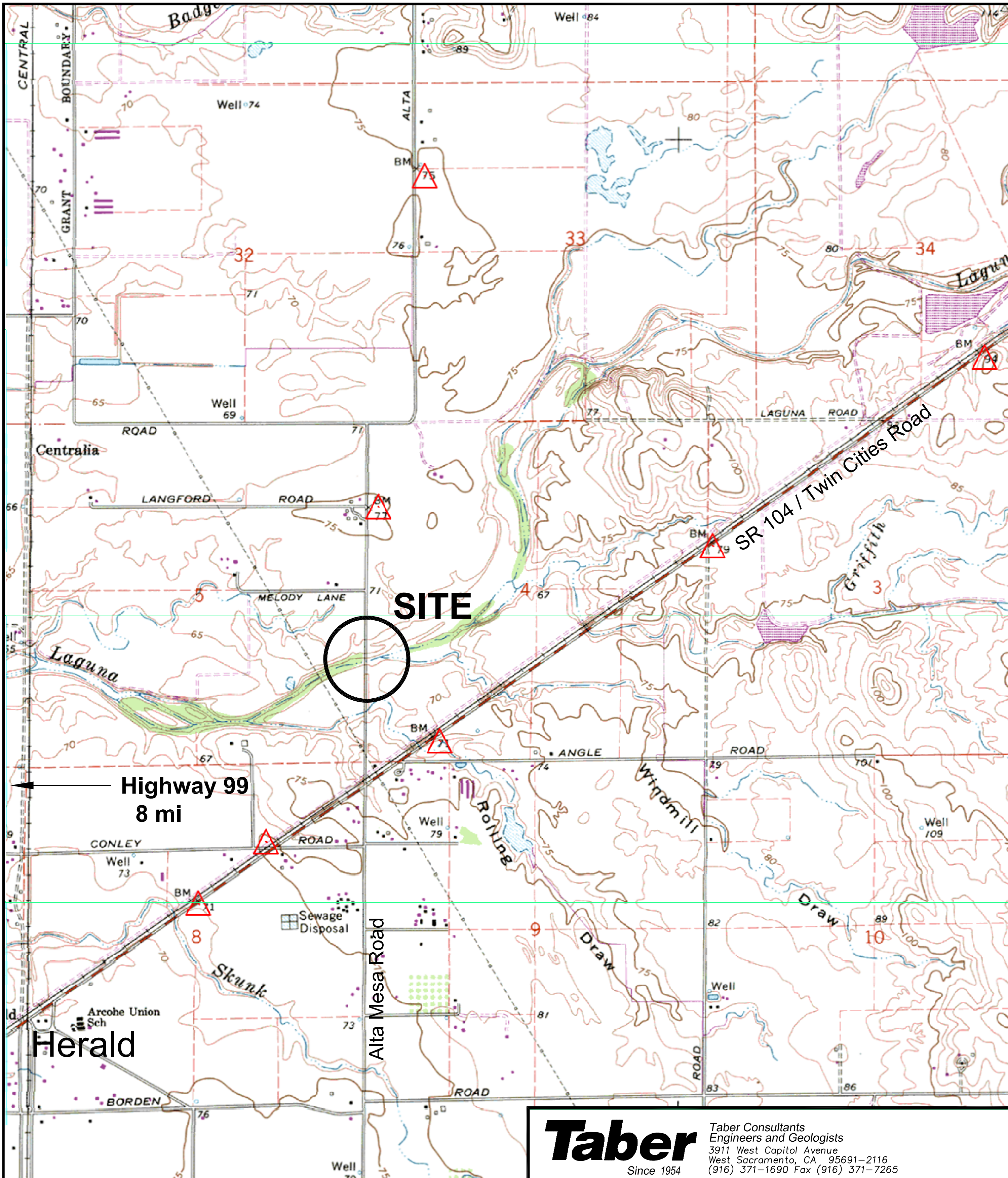
A review by this office of any foundation and/or grading plans and specifications or other work product insofar as they rely upon or implement the content of this report, together with the opportunity to make supplemental recommendations as indicated therefrom is considered an integral part of this study and a condition of recommendations.

Subsequently defined construction observation procedures and/or agencies are an element of work, which may affect supplementary recommendations.

Should there be significant change in the project or should soils conditions different from those described in this report be encountered during construction, this office should be notified for evaluation and supplemental recommendations as necessary or appropriate.

Opinions and recommendations apply to current site conditions and those reasonably foreseeable for the described development--which includes appropriate operation and maintenance thereof. They cannot apply to site changes occurring, made, or induced, of which this office is not aware and has not had opportunity to evaluate.

The scope of this study specifically excluded sampling and/or testing for, or evaluation of the occurrence and distribution of, hazardous substances. No opinion is intended regarding the presence or distribution of any hazardous substances at this or nearby sites.



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Scale: 1:24,000

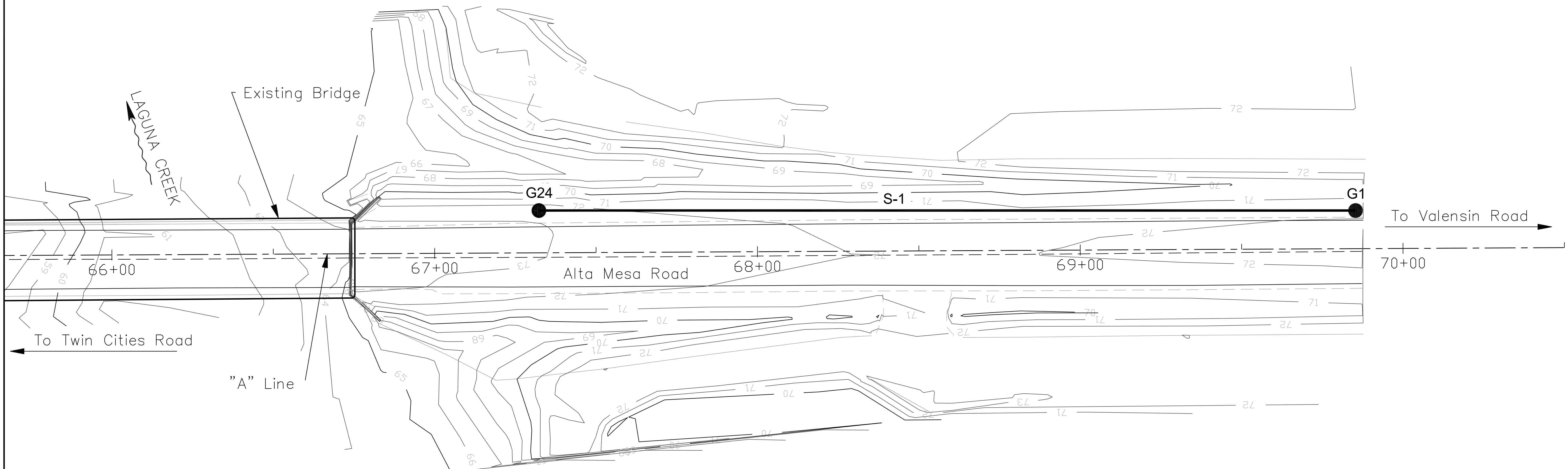
USGS
"Clay" CA
QUADRANGLE 7.5 MINUTE
SERIES (TOPOGRAPHIC),
DATE 1968

Mark Thomas & Company

Alta Mesa Road Bridge at Laguna Creek
Sacramento County, CA

Vicinity Map

2012-0128-1		Figure - 1



LEGEND:

S-1 G1 G24 APPROXIMATE LOCATION OF SEISMIC REFRACTION LINE

Z

PLAN

0 30 ft 60 ft

SCALE

NOTE: All locations are approximate and are referenced from existing site features.
Preliminary Site Plan provided electronically by Mark Thomas & Company on 3/13/2013.

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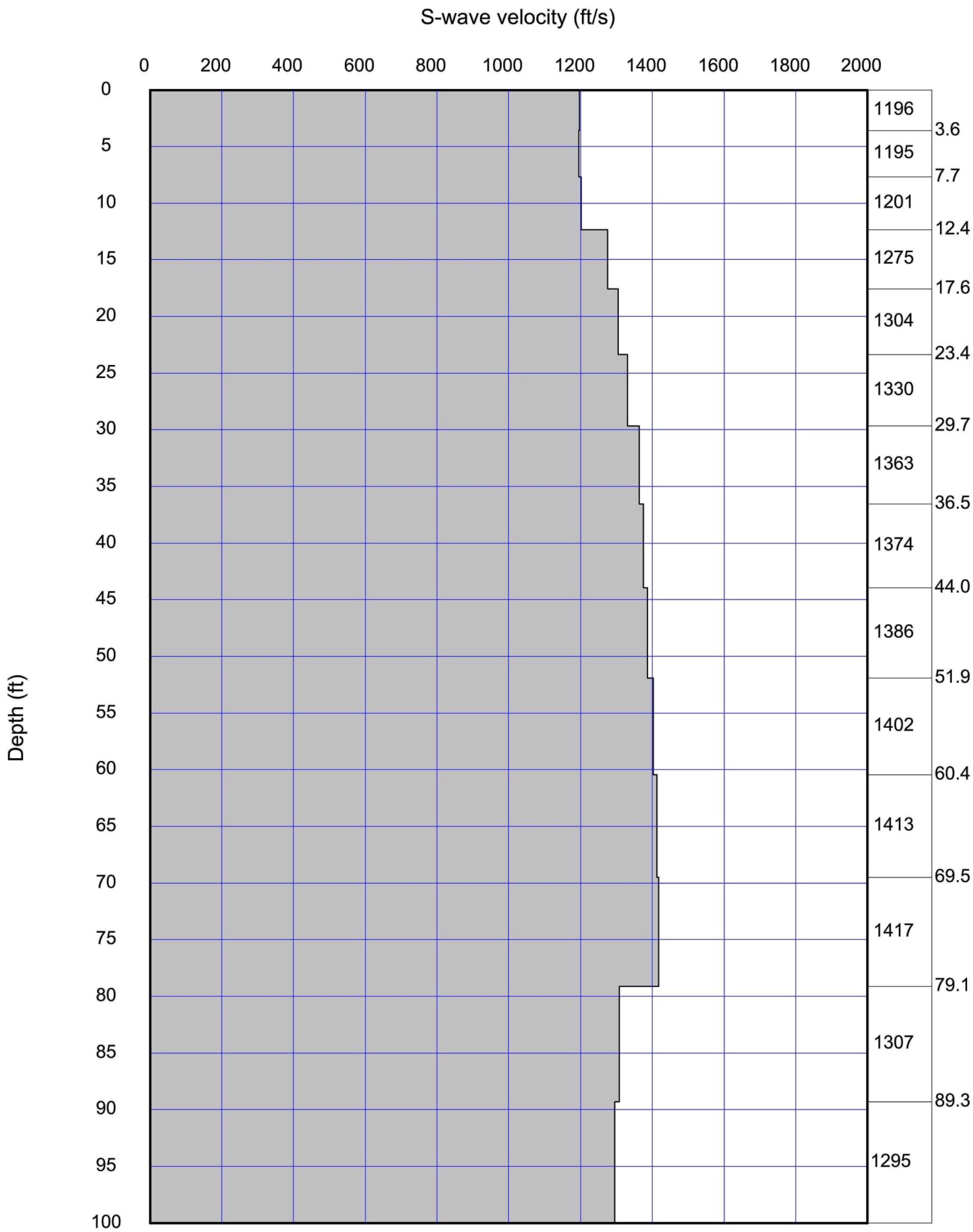
Alta Mesa Road Bridge at Laguna Creek
Sacramento County, California

Location of field Tests

2012-0128-1	June 2013	Figure - 2



Average Vs 100ft = 1332.0 ft/sec 406 meters/sec

Note: Interpreted from data recorded along Seismic Refraction Profile S-1

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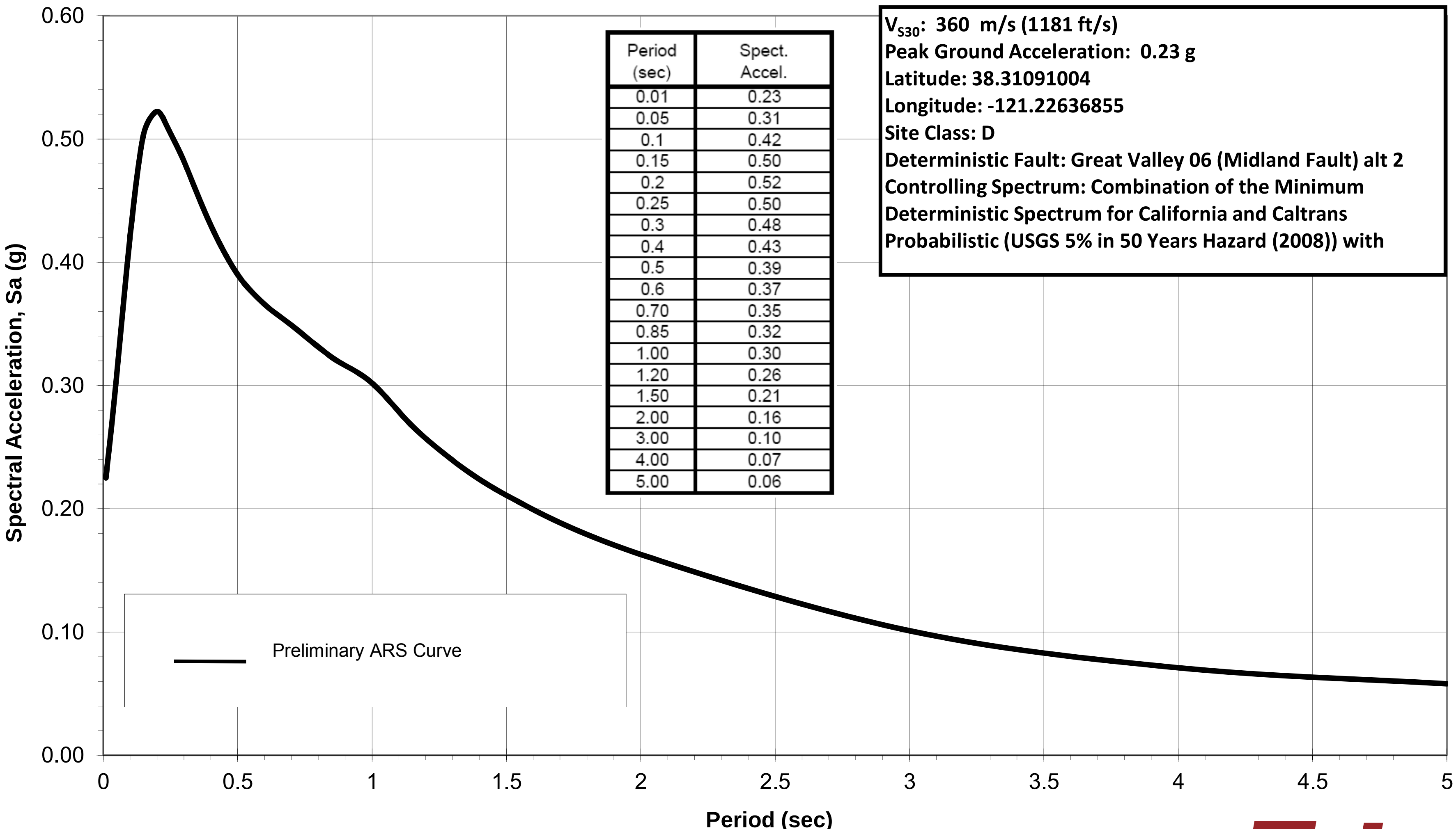
Alta Mesa Road Bridge at Laguna Creek
Sacramento County, California

Shear Wave Velocity vs. Depth

2012-0128-1		Figure - 3

2012-0128-1
May 2013

Preliminary Caltrans ARS Curve
Alta Mesa Road Bridge at Laguna Creek
Sacramento County, California



Period (sec)	Spect. Accel.
0.01	0.23
0.05	0.31
0.1	0.42
0.15	0.50
0.2	0.52
0.25	0.50
0.3	0.48
0.4	0.43
0.5	0.39
0.6	0.37
0.70	0.35
0.85	0.32
1.00	0.30
1.20	0.26
1.50	0.21
2.00	0.16
3.00	0.10
4.00	0.07
5.00	0.06

V_{S30}: 360 m/s (1181 ft/s)
Peak Ground Acceleration: 0.23 g
Latitude: 38.31091004
Longitude: -121.22636855
Site Class: D
Deterministic Fault: Great Valley 06 (Midland Fault) alt 2
Controlling Spectrum: Combination of the Minimum Deterministic Spectrum for California and Caltrans Probabilistic (USGS 5% in 50 Years Hazard (2008)) with

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Figure-4

PRELIMINARY HYDROLOGY & HYDRAULICS

Alta Mesa Road at Laguna Creek Bridge Replacement Project
Sacramento County, California
Existing Bridge No. 24C-0306
Federal Project No. BRLS-5924(182)

Draft Bridge Design Hydraulic Study Report

Prepared for:

Prepared by:

WRECO

July 2013

Alta Mesa Road at Laguna Creek Bridge Replacement Project
Sacramento County, California
Existing Bridge No. 24C-0306
Federal Project No. BRLS-5924(182)

Draft Bridge Design Hydraulic Study Report

Submitted to:
Sacramento County

This report has been prepared by or under the supervision of the following Registered Engineer. The Registered Civil Engineer attests to the technical information contained herein and has judged the qualifications of any technical specialists providing engineering data upon which recommendations, conclusions, and decisions are based.

____
Han-Bin Liang, Ph.D., P.E.
Registered Civil Engineer

____
Date

July 2013

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Figures

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Appendices

Executive Summary

Sacramento County proposes to replace the existing two-lane Alta Mesa Road bridge over Laguna Creek with a new two-lane bridge that will meet the AASHTO minimum standards. The new bridge would be approximately 18 ft longer, and 17 ft wider than the existing bridge. The work area, including the new bridge and reconstructed approaches, would total approximately 1,700 linear feet.

A General Plan for Design Alternatives 2 and 3, provided by Mark Thomas & Company, was used in support of the hydraulic analysis.

The purpose of this Bridge Design Hydraulic Study is to determine the design flow characteristics of the existing condition and the proposed bridge, as well as to determine the scour potential and make recommendations on the need for scour countermeasures for the proposed bridge.

Design flows adopted for this study were generated using a weighted combination of Log Pearson Type III analysis and United States Geological Survey (USGS) Regional Flood-Frequency Equations. USGS stream gage 11336530 was selected in support of the discharge calculations due to its proximity to the Project site. The weighted 100-year discharge was selected for the hydraulic analysis.

Summary of Discharges

Location	50-Year Discharge (cfs)	100-Year Discharge (cfs)
Project Site	15,750	18,260

The hydraulic analyses for the existing and proposed conditions were assessed using the United States Army Corps of Engineers' (USACE) Hydrologic Engineering Center River Analysis System (HEC-RAS) hydraulic modeling software (version 4.1.0). Based on the hydraulic analyses, the 100-year water surface elevations (WSEs) for the proposed bridge design alternatives would be lower than the existing 100-year WSEs. The larger hydraulic opening, caused by an increase in deck elevation and reduction of pier obstruction, allows for greater conveyance, lowering the WSE and resulting in reduced backwater upstream of the bridge crossing.

Summary of Water Surface Elevations

<table>
<tr><th rowspan="2">Condition</th><th rowspan="2">Location</th><th rowspan="2">Minimum Soffit Elevation (ft, NAVD 88[1])</th><th colspan="2">50-Year Storm Event</th><th colspan="2">100-Year Storm Event</th></tr>
<tr><th>WSE (ft, NAVD 88)</th><th>Available Freeboard (ft)</th><th>WSE (ft, NAVD 88)</th><th>Available Freeboard (ft)</th></tr>
<tr><td>Existing</td><td rowspan="2">Just Upstream of Alta Mesa Road bridge at Laguna Creek</td><td>71.6</td><td>71.6</td><td>0.0</td><td>72.4</td><td>-0.8</td></tr>
<tr><td>Proposed</td><td>73.4</td><td>71.4</td><td>2.0</td><td>72.1</td><td>1.3</td></tr>
</table>

[1] North American Vertical Datum of 1988

Scour calculations were performed based on the Federal Highway Administration's (FHWA) Hydraulic Engineering Circular No. 18 (HEC-18), *Evaluating Scour at Bridges*. The design 100-year WSEs and channel flow velocities from the hydraulic analyses, and channel bed particle size information based on soil data obtained from the U.S. Soil Conservation Service, were used as the input parameters for the scour analysis. The total scour measurements for the proposed bridge design during the 100-year storm are summarized in the following table.

Summary of Scour Calculations

Bridge Component	Thalweg Elevation (ft, NAVD 88)	Local Scour Depth (ft)	Contraction Scour Depth (ft)	Long-term Scour Depth (ft)	Total Scour Depth (ft)
Abutment 1 (South)	56.1	4.2	5.7	0.0	9.9
Pier 2	56.1	11.2	5.7	0.0	16.9
Pier 3	56.1	11.2	5.7	0.0	16.9
Abutment 4 (North)	56.1	0.1	5.7	0.0	5.8

According to the *Caltrans Bridge Memo to Designers, 1-23 Hydraulic and Hydrologic Data,* dated October 2003, the bridge footings supported on soil or degradable rock should be embedded below the maximum scour depth. Footing supported on massive, competent rock formations, which are highly resistant to scour, should be placed directly on the cleaned rock surface. The total scour listed in the summary of scour calculations table is a combination of all scour calculations, assuming no RSP has been placed around the abutments, and the abutment footing is not located on competent rock formations.

RSP calculations were performed using the Caltrans' *California Bank and Shore Rock Slope Protection Design* (October 2000). According to the Caltrans design criteria, ¼ ton RSP would be recommended to protect the channel banks from erosion.

Acronyms

BIR	Bridge Inspection Report
BFE	Base Flood Elevation
Caltrans	California Department of Transportation
CFR	Code of Federal Regulations
CFS	cubic feet per second
DWR	Department of Water Resources
FEMA	Federal Emergency Management Agency
FHWA	Federal Highway Administration
FIRM	Flood Insurance Rate Map
FIS	Flood Insurance Study
HEC-18	Hydraulic Engineering Circular No. 18
HEC-23	Hydraulic Engineering Circular No. 23
HEC-RAS	Hydrologic Engineering Center River Analysis System
LPIII	Log Pearson Type III Statistical Distribution
NFIP	National Flood Insurance Program
NAVD 88	North American Vertical Datum of 1988
RSP	rock slope protection
USGS	United States Geological Survey
USACE	United States Army Corps of Engineers
WSE	water surface elevation

1 GENERAL DESCRIPTION

1.1 Project Description

The Alta Mesa Road Bridge Replacement Project (Project) includes the replacement of the existing 16-span bridge. The existing bridge is located on Alta Mesa Road approximately 1,500 feet (ft) north of the intersection with State Route 104 (Twin Cities Road). The Project site is approximately 3.3 miles (mi) northeast of the City of Galt in Sacramento County. See Figure 1 for the Project Location Map and Figure 2 for the Project Vicinity Map.

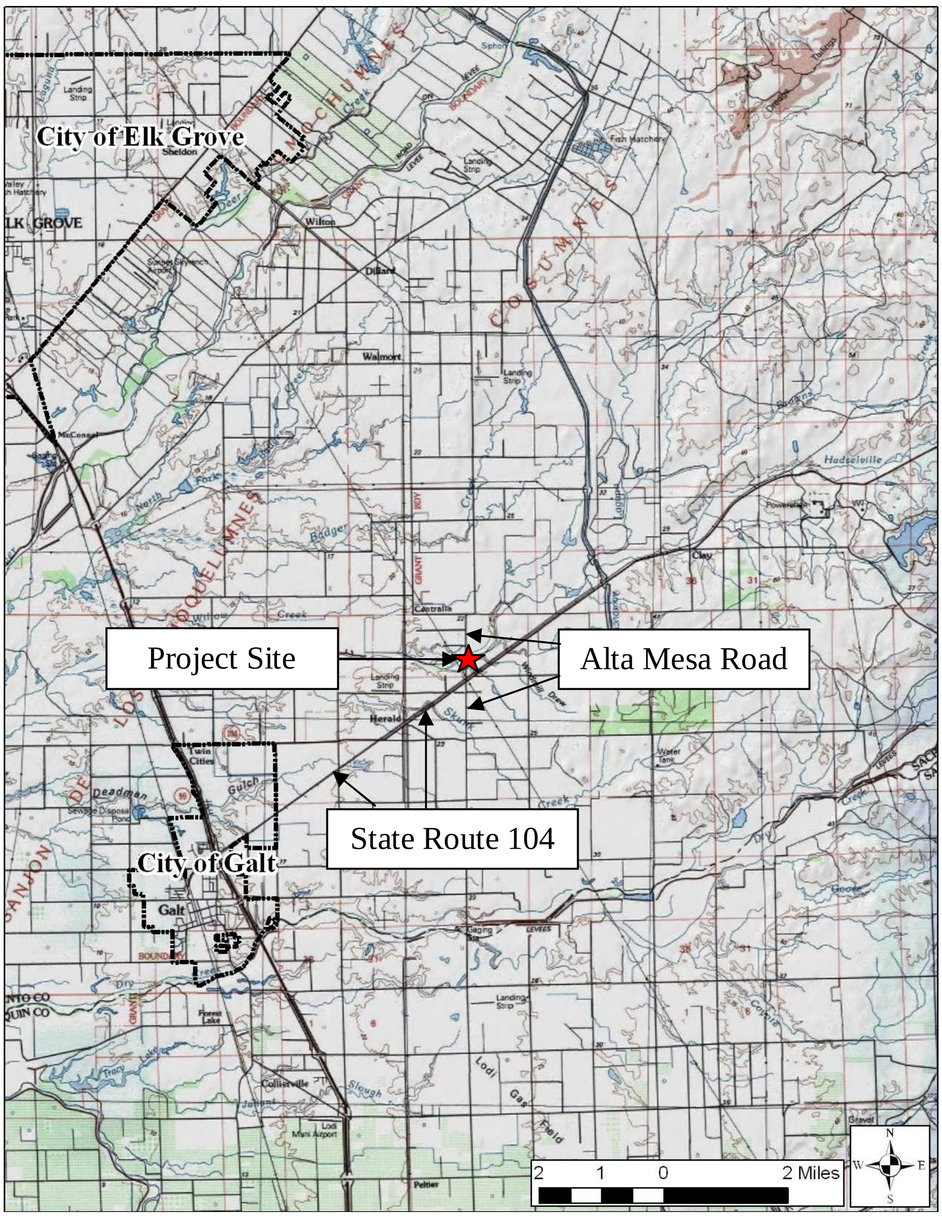

Figure 1. Project Location Map

Source: United States Geological Survey (USGS)

The purpose of this Project is to replace the structurally deficient and functionally obsolete Alta Mesa Road bridge with a two-lane crossing over Laguna Creek that will meet American Association of State Highway and Transportation Officials (AASHTO) minimum standards.

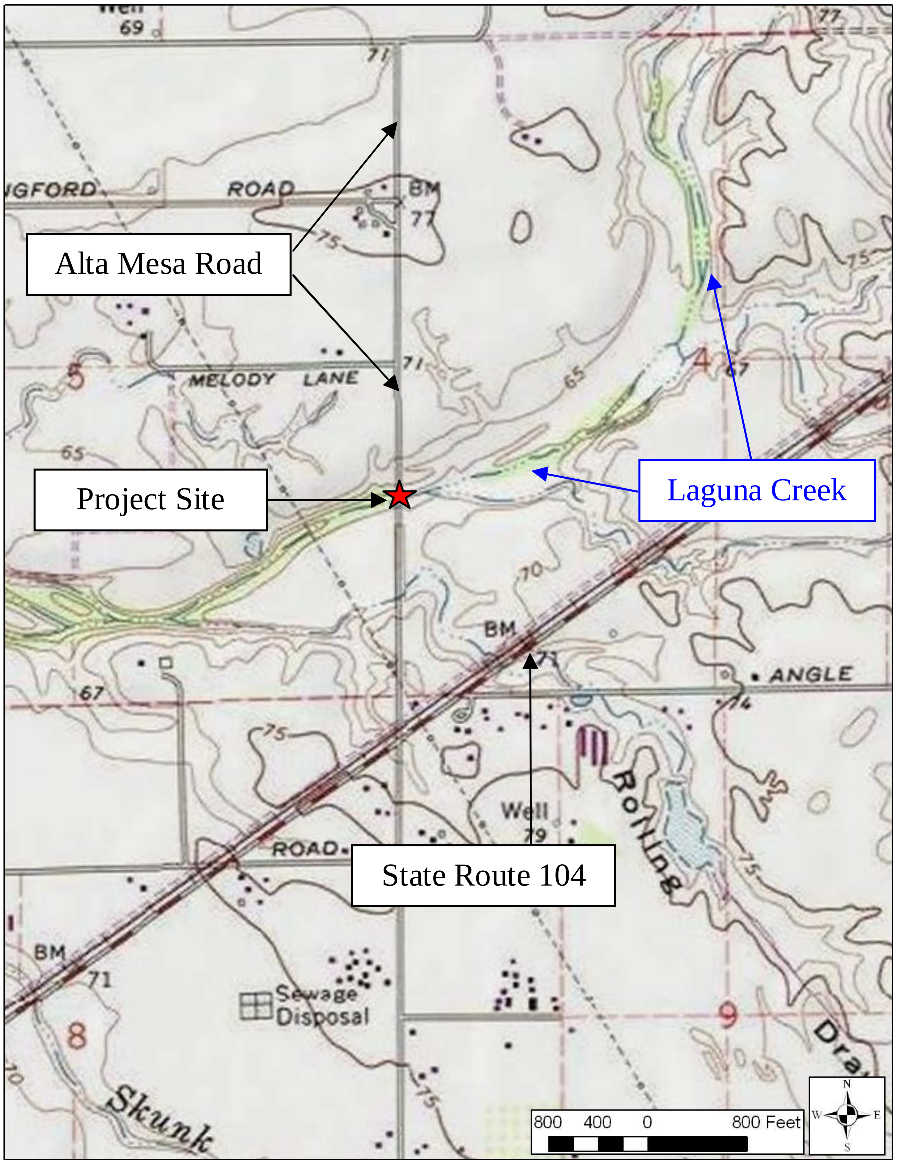

Figure 2. Project Vicinity Map

Source: USGS

The existing bridge is approximately 306 ft long. The proposed alternative replacement design would be 324 ft long, and would have three spans.

This Project would not involve permanent modification or alteration of the streambed, although temporary alterations would occur. The Project would require the removal of existing concrete supports and bents. The new bridge would be designed to allow flood flows to pass without overtopping.

1.2 Need for Project

The existing Alta Mesa Road bridge sufficiency rating is 59.8, and it is functionally obsolete (Caltrans, 2011). The exterior stringers (1 and 15) on either side of the structure in all spans show signs of weathering and dry rot. String 1 in Span 14 was found to have a horizontal split which extends the majority of the member length and measures up to 1.5 in wide (see Photo 1). There is moderate/severe dry rot present in some of the timber lagging boards at Abutment 1. The bridge is eligible and has been programmed for Highway Bridge Program (HBP) funding for replacement.

Photo 1. Crack in Existing Stringer at the Alta Mesa Road bridge over Laguna Creek

Source: Caltrans

1.3 Key Tasks

Key tasks performed for this study include:

1) Review of available hydrologic data;
2) Hydrologic analysis to determine flow rates for Q_{100} and Q_{50};
3) Hydraulic analysis to determine design water surface elevations (WSEs) and flow velocities for the existing and proposed conditions;
4) Scour analysis to estimate potential scour depths; and
5) Rock slope protection (RSP) calculations to estimate required RSP weight and/or diameter.

1.4 Design Criteria

1.4.1 Hydrologic Design Criteria

At least two analysis methods were required to estimate hydrologic design criteria for the Project per the Local Assistance Program Guidelines from Caltrans. WRECO used:

1. USGS weighted discharge eEstimate for an ungaged site along a gaged stream
2. USGS Regional Flood-Frequency Equations

1.4.2 Hydraulic Design Criteria

The hydraulic design of the bridge will follow the Federal Highway Administration's (FHWA's), Caltrans', and Sacramento County's criteria.

The FHWA criterion for the hydraulic design of bridges is that they be designed to pass the 2% probability of annual exceedance flow (50-year recurrence interval design discharge) with adequate freeboard, where practicable, to account for debris and bedload. Two feet of freeboard is commonly used in preliminary bridge designs. The bridge should also be designed to pass the 1% probability of annual exceedance flow (100-year recurrence interval design discharge, or base flood). No freeboard is added to the base flood.

1.4.3 Scour Design Criteria

The evaluation of potential scour at the proposed bridge will follow the criteria described in the FHWA's Hydraulic Engineering Circular No. 18 (HEC-18), *Evaluating Scour at Bridges* (Fifth Edition). The evaluation of potential scour shall be based on the 100-year design discharge hydraulic characteristics. The total scour depth was estimated based upon the cumulative effects of the long-term bed elevation change, general (contraction) scour, and local scour. The life expectancy of the bridge was considered in determining the long-term bed elevation change of the waterway. The long-term bed elevation change was based on an assumed 75-year design life for a new replacement bridge.

1.4.4 Rock Slope Protection Design Criteria

Two procedures for determining RSP design were considered: 1) the FHWA's Hydraulic Engineering Circular No. 23 (HEC-23), *Bridge Scour and Stream Instability Countermeasures: Experience, Selection, and Design Guidance* (Third Edition) (September 2009); and 2) Caltrans' *California Bank and Shore Rock Slope Protection Design* (Third Edition) (October 2000).

2 DESCRIPTION OF WATERSHED

2.1 Geographic Location

The Project site is located along Laguna Creek approximately 9.4 miles upstream of the confluence with the Cosumnes River at latitude 38°18’39.17” North and longitude 121°13’34.86” West, North American Datum of 1983. The Project site is within Sacramento County, in the Sacramento-San Joaquin River Delta. The average elevation of the watershed is approximately 272 ft, North American Vertical Datum of 1988 (NAVD 88). The average elevation in the immediate Project vicinity is approximately 60 ft NAVD 88.

2.2 Watershed Size

The contributing drainage basin to the Project site is approximately 114.9 mi^2 (see Figure 3).

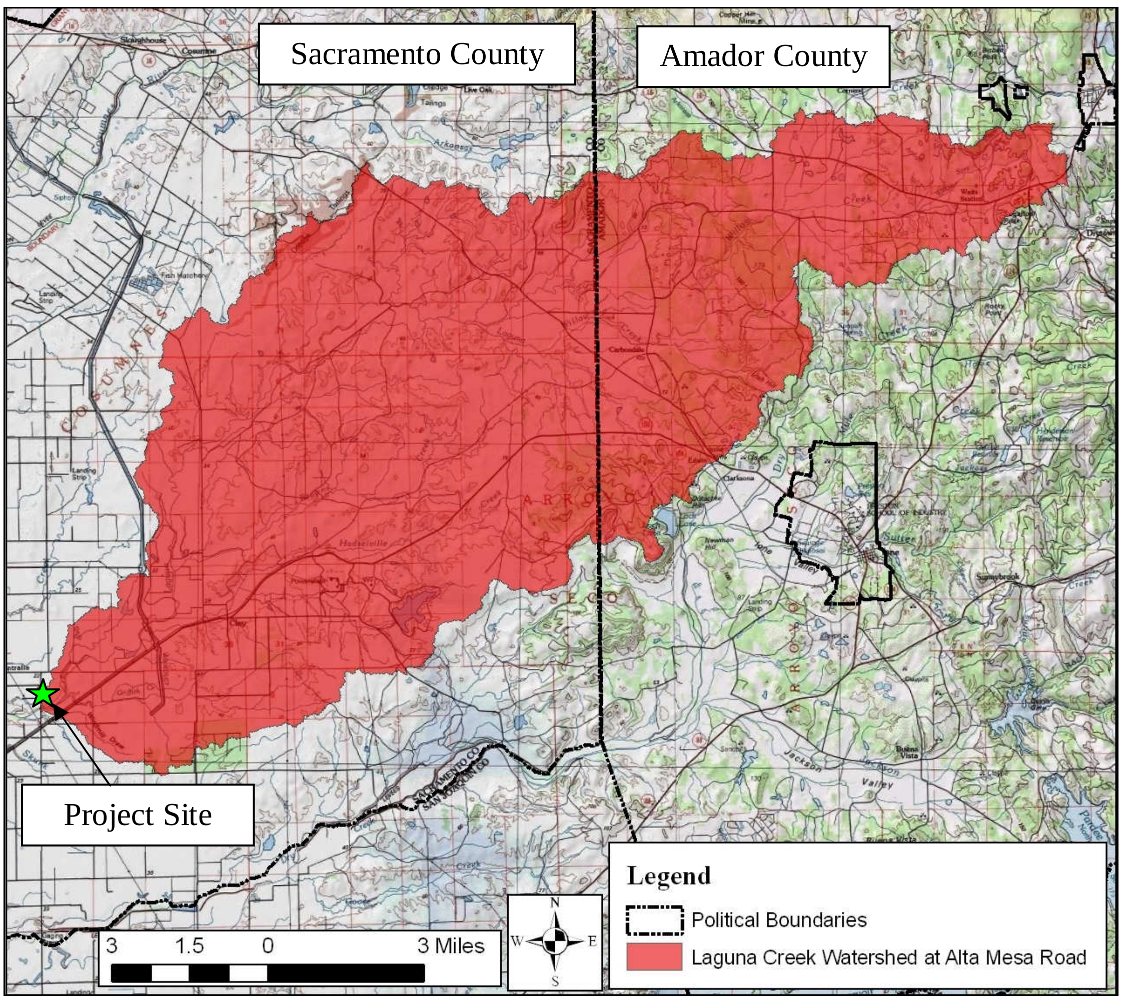

Figure 3. Project Watershed Map

Sources: (Federal Emergency Management Agency) FEMA, USGS

The headwaters of the Laguna Creek watershed are located in Amador County, and flow west into Sacramento County, before draining into the Cosumnes River.

2.3 Precipitation

The centroid of the Project watershed is located approximately at latitude 38°23’40.59” North and longitude 121°4’4.8” West, North American Datum of 1983. Using the USGS Streamstats application, the mean annual precipitation at the centroid of the watershed is approximately 22 inches.

2.4 Land Use

Existing land use conditions for the portion of the watershed in Sacramento County were obtained from the General Plan Land Use Diagram published on November 9, 2011 by Sacramento County. Existing land use conditions for the portion of the watershed in Amador County were obtained from the Amador County General Plan dated October 2007. The Project watershed was superimposed on top of the Land Use Diagrams (see Figure 4), indicating that the watershed consists of mostly agricultural lands. Additionally, mineral resource and industrial zones, as well as public and quasi-public zones were identified within the watershed.

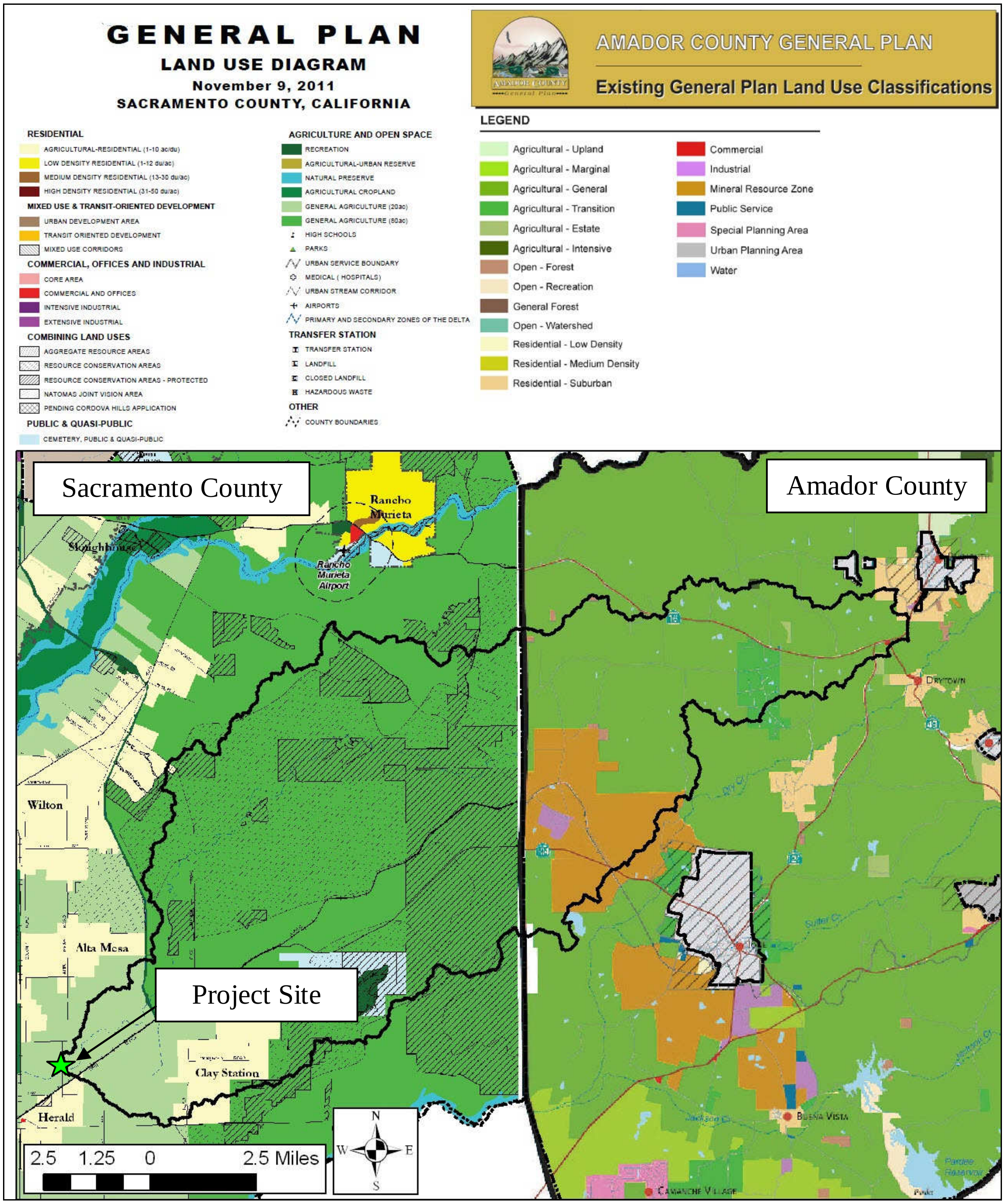

Figure 4. Land Use Map

Sources: Sacramento County, Amador County, USGS

3 DESCRIPTION OF STREAM AND SITE

3.1 Channel Properties

Laguna Creek flows west as it crosses Alta Mesa Road. The Caltrans *Bridge Inspection Report* (BIR) for the Alta Mesa Road bridge at Laguna Creek, dated June 15, 2011, states that the channel consists of an earth-lined, U-shaped pilot stream in a flat watercourse (see Photo 2).

Photo 2. Laguna Creek Low Flow Channel at Alta Mesa Road Bridge Crossing
Source: Caltrans

3.2 Soil and Bed Material

The Natural Resources Conservation Service (NRCS) Web Soil Survey (WSS) was used to determine the soil type in the channel at the Alta Mesa Road crossing of Laguna Creek. According to the WSS, the soil consists of Columbia sandy loam and clayey substratum, with 0 to 2 percent slopes, and is occasionally flooded.

3.3 Existing Bridge

The existing 16-span bridge consists of a Portland cement concrete deck on timber stringer spans and timber cap-and-post bents, with timber bulkhead abutments. The bridge was constructed in 1940.

3.4 Proposed Bridge

The Project proposes to replace the existing bridge with a 324 ft long, 39.5 ft wide three-span bridge. Design plans were provided by Mark Thomas & Company (MTCO) (see Figure 5). The roadway approach embankment would be modified, with a 2:1 (H:V) grade. The approach roadways would modify existing grades with limits located approximately 670 ft south and 710 ft north of Laguna Creek, respectively.

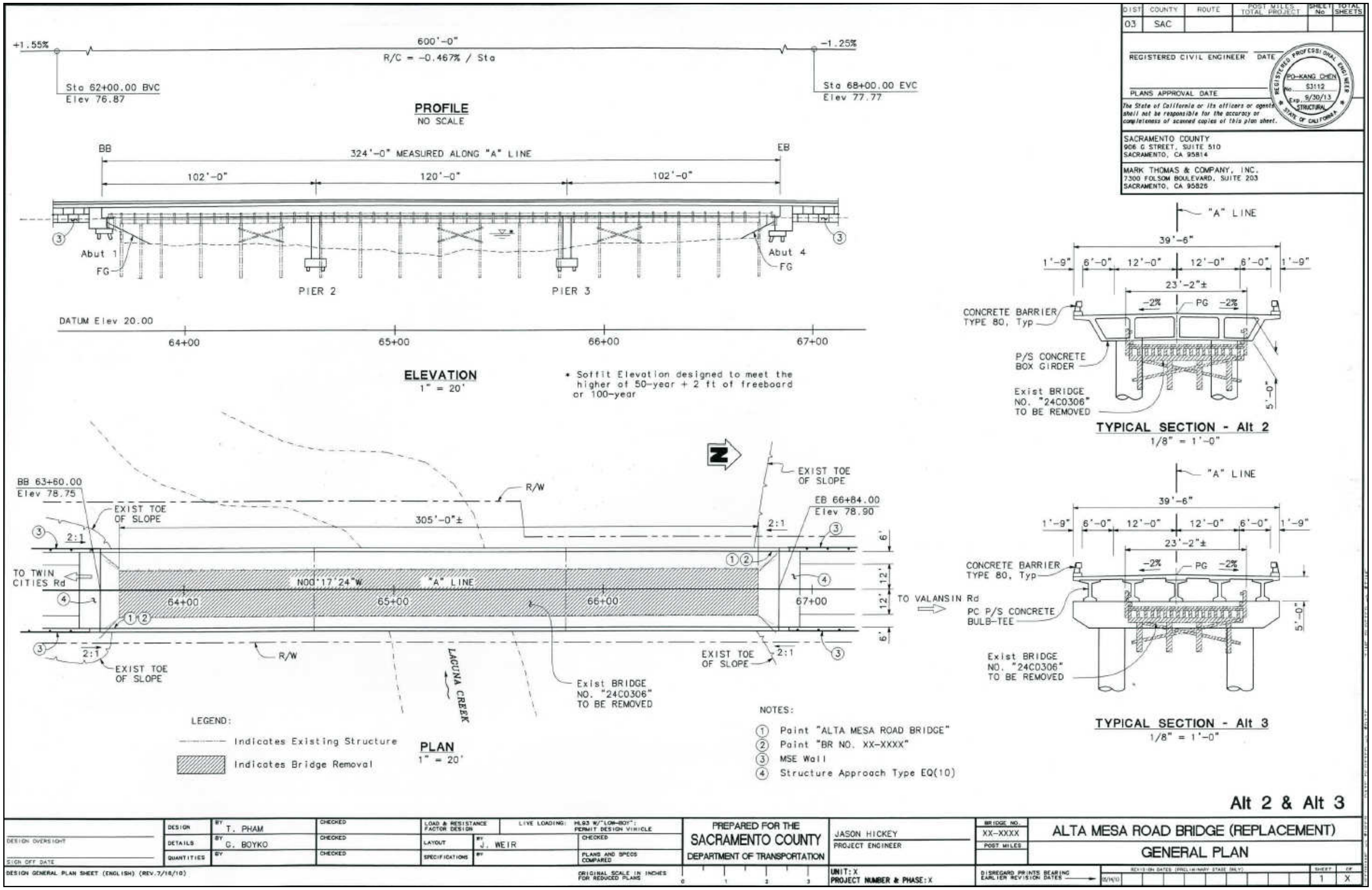

Figure 5. Proposed Alta Mesa Road Bridge Replacement General Plan

Source: MTCO

4 HYDROLOGY

Multiple data sources were considered in support of hydrologic analyses for Laguna Creek at the Alta Mesa Road Bridge, including, but not limited to, USGS Regional Flood-Frequency Equations, USGS stream gage annual peak discharge measurements, and USGS topographic data. Additionally, discharge data for two locations within the Laguna Creek watershed were analyzed, the locations of which are shown in Figure 6.

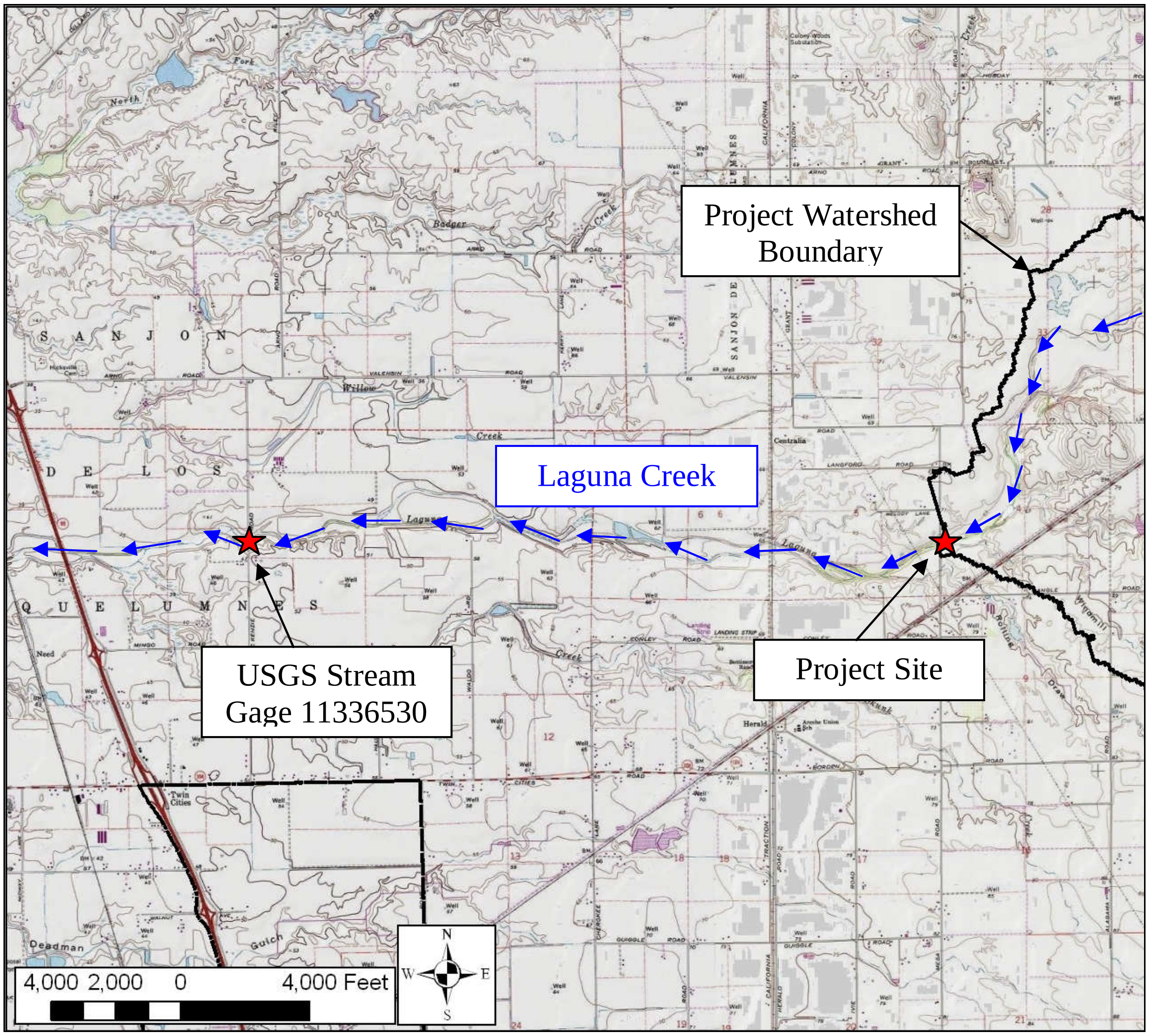

Figure 6. Hydrologic Feature Locations

Source: USGS

4.1 Analysis Methods

Two methodologies were used in support of hydrologic calculations to determine the design storm discharges impacting the Project site including USGS regional flood-frequency equations and USGS weighted discharge estimates for an ungaged site along a gaged stream. Log-Pearson Type III (LPIII) statistical distributions of annual peak discharge measurements were used in support of the USGS weighted discharge estimates.

4.1.1 USGS Regional Flood-Frequency Equations

Flood-frequency equations were developed by the USGS and based on analysis of data from stream gage stations. California is divided into six regions; the Project site is within the Sierra Nevada region. These flood-frequency equations are generally used to estimate stream flow for ungaged sites that are not affected by substantial urban development and that are natural (unregulated) streams.

On July 18, 2012, the USGS released updated regional flood-frequency equations. These equations are based on annual peak-flow data through water year 2006 for 771 streamflow-gaging stations in California having 10 or more years of data. The updated equations were used in support of the Project's hydrologic analysis.

The basin characteristics at the Project site are listed in Table 1.

Table 1. USGS Regional Flood-Frequency Parameters

Location	Parameter	Laguna Creek Watershed	Hydrologic Region
USGS Stream Gage 11336530	Drainage Area	122.7 mi^2	Sierra Nevada
	Mean Annual Precipitation	21.8 in	
	Mean Basin Elevation	259 ft	
Project Site	Drainage Area	114.9 mi^2	
	Mean Annual Precipitation	22.0 in	
	Mean Basin Elevation	272 ft	

Source: USGS

The flood-frequency equations are as follows (*Gotvald, 2012*):

$$Q_{50} = 21.1(DA)^{.879}(EL)^{-.316}(P)^{1.31}$$
$$Q_{100} = 20.6(DA)^{.874}(EL)^{-.250}(P)^{1.24}$$

Where:

Q_n = peak discharge for a storm event with a return period of n-years in cubic feet per second (cfs)
DA = drainage area (mi^2)
EL = mean basin elevation (ft)
P = mean annual precipitation (in)

USGS Gage 11336530 was also selected as a location for hydrologic analysis due to its proximity to the Project site and its availability of peak discharge measurements along Laguna Creek. The resulting design storm discharges are listed in Table 2 on the following page.

Table 2. USGS Regional Flood-Frequency Discharges

Location	50-Year Discharge (cfs)	100-Year Discharge (cfs)
USGS Stream Gage 11336530	14,160	15,700
Project Site	13,320	14,810

4.1.2 USGS Weighted Discharge Estimate

USGS Gage 11336530 is located on Laguna Creek, adjacent to McKenzie Road. The stream gage is approximately 4.3 miles downstream of the Alta Mesa Road bridge crossing of Laguna Creek. Annual peak discharge measurements were acquired between 1972 and 1986.

Sauer (1974) presented the following method to improve flood-frequency estimates for an ungaged site near a streamgage, on the same stream, having 10 or more years of peak-flow record. The weighted flow estimate for the ungaged site ($Q_{P(u)w}$) is computed using the following equation:

$$Q_{P(u)w} = \left[\left(\frac{2\Delta A}{A_{(g)}}\right) + \left(1 - \frac{2\Delta A}{A_{(g)}}\right)\left(\frac{Q_{P(g)w}}{Q_{P(g)r}}\right)\right] Q_{P(u)r}$$

Where

$Q_{P(u)w}$ is the weighted estimate of peak flow for the selected P-percent annual exceedance probability at the ungaged site, *u*, in cubic feet per second (cfs)

ΔA is the absolute value of the difference between the drainage areas of the streamgage and ungaged site, in mi^2

$A_{(g)}$ is the drainage area for the streamgage, in mi^2

$Q_{P(g)w}$ is the weighted estimate of peak flow for any P-percent annual exceedance probability for a streamgage, in cfs

$Q_{P(u)r}$ is the peak-flow estimate derived from the applicable regional equations, for the selected P-percent annual, exceedance probability at the ungaged site, *u* in cfs

$Q_{P(g)r}$ is the peak-flow estimate for the P-percent annual exceedance probability at the streamgage derived from the applicable regression equations, in cfs

In order to determine the weighted discharge estimates at the Project site, weighted estimates at USGS gage 11336530 are required. The Interagency Advisory Committee on Water Data recommends that better estimates of flood-frequency statistics for a streamgage can be obtained by combining (weighting) at-site flow estimates determined from LPIII analysis of the annual peak flows with flow estimates obtained for the streamgage from regression equations (determined in Section 4.1.1). Peak flow estimates derived from LPIII analyses at USGS stream gages considered for use in the USGS regression equations for California were obtained from Table 4 of the USGS *Methods for Determining Magnitude and Frequency of Floods in California, Based on Data through Water Year 2006.* Variances of prediction for streamgage LPIII analysis and regional regression estimates are found in tables 6 and 10 of the same report. The weighted discharge estimates at the gage were determined using the following equation:

$$\log Q_{P(g)w} = \frac{VP_{P(g)r} \log Q_{P(g)s} + VP_{P(g)s} \log Q_{P(g)r}}{VP_{P(g)s} + VP_{P(g)r}}$$

Where

$Q_{P(g)w}$ is the weighted estimate of peak flow for any P-percent annual exceedance probability, for a streamgage, g, in cfs;

$VP_{P(g)r}$ is the variance of prediction at the streamgage derived from the applicable regression equations for the selected P-percent annual exceedance probability, in log units;

$Q_{P(g)s}$ is the estimate of peak flow at the streamgage from the LPIII analysis for the selected P-percent annual exceedance probability, in cfs

$VP_{P(g)s}$ is the variance of the prediction at the streamgage from the LPIII analysis for the selected P-percent annual exceedance probability, in log units

The weighted design storm discharges were calculated at the USGS stream gage downstream of the Project site, using both LPIII and regression methods and associated variances. The LPIII discharge values, and other parameters obtained from the USGS *Methods for Determining Magnitude and Frequency of Floods in California, Based on Data through Water Year 2006*, are included in Table 3. This information was then used to determine the weighted discharge at the Project location; the results are summarized in Table 4.

Table 3. USGS Determined Parameters at Stream Gage 11336530

USGS Information at Stream Gage 11336530			
Recurrence Interval	LPIII Peak Q (cfs)	Variance of Regression Peak Flow Estimate (log units)	Variance of LPIII Peak Flow Estimate (log units)
50-Year	20,800	0.049	0.055
100-Year	26,000	0.0503	0.0624

Source: USGS

Table 4. Weighted Discharge Estimates

Location	Weighted Q50 (cfs)	Weighted Q_{100} (cfs)
USGS Stream Gage 11336530	17,120	19,880
Project Site	15,750	18,260

4.2 Design Discharge Summary

The flow measurements for Laguna Creek from the USGS stream gage are approximately 30 to 40 years old. Due to the age of the stream gage data, historical aerial imagery was examined and indicates no significant development within the watershed. Flows developed using the USGS Weighted Discharge Method are more detailed, and provide more conservative results when compared to the Regional Flood-Frequency Equations. Therefore, WRECO selected 15,.750 cfs for the 50-year discharge and 18,260 cfs for the 100-year discharge.

4.3 Hydrologic Stability

According to the Sacramento and Amador counties' general plans, the vast majority of the Project watershed consists of agricultural and mineral resource areas. Therefore, significant development that would impact watershed runoff rates is not anticipated.

5 HYDRAULIC ANALYSIS

The following sections discuss the developments of the hydraulic models and summarize the results for the existing and proposed alternative conditions. The water surface profile plots, hydraulic summary tables, and channel cross sections are included in Appendix A for the existing bridge and Appendix B for the proposed bridge.

5.1 Design Tools

The hydraulic conditions at the Project site were evaluated using the HEC-RAS Version 4.1.0., which is hydraulic modeling software that was developed by the USACE. The analyses were performed for the existing and proposed replacement bridges for Alta Mesa Road.

5.2 Cross Section Data

Nine cross sections along a 970 ft reach of Laguna Creek were used for the hydraulic analyses. Surveyed elevation data points provided by MTCO referencing the vertical datum NAVD 88 were used as the basis for the cross section geometries. The upstream and downstream limits of the hydraulic model are approximately 420 ft and 550 ft away from the existing Alta Mesa Road bridge over Laguna Creek, respectively.

5.2.1 Existing and Proposed Bridges

The geometry of the existing Alta Mesa Road bridge in the hydraulic model is based on the information found in the Caltrans BIR as well as survey information provided by MTCO. The structural design and roadway profile for the proposed bridge alternative is based on information provided by MTCO.

5.3 Model Boundary Condition

There are two Laguna Creeks within Sacramento County. No WSE was published by FEMA along the Laguna Creek that drains into the Cosumnes River. Therefore, normal depth was used as the downstream reach boundary condition rather than a known WSE. A slope of 0.0022 ft/ft was used as the downstream boundary condition. No upstream boundary condition was used in the hydraulic analyses.

5.4 Manning’s Roughness Coefficients

Manning’s n values were used in the hydraulic model to estimate energy losses in the flow due to friction. Manning’s n values were selected to best describe the existing and proposed channel characteristics of Laguna Creek based on site descriptions in the Caltrans BIR, and photos from a field inspection. A roughness coefficient of 0.055 was selected for the left and right overbank areas. Within the channel banks, a roughness coefficient of 0.035 was selected for low flow channels, and 0.06 was selected for the remaining portions within the channel banks that contain dense vegetation.

5.5 Expansion and Contraction Coefficients

Expansion and contraction coefficients were used in the hydraulic model to estimate hydraulic losses at transitions between cross sections. The expansion and contraction coefficients used in the channel were 0.3 and 0.1, respectively. These values represent a channel with gradual transitions between cross sections. The expansion and contraction coefficients used in the vicinity of the bridge were 0.5 and 0.3, respectively. These values represent the flow interference caused by the bridge.

5.6 Water Surface Elevations

The design 50- and 100-year WSEs at the existing and proposed conditions in the vicinity of the Alta Mesa Road bridge at Laguna Creek are summarized in Table 5 and Figure 7 through Figure 10.

Table 5. Summary of Water Surface Elevations, Existing and Proposed Conditions

Location	Water Surface Elevation (ft, NAVD 88)			
	50-Year Storm		100-Year Storm	
	Existing	Proposed	Existing	Proposed
392 ft Upstream of Existing Bridge	72.4	72.3	73.3	73.1
250 ft Upstream of Existing Bridge	71.5	71.3	72.3	72.0
122 ft Upstream of Existing Bridge	71.4	71.2	72.2	71.9
Just Upstream of Existing Bridge	71.6	71.4	72.4	72.1
Just Downstream of Existing Bridge	71.3	71.3	72.0	72.0
116 ft Downstream of Existing Bridge	71.3	71.3	72.0	72.0
257 ft Downstream of Existing Bridge	71.2	71.2	71.9	71.9

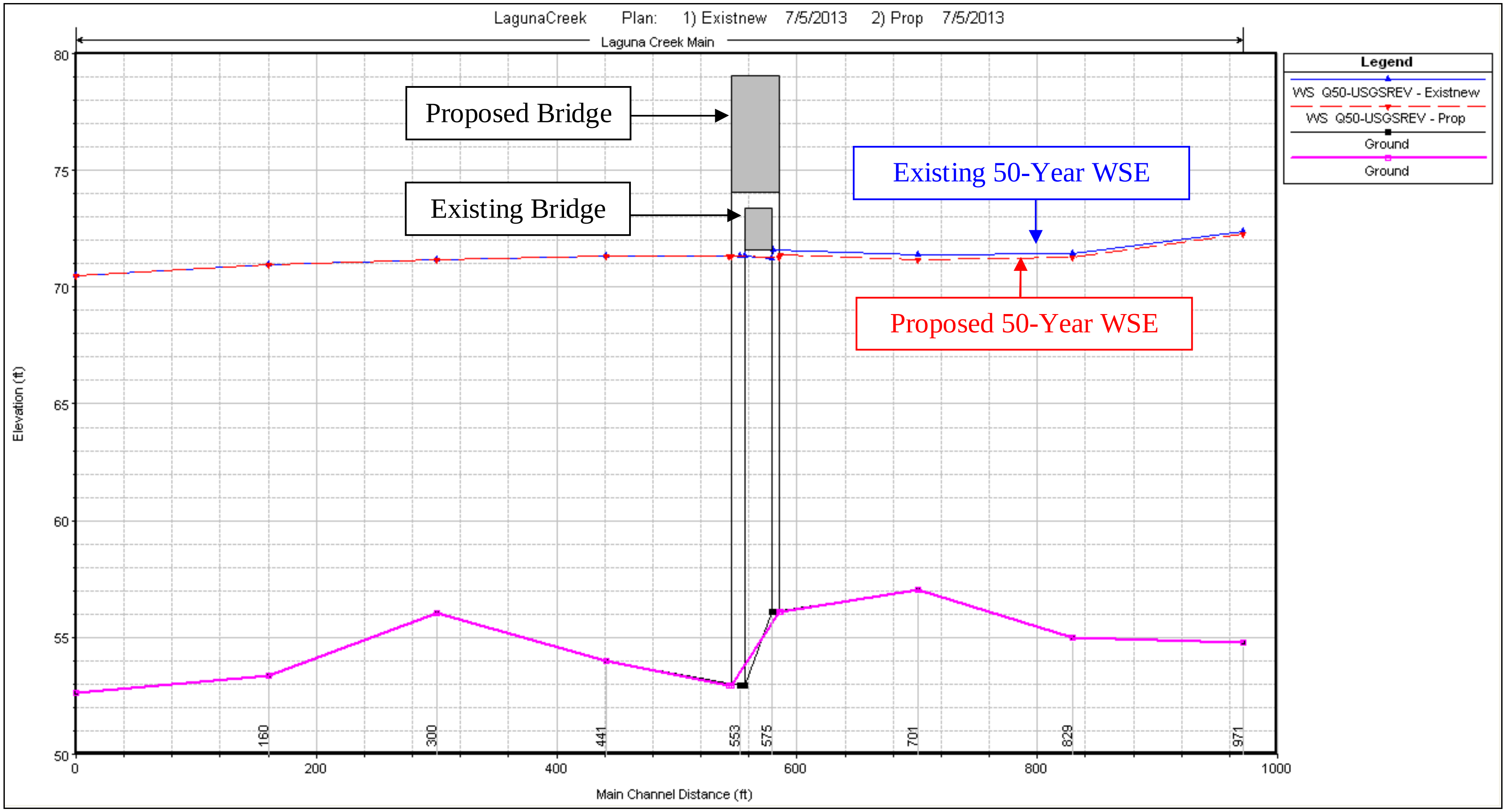

Figure 7. 50-Year Water Surface Profiles for Existing and Proposed Conditions

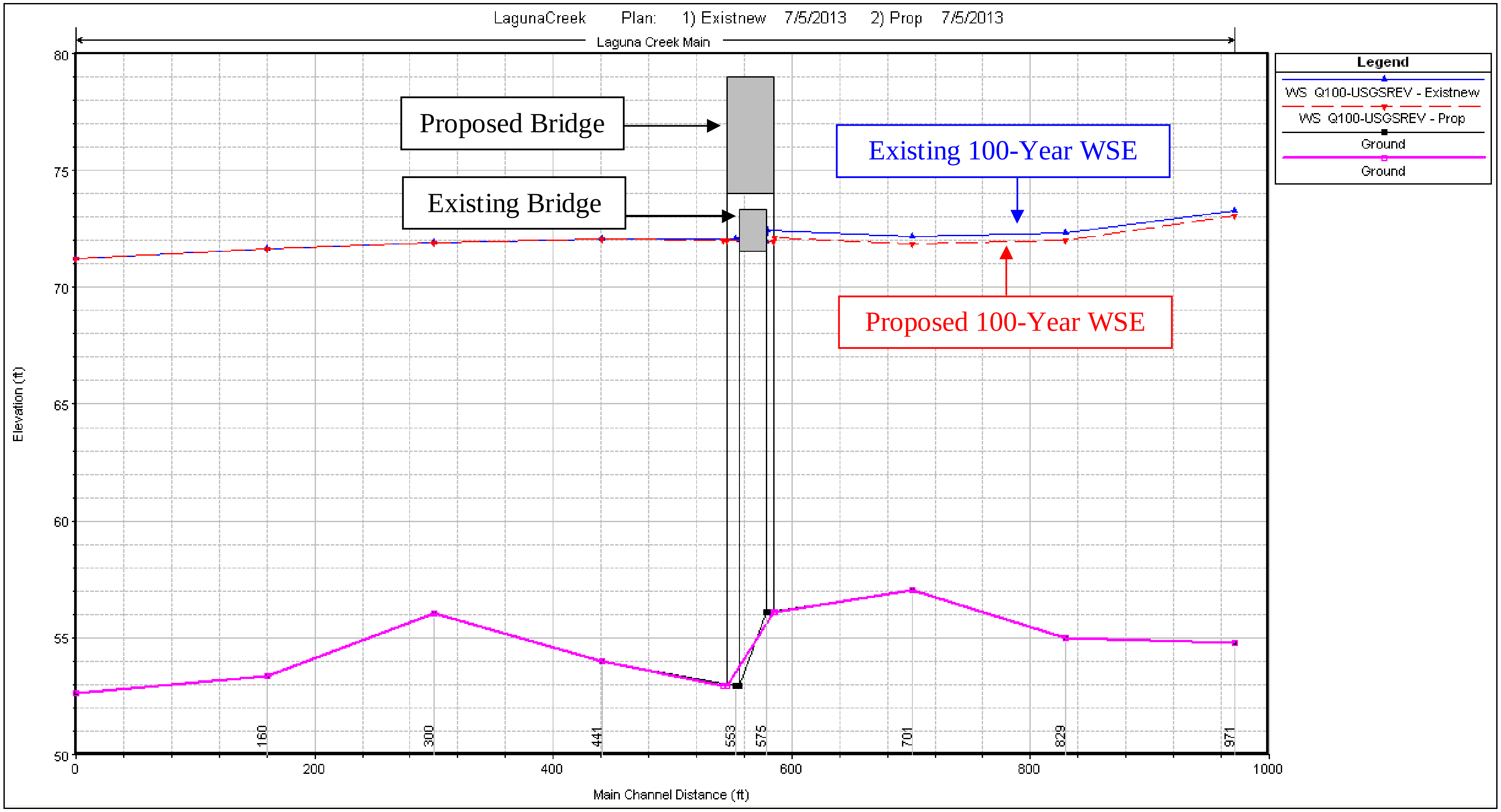

Figure 8. Existing 100-Year Water Surface Profiles for Existing and Proposed Conditions

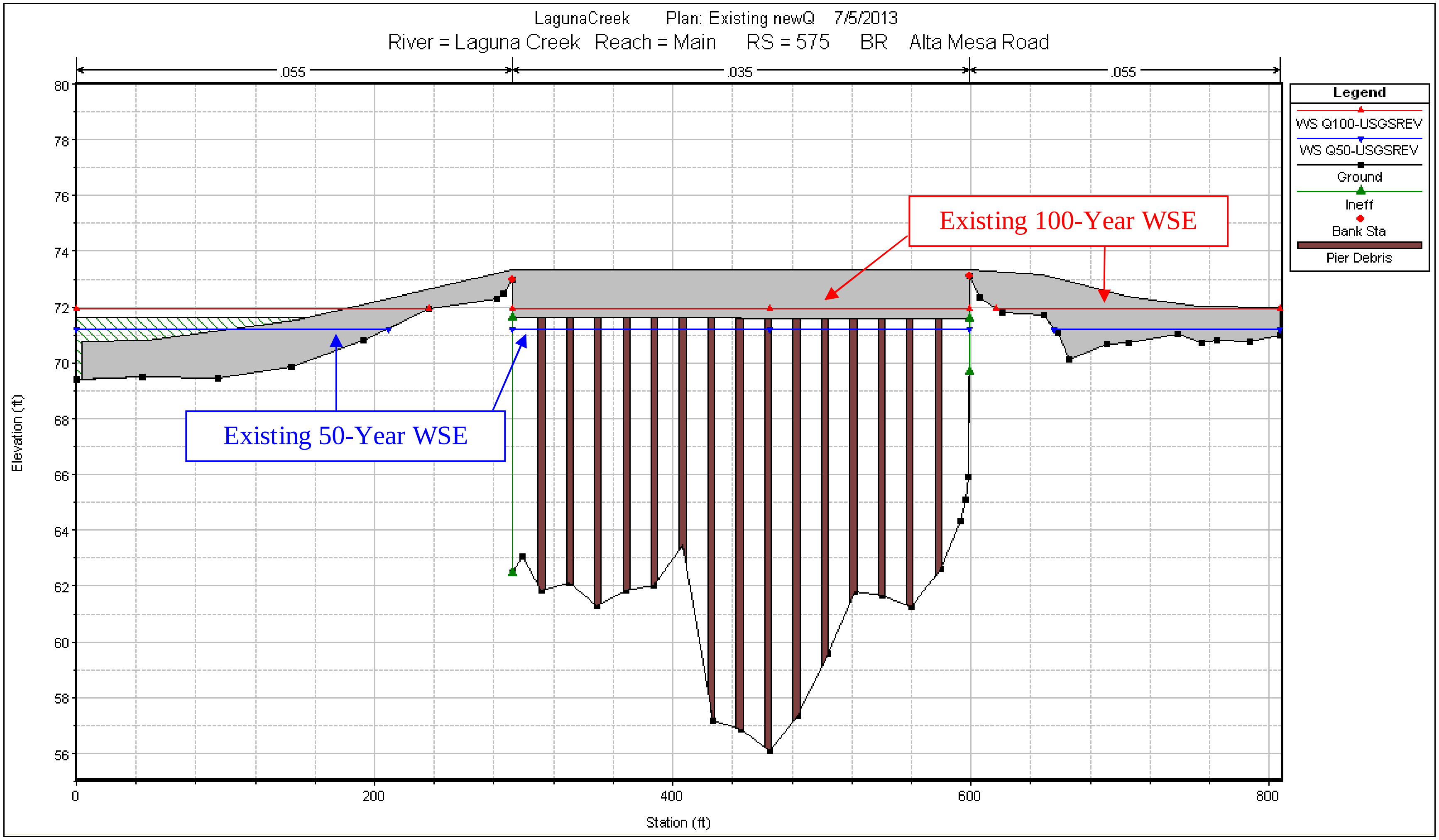

Figure 9. Upstream Face of Existing Alta Mesa Road Bridge over Laguna Creek

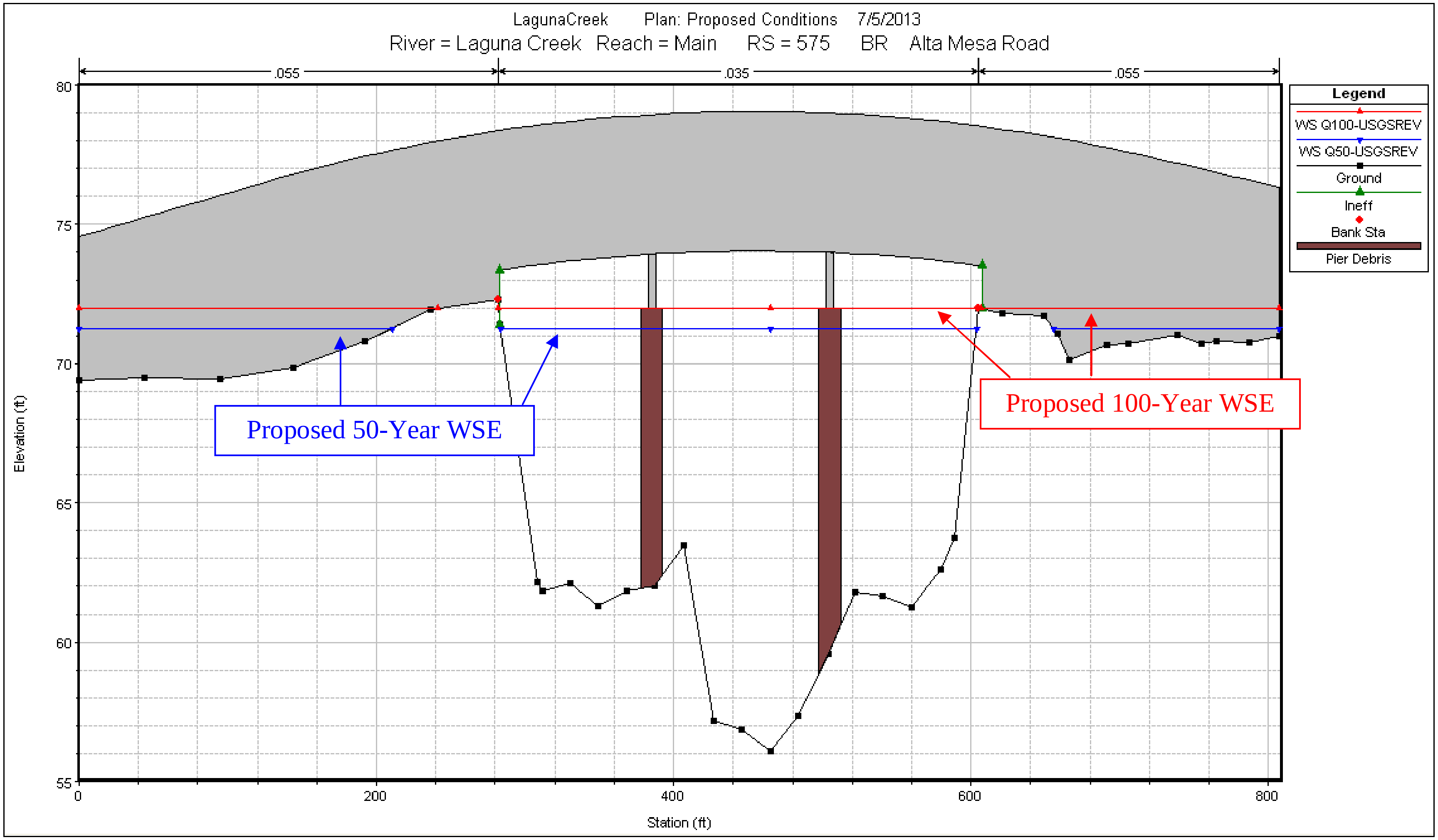

Figure 10. Upstream Face of Proposed Alta Mesa Road Bridge over Laguna Creek

5.7 Freeboard

The available freeboard for the proposed bridge during the 50-year storm event is greater than the Caltrans recommended freeboard of 2 ft for preliminary bridge design. See Table 5 for a summary of available freeboard of the existing and proposed bridges.

Table 6. Summary of Freeboard

Condition	Location	Minimum Soffit Elevation (ft, NAVD 88[1])	50-Year Storm Event		100-Year Storm Event	
			WSE (ft, NAVD 88)	Available Freeboard (ft)	WSE (ft, NAVD 88)	Available Freeboard (ft)
Existing	Just Upstream of Alta Mesa Road bridge at Laguna Creek	71.6	71.6	0.0	72.4	-0.8
Proposed		73.4	71.4	2.0	72.1	1.3

5.8 Upstream Flow Velocities

The average channel flow velocities through Laguna Creek in the Project vicinity during the 50- and 100-year storm events are summarized in Table 6. The results of the hydraulic analyses indicated that the average channel flow velocities would increase upstream of the proposed Alta Mesa Road bridge replacement because of the increased conveyance capacity of the proposed replacement bridge.

Table 7. Summary of Average Channel Flow Velocities

Location (in relation to existing bridge)	Average Channel Flow Velocity (ft/sec)			
	50-Year Storm		100-Year Storm	
	Existing	Proposed	Existing	Proposed
392 ft Upstream	7.3	7.4	7.5	7.7
250 ft Upstream	11.5	11.8	11.9	12.4
122 ft Upstream	9.2	9.4	9.7	10.1
Just Upstream	4.7	4.9	5.0	5.3
Just Downstream	4.4	4.4	4.8	4.8
116 ft Downstream	3.6	3.6	3.8	3.8
257 ft Downstream	3.9	3.9	4.2	4.2

6 SCOUR ANALYSIS

The scour calculations assume that the channel bed material is erodible. The following sub-sections summarize the results of the analysis, and the detailed calculations are included in Appendix C.

6.1 Design Criteria

WRECO evaluated bridge scour per the criteria described in the FHWA HEC-18, Fifth Edition. The evaluation of potential scour was based on the 100-year design discharge hydraulic characteristics. The total scour was estimated based upon the cumulative effects of the long-term bed elevation change, general (contraction) scour, and local scour. The life expectancy of the bridge was considered in determining the long-term bed elevation change of the waterway. The long-term bed elevation change was based on an assumed 75-year design life for a new replacement bridge. Scour prevention methods are necessary to protect channel and bridge stability.

6.2 Caltrans Bridge Inspection Reports

The Caltrans BIRs for the existing Alta Mesa Road Bridge were reviewed in support of the scour analysis. According to the BIR dated February 26, 2002, local pier scour was present at bent number 9 on the upstream end. The scour hole measured 6 meters in diameter and 1 meter in depth. The bridge foundation elevations are unknown. Multiple surveyed cross sections of the channel were included. These cross sections were used to support long-term bed elevation change calculations.

6.3 Existing Channel Bed

The existing channel bed soil consists of Columbia sandy loam and clayey substratum, with 0 to 2 percent slopes, and is occasionally flooded. According to the Soil Conservation Service (SCS) Soil Survey of Sacramento County (1993), the corresponding sieve size that approximately 50 percent of the Columbia sandy loam bed material passes through is a 200 sieve. Therefore, the mass-median-diameter (D_{50}) is 0.075 mm.

6.4 Long-Term Bed Elevation Change

The channel bed elevation may fluctuate over time as a result of changes in local sediment transport capacity and availability. The 2002 Caltrans BIR provides the distance between the creek bottom and the top of the existing deck. An existing deck elevation of 73.3 ft NAVD 88 and surveyed cross section at the existing bridge were obtained from the data provided by Sacramento County (2013). The elevation data was plotted in order to quantify long-term bed elevation change; see Figure 11.

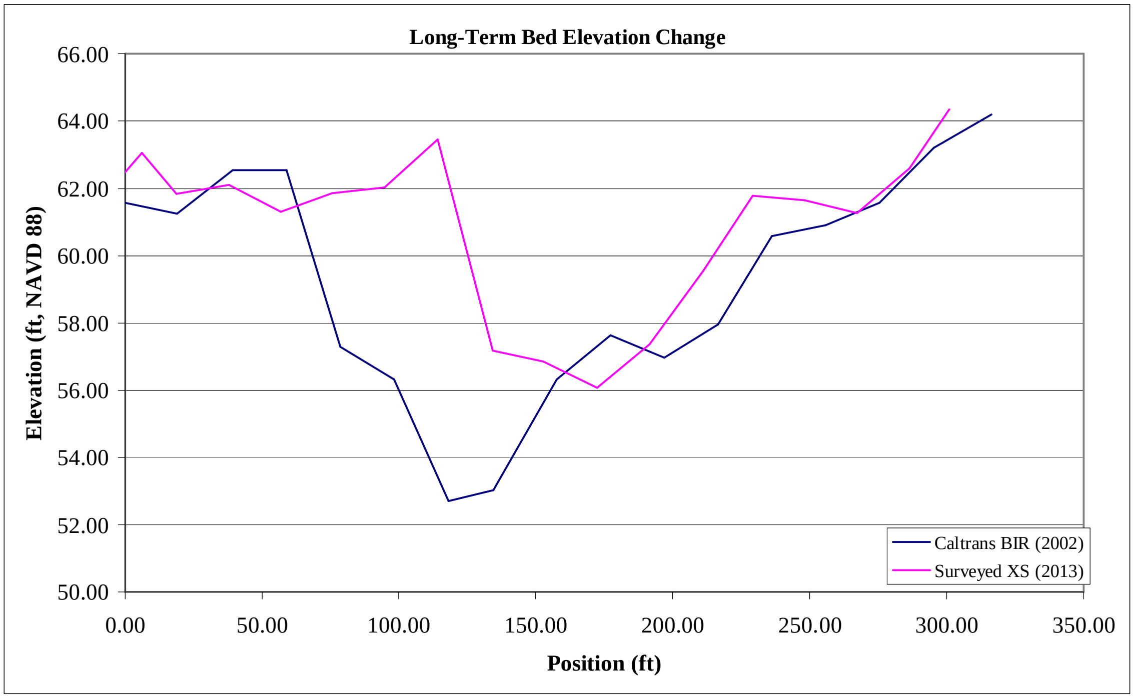

Figure 11. Channel Cross Sections of Laguna Creek at the Alta Mesa Road Bridge
Sources: Caltrans, Sacramento County

A comparison of the channel cross sections from 2002 and 2013 indicates channel aggradation has occurred from 2002 to 2013. Therefore, over the design life of the Project, long-term bed elevation change is assumed to be negligible.

6.5 Contraction Scour

Contraction scour occurs when the flow area of a stream is reduced either by: 1) the natural contraction of the stream channel; 2) a bridge structure; or 3) the overbank flow forced back to the channel.

From the continuity equation, a decrease in flow area results in an increase in average velocity and bed shear stress through the contraction. Hence, there is an increase in erosive forces in the contracted section, and more bed material is removed from the contracted reach than is transported into the reach. This increase in transport of bed material from the reach lowers the natural bed elevation. As the bed elevation is lowered, the flow area increases. Thus, the velocity and shear stress decrease until relative equilibrium is reached; i.e., the quantity of bed material that is transported into the reach is equal to that removed from the reach, or the bed shear stress is decreased to a value such that no sediment is transported out of the reach. Contraction scour, in a natural channel or at a bridge crossing, involves removal of material from the bed across all or most of the channel width (FHWA 2001).

As described in Section 6.3, a D_{50} of 0.075 mm was obtained from the NRCS WSS. Because of the small grain size at the Project site, the ultimate scour for cohesive (silt and clay) materials was estimated using a relationship between the median grain size and the critical shear stress.

The equation for estimating contraction scour, as presented in HEC-18, is as follows:

$$y_{s-ult} = 0.94 y_1 \left(\frac{1.83 V_2}{g y_1} - \frac{K_u \sqrt{\frac{\tau_c}{\rho}}}{g n y_1^{1/3}} \right)$$

y_{s-ult} = scour depth for cohesive soils, ft
y_1 = average depth in the upstream main channel, ft
V_2 = average flow velocity in the contracted section, ft/s
g = gravitational acceleration, 32.2 ft/s^2
K_u = 1.486 for U.S. Customary units, and 1.0 for S.I. units
τ_c = critical shear stress, lbs/ft^2
ρ = density of sediment, slugs/ft^3
n = Manning's roughness coefficient, unitless

The contraction scour was calculated to be 5.2 ft for the proposed bridge.

6.6 Pier Scour

Pier scour is caused by the formation of vortices (known as a horseshoe vortex) at the pier base. The horseshoe vortex results from the pileup of water on the upstream surface of the pier and subsequent acceleration of the flow around the base of the pier.

The scour depth at the pier is estimated based on the pier design (shape and dimensions), flow characteristics (flow rate, local flow velocity at the pier, and local flow depth at the pier), and sediment particle size distribution. Scour depths at the piers were estimated assuming the pile caps and footings will not be exposed. An equation based on the Colorado State University (CSU) equation was used to estimate local pier scour. The equation predicts maximum scour depths.

The equation for estimating local pier scour, as presented in HEC-18, is as follows:

$$\frac{y_s}{a} = 2.0 y_1 K_1 K_2 K_3 \left(\frac{y_1}{a} \right)^{0.35} Fr_1^{0.43}$$

$$Fr_1 = \frac{V_1}{\sqrt{g y_1}}$$

Where:

y_s = scour depth (ft)
y_1 = flow depth directly upstream of the pier (ft)
K_1 = correction factor for pier nose shape, 1.1 for square nose, 1.0 for round nose, circular cylinder, and group of cylinders, and 0.9 for sharp nose
K_2 = correction factor for angle of attack (see below for details)

$$K_2 = [Cos\theta + (L/a)Sin\theta]^{0.65}$$

L = lengths of pier (ft)
θ = skew angle of flow (degrees)
K_2 = 1.0 when skew angle of flow is 0 degrees

K_3 = correction factor for bed condition, 1.1 for clear-water scour and small dunes
a = pier width (ft)
Fr_1 = Froude number directly upstream of the pier
V_1 = mean velocity of flow directly upstream of the pier (ft/s)
g = gravitational acceleration (32.2 ft/s^2)

The length, widths, and skew angle of the piers for the proposed three-span bridge over Laguna Creek are based on the General Plan provided by MTCO. The local scour at the piers for the proposed three-span bridge is summarized in Table 8.

Table 8. Pier Scour Depth Summary

Pier Number	Scour Depth (ft)
2 (South)	11.2
3 (North)	11.2

6.7 Abutment Scour

Local scour occurs at abutments when the abutment and roadway embankment obstruct the flow. According to survey data provided by MTCO, as well as available aerial imagery, the Laguna Creek channel increases in size downstream of the Alta Mesa Road crossing when compared to the upstream channel. Therefore, the abutments do not constitute a significant obstruction to flow in the main active channel.

However the Laguna Creek channel has insufficient capacity to convey the 100-year design storm discharge in the main channel resulting in significant inundation for overbank areas near the Alta Mesa Road bridge crossing. Because the approach roadways and embankments would obstruct the overbank flows, local scour at the abutments was analyzed at the Project site for the overbank flows.

Based on information in the General Plan provided by MTCO for the Alta Mesa Road bridge replacement, the approach roadways would extend laterally approximately 500 ft beyond the beginning and ending of the replacement bridge. However, the depth of flow obstructed by the roadway is minimal.

Abutment scour is commonly evaluated using either the Froehlich or HIRE live-bed scour equation. The HIRE equation is applicable when the ratio of the projected abutment length (the L parameter) to the flow depth (the y_1 parameter) is greater than 25. The HIRE equation was used for the scour analysis because the ratio of the projected abutment length to the flow depth was greater than 25 for the abutments.

The HIRE equation is given below:

$$\frac{y_s}{y_1} = 4Fr^{0.33}\frac{K_1}{0.55}K_2$$

Where:

y_s = Scour depth, ft

y_1 = Depth of flow at the abutment on the overbank or in the main channel, ft

Fr = Froude number, based on the velocity and depth adjacent to and upstream of the abutment

K_1 = Abutment shape coefficient (from Table 8.1 of HEC-18)

K_2 = Coefficient for skew angle of abutment to flow calculated as for Froehlich's equation (Section 8.7.1 of HEC-18)

The local abutment scour is summarized in Table 9.

Table 9. Abutment Scour Summary

Location	Bridge Component	Scour Depth (ft)
Left Abutment (South)	Abutment 1	4.2
Right Abutment (North)	Abutment 2	0.1

6.8 Total Scour and Scour Countermeasures

Total scour is the sum of the long-term bed elevation change, contraction scour, pier scour, and local scour. The total scour depths at the proposed bridges are shown in Table 10.

Table 10. Total Scour Summary

Bridge Component	Thalweg Elevation (ft, NAVD 88)	Local Scour Depth (ft)	Contraction Scour Depth (ft)	Long-term Scour Depth (ft)	Total Scour Depth (ft)
Abutment 1 (South)	56.1	4.2	5.7	0.0	9.9
Pier 2	56.1	11.2	5.7	0.0	16.9
Pier 3	56.1	11.2	5.7	0.0	16.9
Abutment 4 (North)	56.1	0.1	5.7	0.0	5.8

According to the *Caltrans Bridge Memo to Designers*, bridge footings supported on soil or degradable rock should be embedded below the maximum computed scour depth. If the footing is supported by massive, competent rock formations resistant to scour, the footing should be placed directly on the cleaned rock surface. According to HEC-18, abutment foundations should be designed for contraction scour and long-term bed elevation change if the proposed bridge abutments are protected by RSP, built on bedrock, or alternative bank protection measures are taken.

7 ROCK SLOPE PROTECTION

The proposed abutments of the proposed Alta Mesa Road bridge replacement alternative do not obstruct the 100-year design discharge; therefore, procedures for determining RSP design were based on Caltrans' *California Bank and Shore Rock Slope Protection Design* (October 2000).

7.1 California Bank and Shore RSP Design

A nomograph of stream-bank rock slope protection was established by Caltrans for stream scour that is governed by flow velocity (as opposed to wave action). The nomograph is derived from the following formula:

$$W = \frac{0.00002 \times V^6 \times SG_R}{(SG_R - 1)^3 \times \sin^3(r - a)}$$

Where:

V = 2/3 average stream velocity for flows parallel to the bank, fps
4/3 average stream velocity for flows which impinge the bank, fps
SG_R = specific gravity of the stones, 2.65
r = 70° for broken rock, constant
a = angle of face slope from the horizontal
W = RSP weight, lbs.

The Laguna Creek flow path is perpendicular to the proposed bridge alignment; therefore, the velocity is considered parallel to the bank. The average channel flow velocities during the design 100-year storm event from the hydraulic analysis and the grading plan from MTCO were used to determine minimum RSP weights. The calculated and recommended RSP classes for the proposed bridge abutments are summarized in Table 11.

Table 11. California Bank and Shore RSP Sizing Summary

Location	2/3 Flow Velocity (ft/sec)	Angle of Face Slope from the Horizontal	RSP Weight (lbs.)	RSP Class Calculated	RSP Class Recommended
Immediately Upstream of the Proposed Bridge	3.5	26.6	0.1	Facing	1/4 Ton
Proposed Bridge Upstream Face	3.9	26.6	0.1	Facing	1/4 Ton
Proposed Bridge Downstream Face	3.3	26.6	0.1	Facing	1/4 Ton
Immediately Downstream of the Proposed Bridge	3.2	26.6	0.0	Facing	1/4 Ton

Although the recommended RSP class per the Caltrans guidelines indicates facing class RSP would be sufficient to protect against scour, we recommend the minimum RSP for the Project be ¼ ton.

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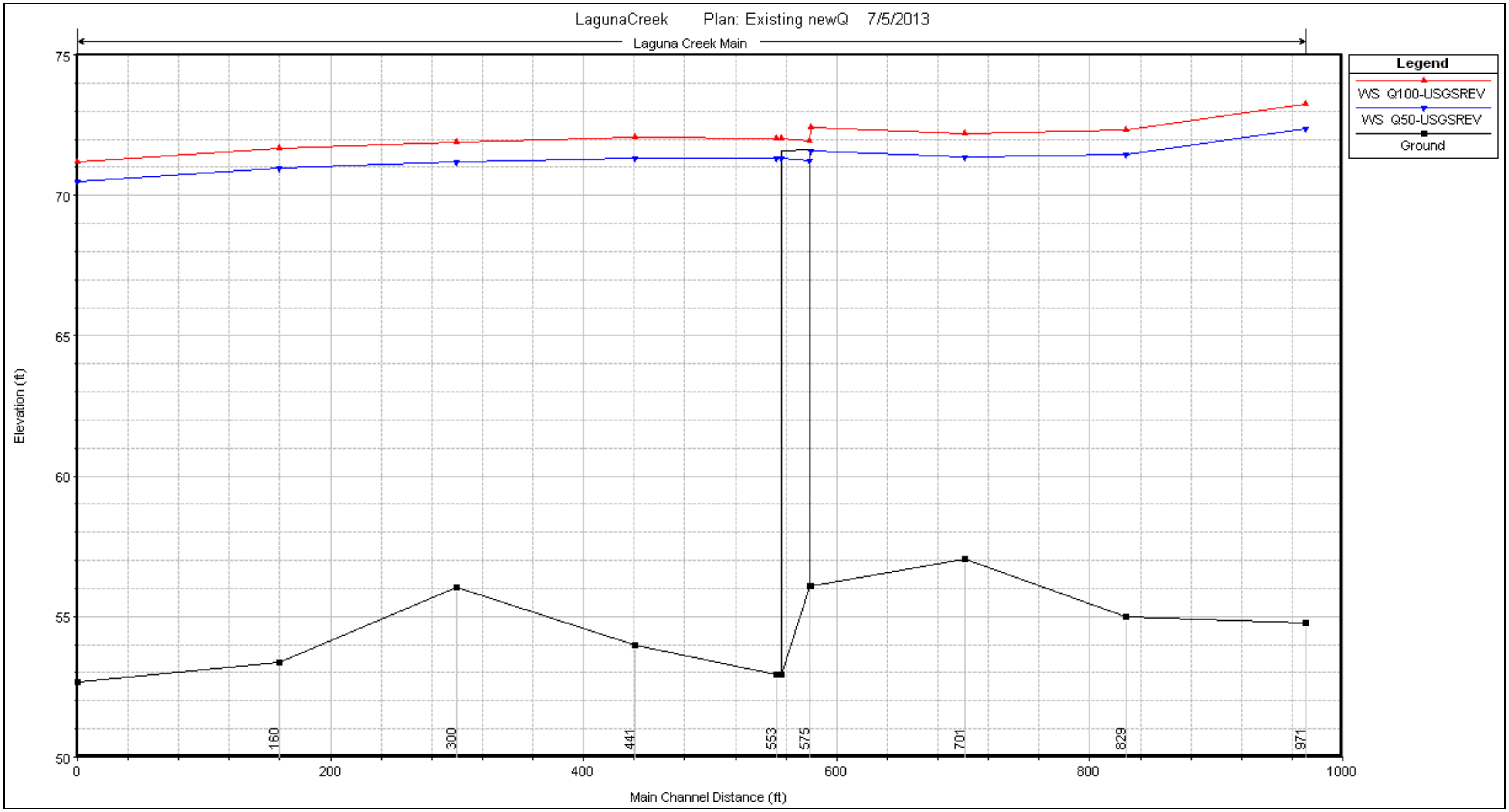
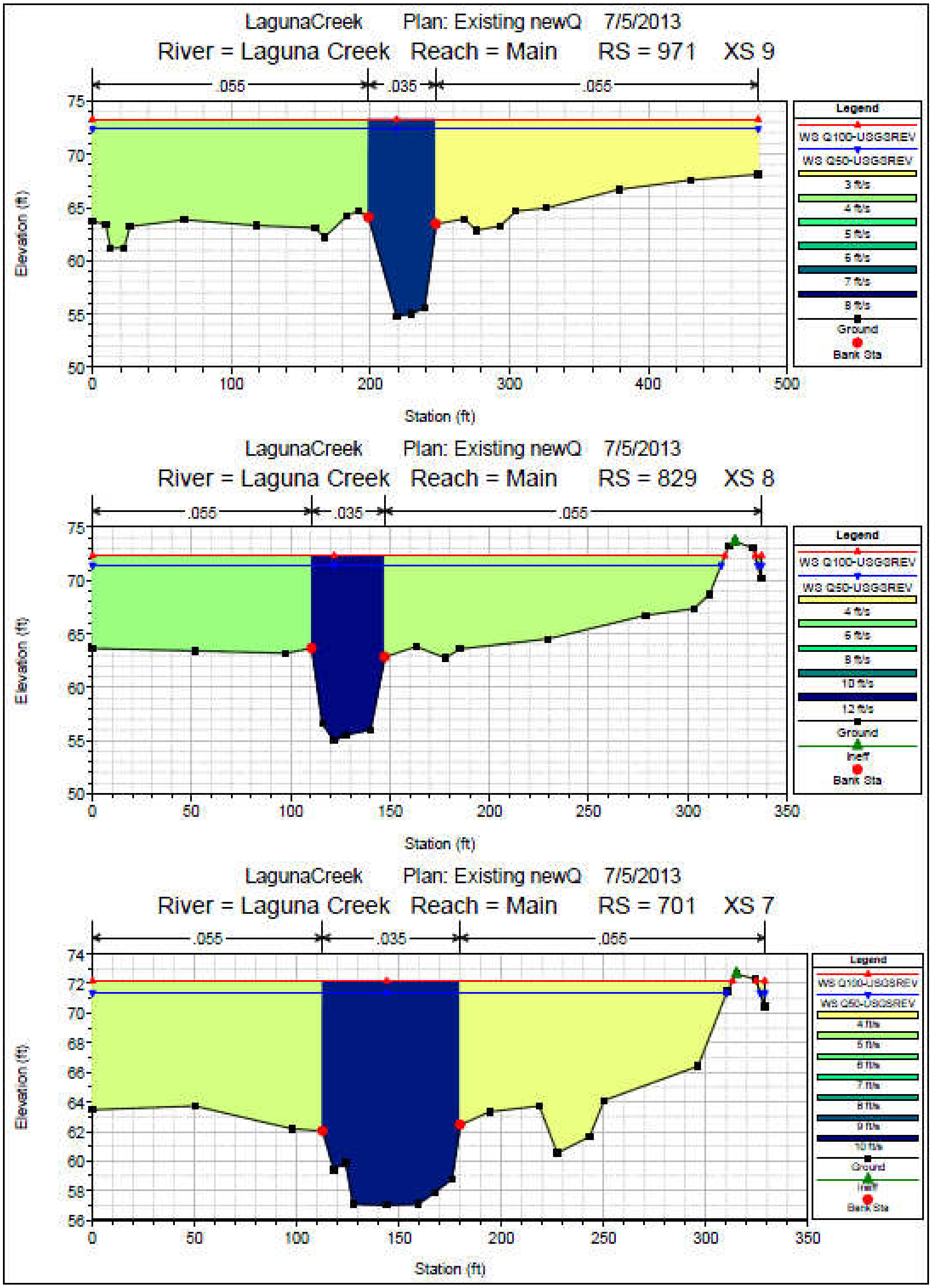
Appendix A HEC-RAS Results: Existing Conditions

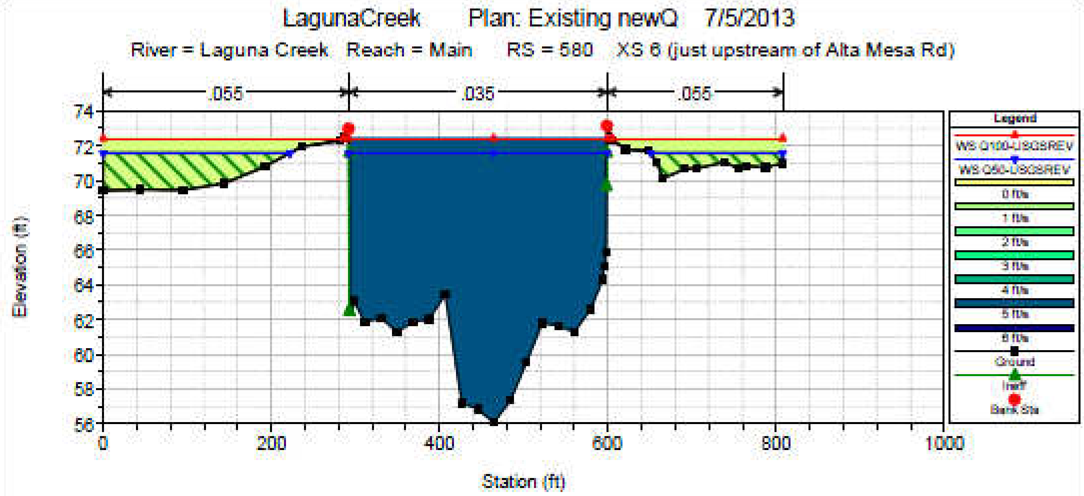
Reach	River Station	Profile	Q Total	Min Ch El	W.S. Elev	Crit W.S.	E.G. Elev	E.G. Slope	Vel Chnl	Flow Area	Top Width	Froude # Chl
			(cfs)	(ft)	(ft)	(ft)	(ft)	(ft/ft)	(ft/s)	(sq ft)	(ft)	
Main	971	Q100-USGSREV	18260	54.79	73.27		73.67	0.000925	7.54	4454.59	479.29	0.34
Main	971	Q50-USGSREV	15750	54.79	72.38		72.76	0.000931	7.27	4029.62	479.29	0.33
Main	829	Q100-USGSREV	18260	54.99	72.31		73.39	0.002573	11.89	2755.42	321.67	0.54
Main	829	Q50-USGSREV	15750	54.99	71.46		72.49	0.002595	11.49	2483.47	318.6	0.53
Main	701	Q100-USGSREV	18260	57.04	72.19		73.08	0.001631	9.72	3010.66	317.51	0.46
Main	701	Q50-USGSREV	15750	57.04	71.36		72.16	0.001577	9.18	2751.59	312.32	0.44
Main	580	Q100-USGSREV	18260	56.08	72.43	65.48	72.82	0.000568	5.03	3950.31	795.1	0.26
Main	580	Q50-USGSREV	15750	56.08	71.57	65.02	71.92	0.000527	4.73	3328.72	685.56	0.25
Main	575		Bridge									
Main	553	Q100-USGSREV	18260	52.93	72.04		72.39	0.000464	4.75	3968.67	658.97	0.24
Main	553	Q50-USGSREV	15750	52.93	71.33		71.62	0.000403	4.35	3616.86	525.26	0.22
Main	441	Q100-USGSREV	18260	53.99	72.04		72.27	0.000621	3.82	4796.34	668.37	0.22
Main	441	Q50-USGSREV	15750	53.99	71.32		71.52	0.00058	3.55	4430.93	659.53	0.21
Main	300	Q100-USGSREV	18260	56.03	71.89		72.16	0.000986	4.18	4660.3	757.33	0.23
Main	300	Q50-USGSREV	15750	56.03	71.17		71.41	0.000966	3.94	4167.86	750.65	0.23
Main	160	Q100-USGSREV	18260	53.35	71.66		71.99	0.001321	4.71	4228.68	740.03	0.26
Main	160	Q50-USGSREV	15750	53.35	70.95		71.25	0.001297	4.44	3770.02	728.36	0.26
Main	0	Q100-USGSREV	18260	52.65	71.21	66.31	71.71	0.002203	5.79	3464.78	693.36	0.33
Main	0	Q50-USGSREV	15750	52.65	70.51	65.82	70.97	0.002202	5.5	3054.28	627.53	0.33

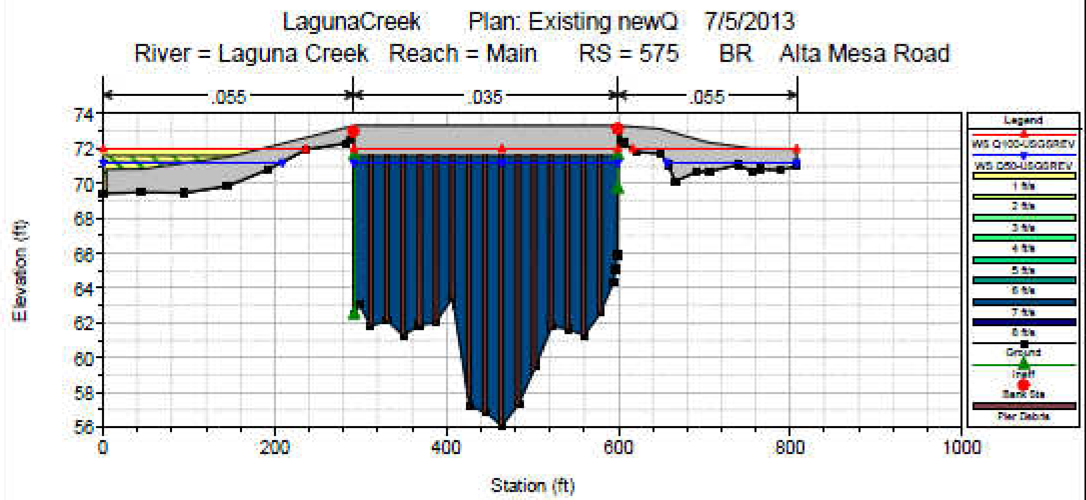
Plan: Existnew Laguna Creek Main RS: 575 Profile: Q50-USGSREV				
E.G. US. (ft)	71.92	Element	Inside BR US	Inside BR DS
W.S. US. (ft)	71.57	E.G. Elev (ft)	71.84	71.65
Q Total (cfs)	15750	W.S. Elev (ft)	71.22	71.31
Q Bridge (cfs)	15750	Crit W.S. (ft)	65.88	64.04
Q Weir (cfs)		Max Chl Dpth (ft)	15.15	18.35
Weir Sta Lft (ft)		Vel Total (ft/s)	6.31	4.72
Weir Sta Rgt (ft)		Flow Area (sq ft)	2496.11	3337.48
Weir Submerg		Froude # Chl	0.29	0.24
Weir Max Depth (ft)		Specif Force (cu ft)	16788.14	23411.3
Min El Weir Flow (ft)	71.62	Hydr Depth (ft)	10.47	11.77
Min El Prs (ft)	71.61	W.P. Total (ft)	768.02	653.47
Delta EG (ft)	0.3	Conv. Total (cfs)	232516.5	420220.1
Delta WS (ft)	0.25	Top Width (ft)	238.52	283.52
BR Open Area (sq ft)	2585.95	Frctn Loss (ft)	0.05	0
BR Open Vel (ft/s)	6.31	C & E Loss (ft)	0.14	0.03
Coef of Q		Shear Total (lb/sq ft)	0.93	0.45
Br Sel Method	Energy only	Power Total (lb/ft s)	0	0

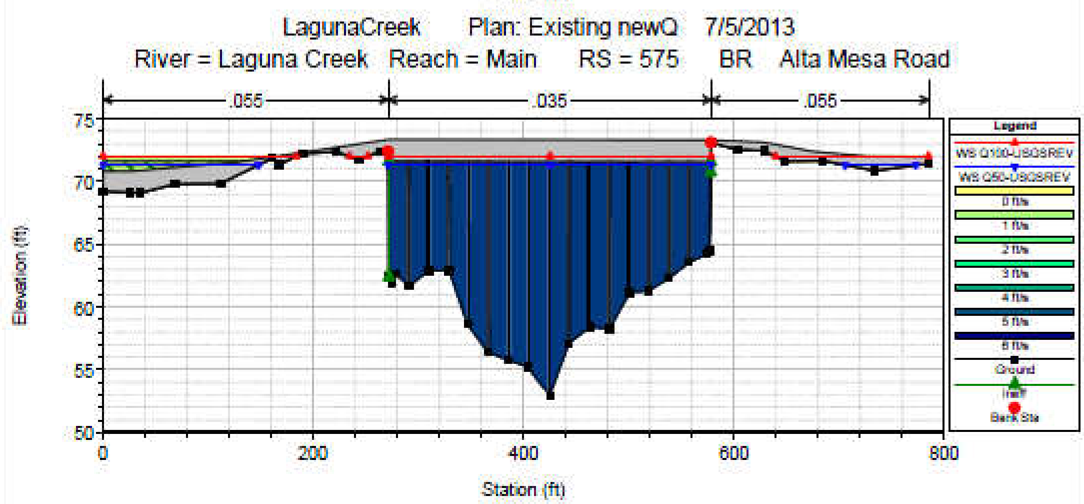
Plan: Existnew Laguna Creek Main RS: 575 Profile: Q100-USGSREV				
E.G. US. (ft)	72.82	Element	Inside BR US	Inside BR DS
W.S. US. (ft)	72.43	E.G. Elev (ft)	72.71	72.44
Q Total (cfs)	18260	W.S. Elev (ft)	71.94	72
Q Bridge (cfs)	18200.84	Crit W.S. (ft)	66.41	64.57
Q Weir (cfs)		Max Chl Dpth (ft)	15.86	19.04
Weir Sta Lft (ft)		Vel Total (ft/s)	6.92	5.25
Weir Sta Rgt (ft)		Flow Area (sq ft)	2640.45	3479.17
Weir Submerg		Froude # Chl	0.31	0.21
Weir Max Depth (ft)		Specif Force (cu ft)	19529.65	26495.69
Min El Weir Flow (ft)	71.62	Hydr Depth (ft)	14.76	16.68
Min El Prs (ft)	71.61	W.P. Total (ft)	1217.28	1165.52
Delta EG (ft)	0.43	Conv. Total (cfs)	202867	340544.1
Delta WS (ft)	0.39	Top Width (ft)	178.86	208.55
BR Open Area (sq ft)	2585.95	Frctn Loss (ft)	0.1	0
BR Open Vel (ft/s)	7.04	C & E Loss (ft)	0.16	0.04
Coef of Q		Shear Total (lb/sq ft)	1.1	0.54
Br Sel Method	Energy only	Power Total (lb/ft s)	0	0

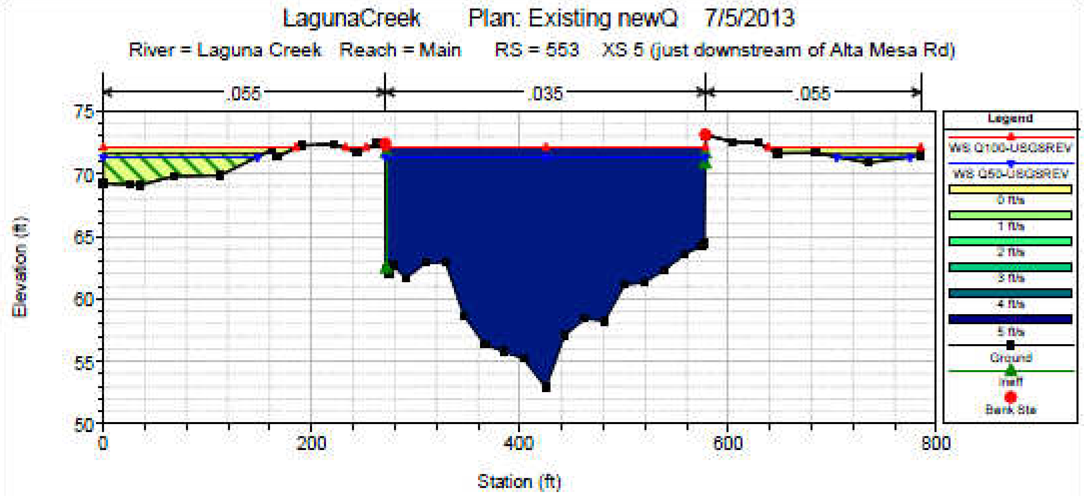

LagunaCreek Plan: Existing newQ 7/5/2013
Laguna Creek Main
Legend
WS Q100-USGSREV
WS Q50-USGSREV
Ground
Elevation (ft)
Main Channel Distance (ft)
75
70
65
60
55
50
0
200
400
600
800
1000
160
300
441
553
575
701
829
971

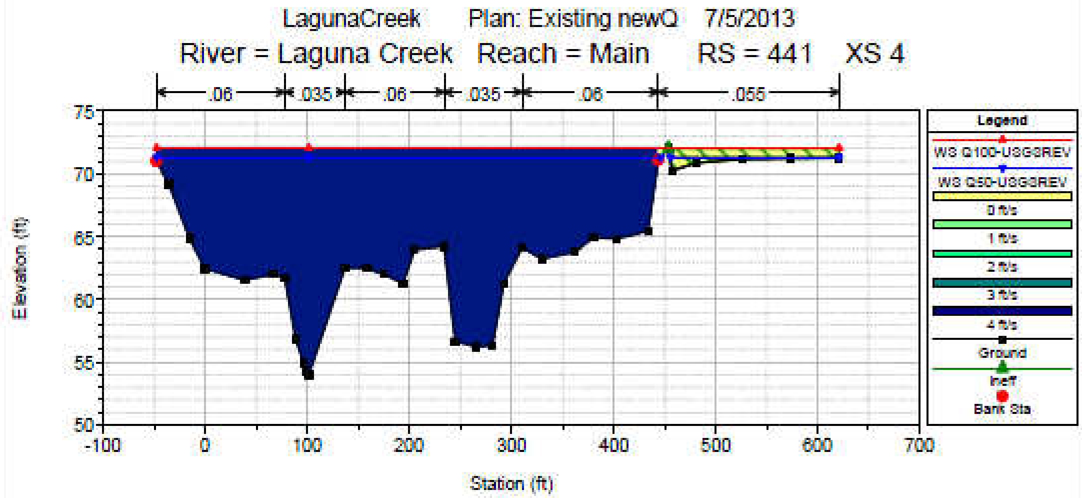

LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 971 XS 9
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
3 ft/s
4 ft/s
5 ft/s
6 ft/s
7 ft/s
8 ft/s
Ground
Bank Sta
LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 829 XS 8
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
4 ft/s
6 ft/s
8 ft/s
10 ft/s
12 ft/s
Ground
Ineff
Bank Sta
LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 701 XS 7
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
4 ft/s
5 ft/s
6 ft/s
7 ft/s
8 ft/s
9 ft/s
10 ft/s
Ground
Ineff
Bank Sta

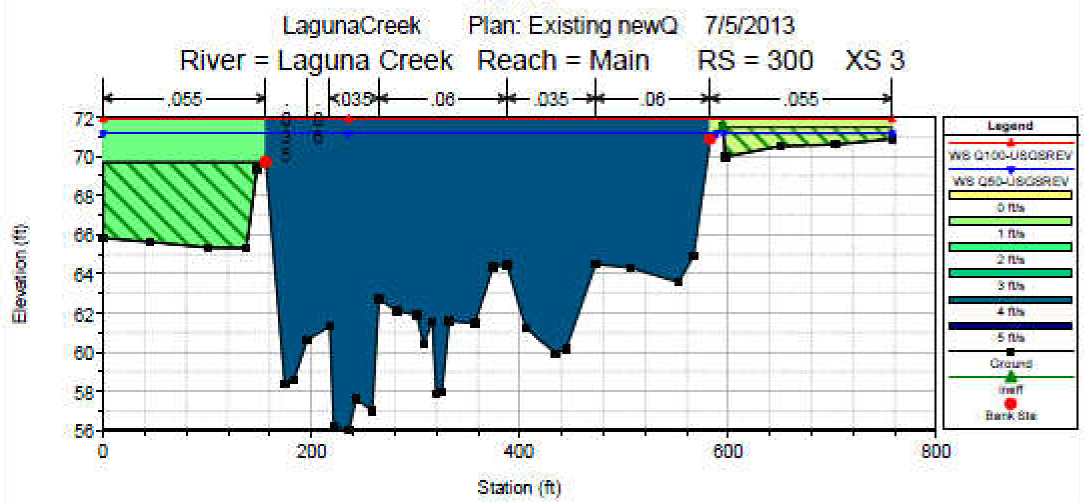

LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 580 XS 6 (just upstream of Alta Mesa Rd)
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
6 ft/s
Ground
Ineff
Bank Sta

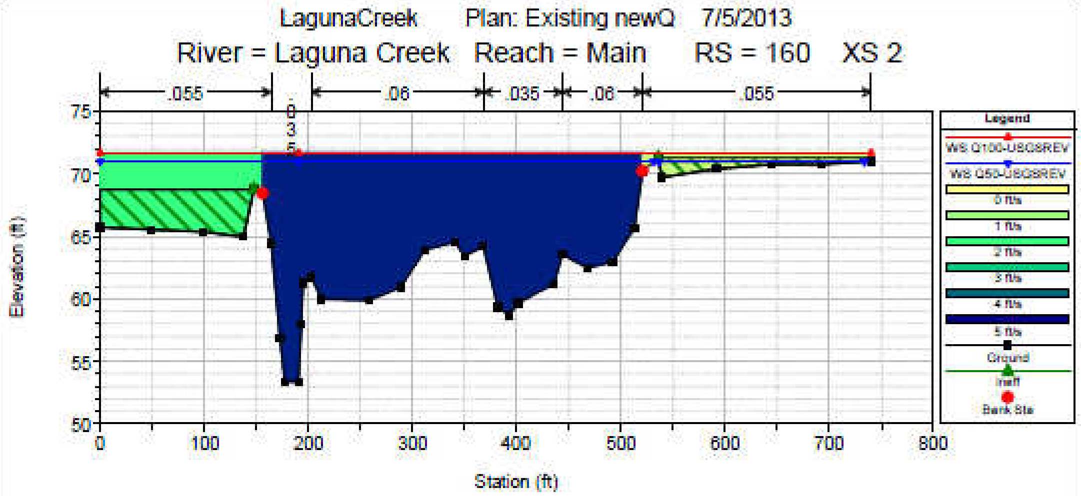

LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 575 BR Alta Mesa Road
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
6 ft/s
7 ft/s
8 ft/s
Ground
Ineff
Bank Sta
Pier Debris

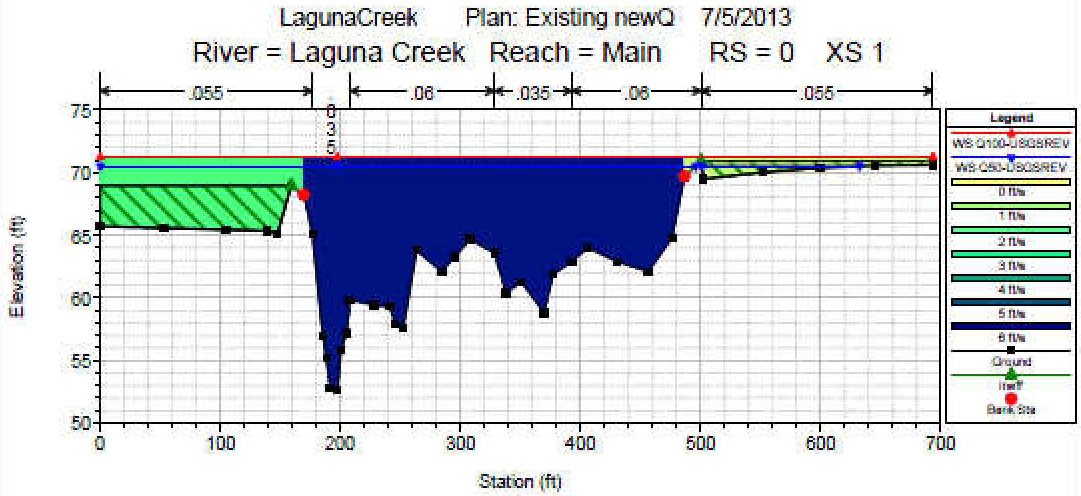

LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 575 BR Alta Mesa Road
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
6 ft/s
Ground
Ineff
Bank Sta


LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 553 XS 5 (just downstream of Alta Mesa Rd)
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
Ground
Ineff
Bank Sta


LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 441 XS 4
.06
.035
.06
.035
.06
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
Ground
Ineff
Bank Sta


LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 300 XS 3
.055
.035
.06
.035
.06
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
Ground
Ineff
Bank Sta


LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 160 XS 2
.055
.06
.035
.06
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
Ground
Ineff
Bank Sta

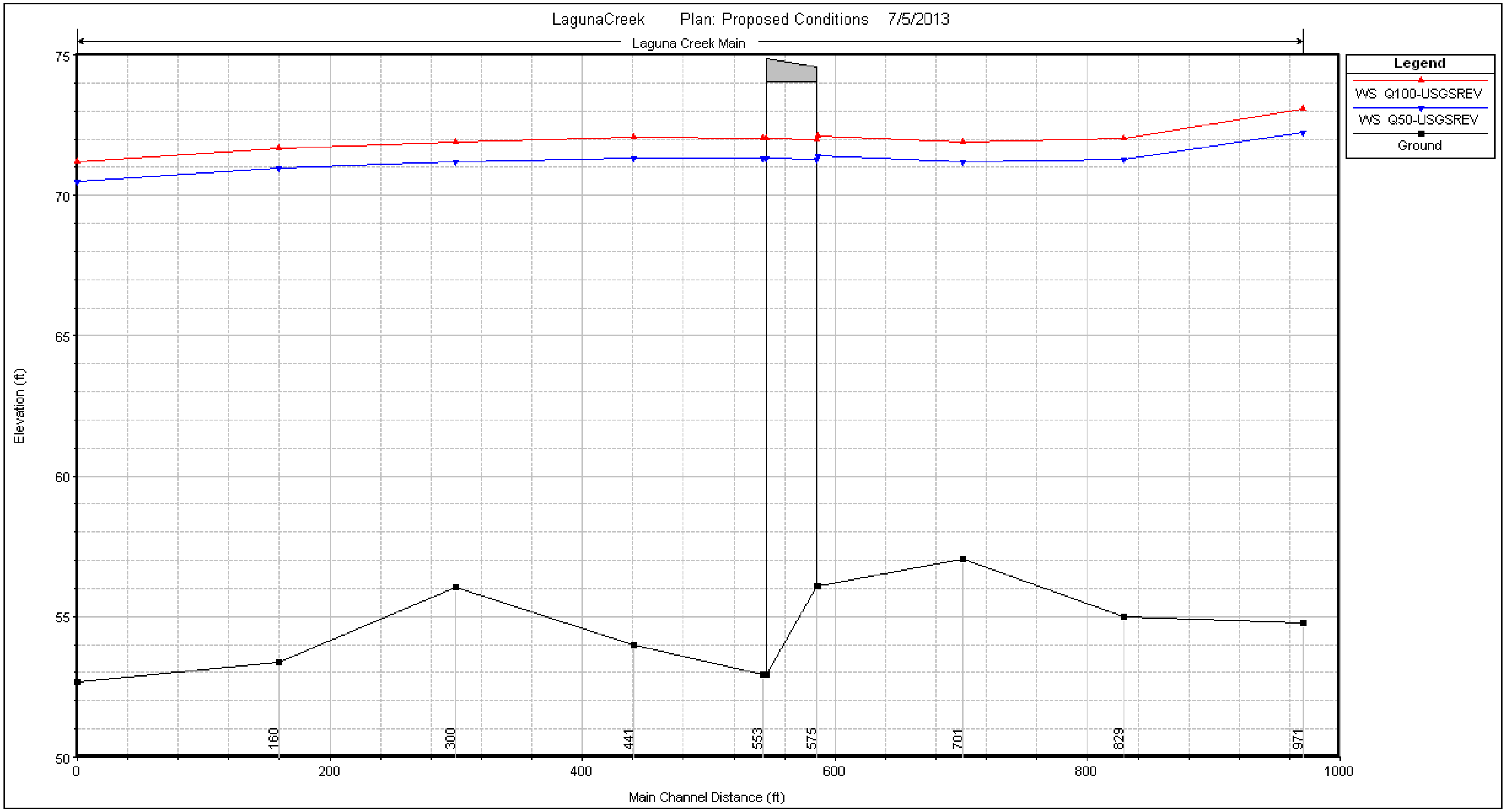

LagunaCreek Plan: Existing newQ 7/5/2013
River = Laguna Creek Reach = Main RS = 0 XS 1
.055
.06
.035
.06
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
6 ft/s
Ground
Ineff
Bank Sta

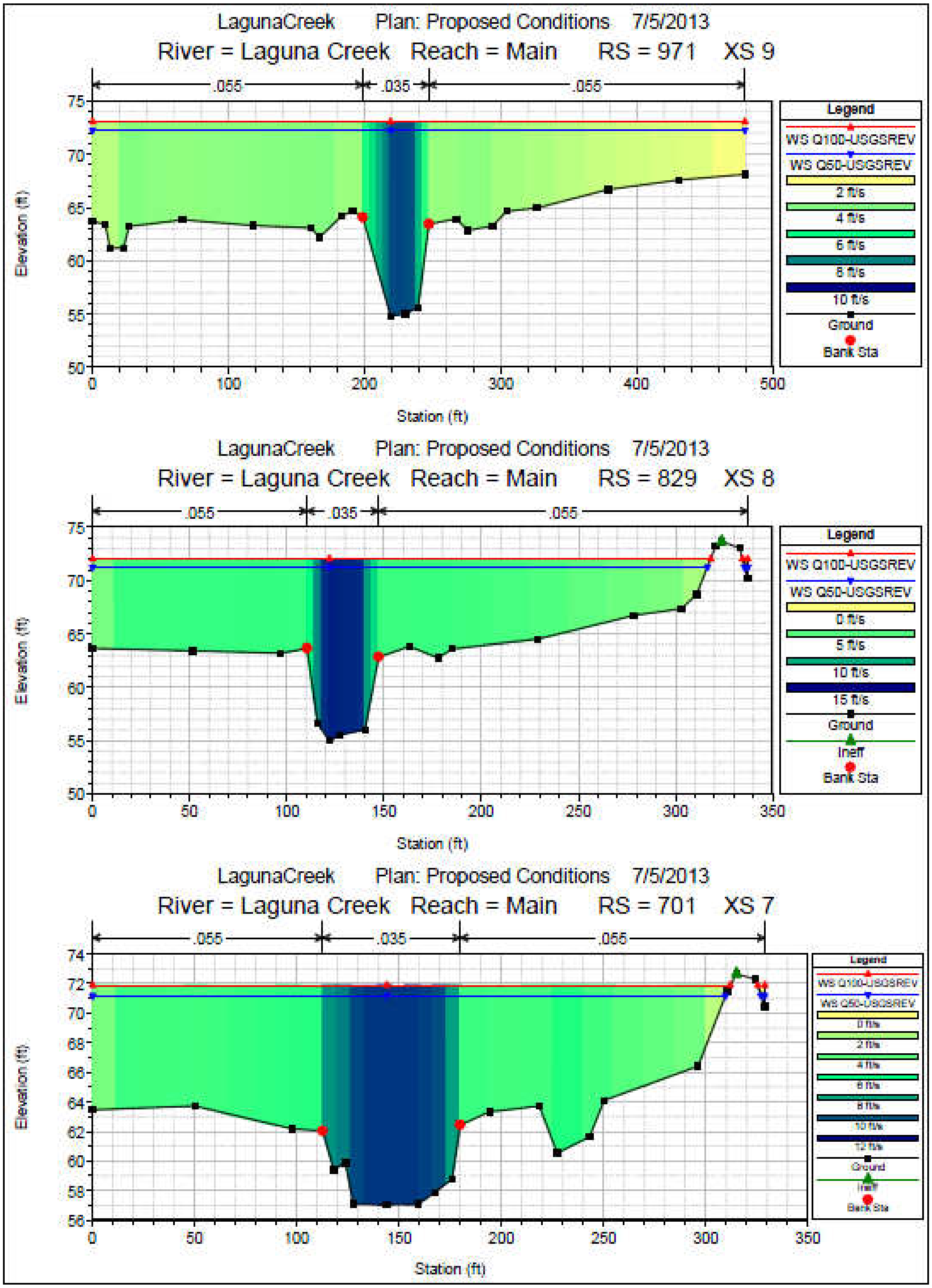
Appendix B HEC-RAS Results: Proposed Conditions

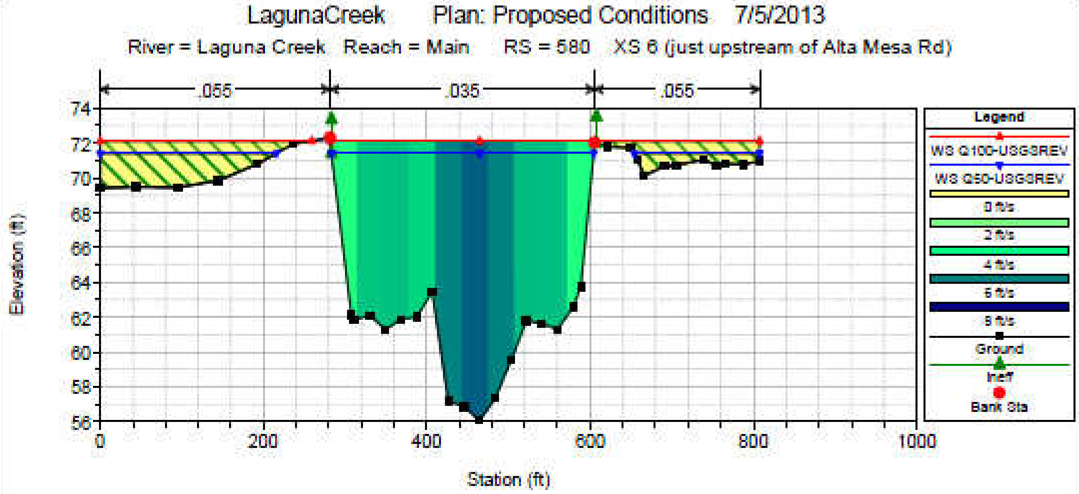
Reach	River Station	Profile	Q Total	Min Ch El	W.S. Elev	Crit W.S.	E.G. Elev	E.G. Slope	Vel Chnl	Flow Area	Top Width	Froude # Chl
			(cfs)	(ft)	(ft)	(ft)	(ft)	(ft/ft)	(ft/s)	(sq ft)	(ft)	
Main	971	Q100-USGSREV	18260	54.79	73.05		73.48	0.000992	7.74	4353.56	479.29	0.35
Main	971	Q50-USGSREV	15750	54.79	72.26		72.65	0.000972	7.39	3971.17	479.29	0.34
Main	829	Q100-USGSREV	18260	54.99	72		73.18	0.002866	12.38	2656.18	320.55	0.56
Main	829	Q50-USGSREV	15750	54.99	71.27		72.36	0.002781	11.8	2424.94	317.94	0.55
Main	701	Q100-USGSREV	18260	57.04	71.87		72.83	0.0018	10.05	2910.82	315.44	0.48
Main	701	Q50-USGSREV	15750	57.04	71.17		72.01	0.001678	9.38	2693.76	311.34	0.46
Main	580	Q100-USGSREV	18260	56.08	72.12	65.52	72.55	0.000653	5.25	3480.19	784.91	0.28
Main	580	Q50-USGSREV	15750	56.08	71.4	65.05	71.76	0.000609	4.85	3246.12	690.04	0.27
Main	575		Bridge									
Main	553	Q100-USGSREV	18260	52.93	72.04		72.39	0.000492	4.75	4011.25	699.87	0.25
Main	553	Q50-USGSREV	15750	52.93	71.32		71.62	0.000426	4.37	3601.74	546.21	0.23
Main	441	Q100-USGSREV	18260	53.99	72.04		72.27	0.000621	3.82	4796.34	668.37	0.22
Main	441	Q50-USGSREV	15750	53.99	71.32		71.52	0.00058	3.55	4430.93	659.53	0.21
Main	300	Q100-USGSREV	18260	56.03	71.89		72.16	0.000986	4.18	4660.3	757.33	0.23
Main	300	Q50-USGSREV	15750	56.03	71.17		71.41	0.000966	3.94	4167.86	750.65	0.23
Main	160	Q100-USGSREV	18260	53.35	71.66		71.99	0.001321	4.71	4228.68	740.03	0.26
Main	160	Q50-USGSREV	15750	53.35	70.95		71.25	0.001297	4.44	3770.02	728.36	0.26
Main	0	Q100-USGSREV	18260	52.65	71.21	66.31	71.71	0.002203	5.79	3464.78	693.36	0.33
Main	0	Q50-USGSREV	15750	52.65	70.51	65.83	70.97	0.002202	5.5	3054.28	627.53	0.33

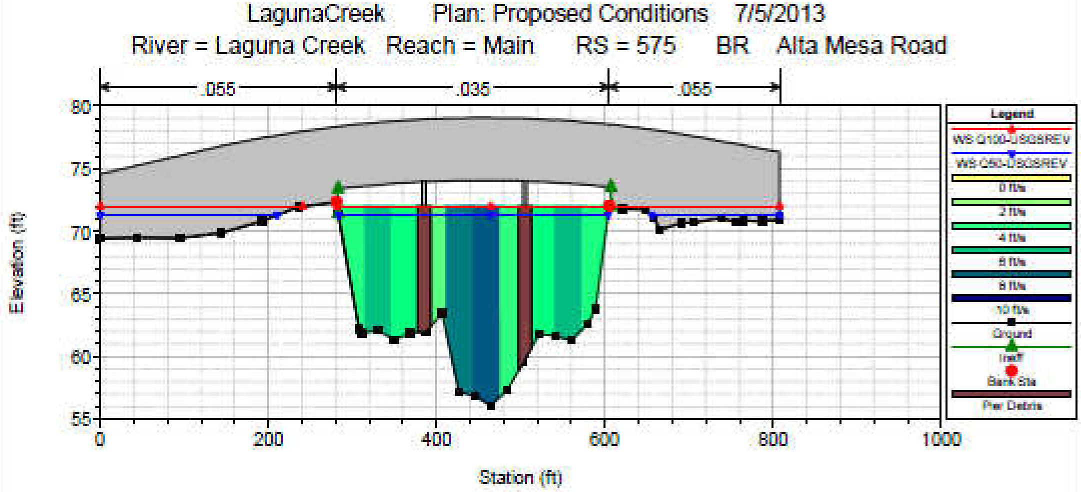
Plan: Prop Laguna Creek Main RS: 575 Profile: Q50-USGSREV				
E.G. US. (ft)	71.76	Element	Inside BR US	Inside BR DS
W.S. US. (ft)	71.4	E.G. Elev (ft)	71.73	71.64
Q Total (cfs)	15750	W.S. Elev (ft)	71.28	71.31
Q Bridge (cfs)	15750	Crit W.S. (ft)	65.36	63.96
Q Weir (cfs)		Max Chl Dpth (ft)	15.2	18.38
Weir Sta Lft (ft)		Vel Total (ft/s)	5.43	4.55
Weir Sta Rgt (ft)		Flow Area (sq ft)	2898.78	3461.69
Weir Submerg		Froude # Chl	0.3	0.19
Weir Max Depth (ft)		Specif Force (cu ft)	18681.2	23914.96
Min El Weir Flow (ft)	74.89	Hydr Depth (ft)	9.99	11.25
Min El Prs (ft)	74.05	W.P. Total (ft)	377.86	367.3
Delta EG (ft)	0.14	Conv. Total (cfs)	478699.8	655725.9
Delta WS (ft)	0.07	Top Width (ft)	290.21	307.76
BR Open Area (sq ft)	3648.32	Frctn Loss (ft)	0.03	0
BR Open Vel (ft/s)	5.43	C & E Loss (ft)	0.07	0.01
Coef of Q		Shear Total (lb/sq ft)	0.52	0.34
Br Sel Method	Energy only	Power Total (lb/ft s)	0	0

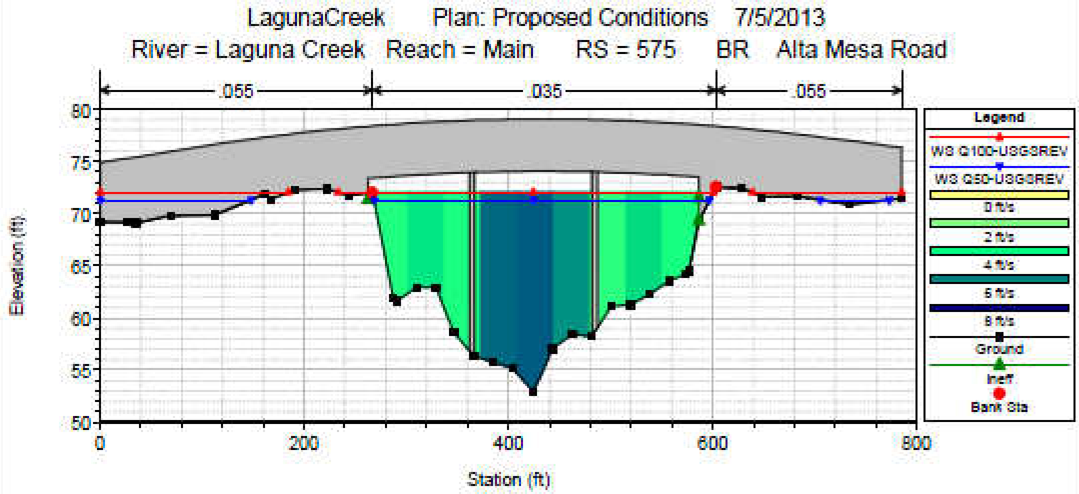
Plan: Prop Laguna Creek Main RS: 575 Profile: Q100-USGSREV				
E.G. US. (ft)	72.55	Element	Inside BR US	Inside BR DS
W.S. US. (ft)	72.12	E.G. Elev (ft)	72.52	72.41
Q Total (cfs)	18260	W.S. Elev (ft)	71.98	72.02
Q Bridge (cfs)	18260	Crit W.S. (ft)	65.88	64.49
Q Weir (cfs)		Max Chl Dpth (ft)	15.9	19.09
Weir Sta Lft (ft)		Vel Total (ft/s)	5.88	4.96
Weir Sta Rgt (ft)		Flow Area (sq ft)	3104.13	3681.32
Weir Submerg		Froude # Chl	0.26	0.2
Weir Max Depth (ft)		Specif Force (cu ft)	21475.8	27042.36
Min El Weir Flow (ft)	74.89	Hydr Depth (ft)	10.61	11.72
Min El Prs (ft)	74.05	W.P. Total (ft)	384.68	376.56
Delta EG (ft)	0.16	Conv. Total (cfs)	530784.1	720720.1
Delta WS (ft)	0.08	Top Width (ft)	292.6	314
BR Open Area (sq ft)	3648.32	Frctn Loss (ft)	0.03	0
BR Open Vel (ft/s)	5.88	C & E Loss (ft)	0.08	0.02
Coef of Q		Shear Total (lb/sq ft)	0.6	0.39
Br Sel Method	Energy only	Power Total (lb/ft s)	0	0

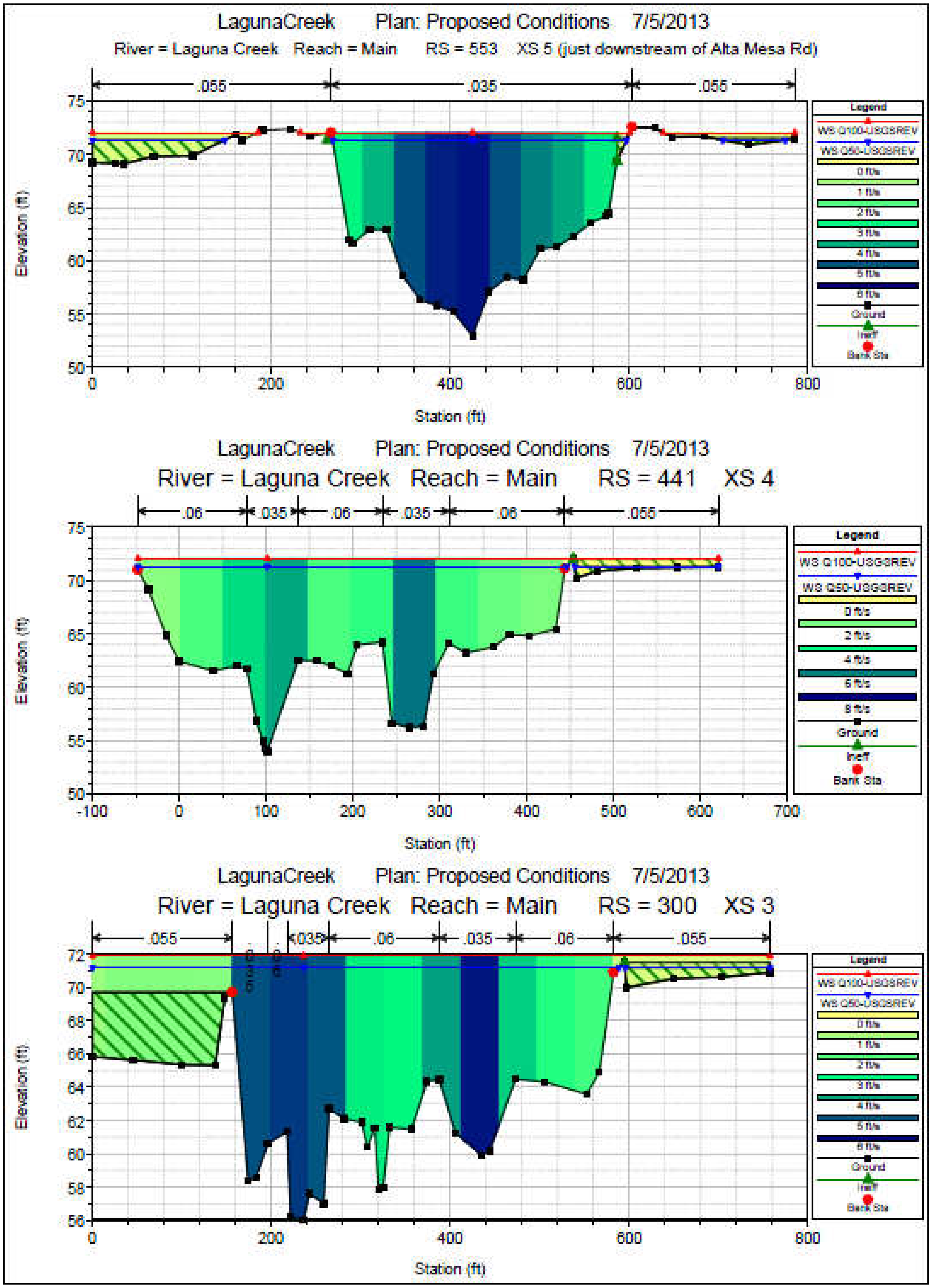

LagunaCreek Plan: Proposed Conditions 7/5/2013
Laguna Creek Main
Legend
WS Q100-USGSREV
WS Q50-USGSREV
Ground
Elevation (ft)
Main Channel Distance (ft)
75
70
65
60
55
50
0
200
400
600
800
1000
160
300
441
553
575
701
829
971

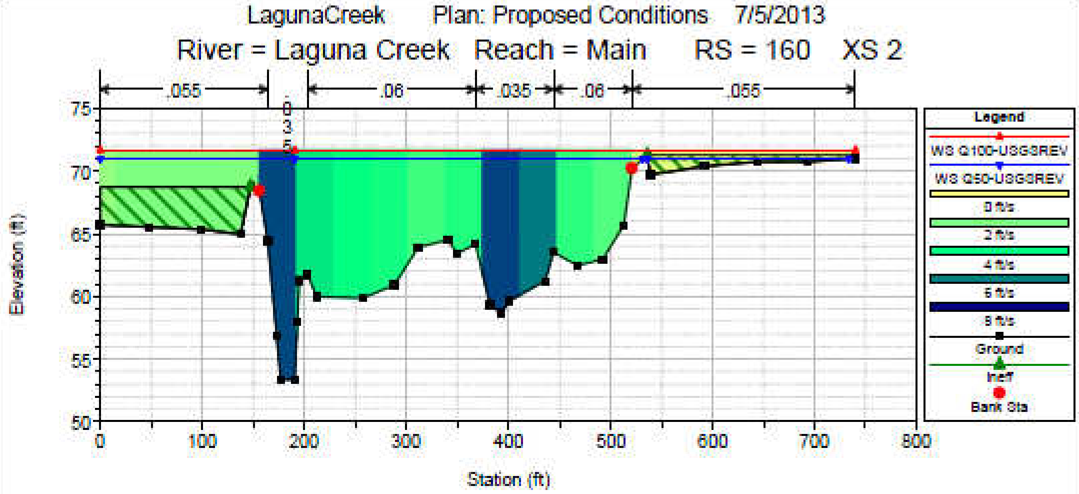

LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 971 XS 9
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
2 ft/s
4 ft/s
6 ft/s
8 ft/s
10 ft/s
Ground
Bank Sta
LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 829 XS 8
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
5 ft/s
10 ft/s
15 ft/s
Ground
Ineff
Bank Sta
LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 701 XS 7
.055
.035
.055
Elevation (ft)
Station (ft)

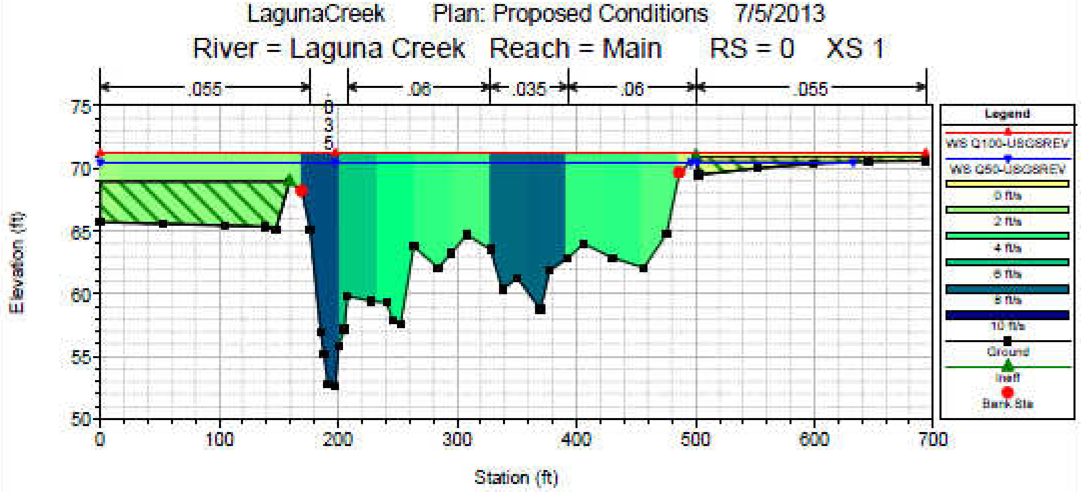

LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 580 XS 6 (just upstream of Alta Mesa Rd)
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
2 ft/s
4 ft/s
6 ft/s
8 ft/s
Ground
Ineff
Bank Sta


LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 575 BR Alta Mesa Road
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
2 ft/s
4 ft/s
6 ft/s
8 ft/s
10 ft/s
Ground
Ineff
Bank Sta
Pier Debris


LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 575 BR Alta Mesa Road
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
2 ft/s
4 ft/s
6 ft/s
8 ft/s
Ground
Ineff
Bank Sta


LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 553 XS 5 (just downstream of Alta Mesa Rd)
.055
.035
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
6 ft/s
Ground
Ineff
Bank Sta
LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 441 XS 4
.06
.035
.06
.035
.06
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
2 ft/s
4 ft/s
6 ft/s
8 ft/s
Ground
Ineff
Bank Sta
LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 300 XS 3
.055
.035
.06
.035
.06
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
1 ft/s
2 ft/s
3 ft/s
4 ft/s
5 ft/s
6 ft/s
Ground
Ineff
Bank Sta


LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 160 XS 2
.055
.035
.06
.035
.06
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
2 ft/s
4 ft/s
6 ft/s
8 ft/s
Ground
Ineff
Bank Sta


LagunaCreek Plan: Proposed Conditions 7/5/2013
River = Laguna Creek Reach = Main RS = 0 XS 1
.055
.035
.06
.035
.06
.055
Elevation (ft)
Station (ft)
Legend
WS Q100-USGSREV
WS Q50-USGSREV
0 ft/s
2 ft/s
4 ft/s
6 ft/s
8 ft/s
10 ft/s
Ground
Ineff
Bank Sta

Appendix C Scour Calculations

Alta Mesa Road at Laguna Creek
Sacramento County, California
Ultimate (Contraction) Scour

100-year Flow
Calculation guideline from HEC-18 5th Edition
Input from HEC-RAS for Proposed Alternatives 2 and 3

Page 6.15, Page 151 / 340 , Section 6.7 Contraction Scour in Cohesive Materials (6.7.1 Ultimate Scour)

$$y_{s-ult} = 0.94 y_1 \left(\frac{1.83 V_2}{\sqrt{g y_1}} - \frac{K_u \sqrt{\frac{\tau_c}{\rho}}}{g n y_1^{1/3}} \right)$$

	English Units		Metric Units					
y1	10.8	ft	3.3	m	Upstream depth			
V2	5.9	ft/s	1.8	m/s	Average velocity in contracted section			
tc					Critical shear stress			
n	0.035		0.035		Manning's n			
Ku	1.486		1		1.486 fot U.S. Customary, and 1.0 for S.I.			
r		slugs/ft3			Density	1000 kg/m3 =	1.94	slugs/ft3
g	32.2	ft/s^2	9.81	m/s^2				
D50			0.075	mm				

Density, rho
http://www.mo10.nrcs.usda.gov/references/guides/properties/moistbulkdensity.html

Material	Density			
	Metric Units		English Units	
Sandy Loam	1500	kg/m3	2.91	slugs/ft3
Sandy Loam	1600	kg/m3	3.10	slugs/ft3
Water, sea	1026	kg/m3	1.99	slugs/ft3
Water, pure	1000	kg/m3	1.94	slugs/ft3

Critical Shear Stress Tc	Tc (N/m2)	Tc (lb/ft2)
Tc=0.05(D50)^-0.4	0.1	
Tc=0.006(D50)^-2	1.1	

Scour Depths

Sandy Loam ,Tc=0.05(D50)^-0.4					**Sandy Loam ,Tc=0.05(D50)^-0.4**				
1.73	m	5.7	ft	ys	1.73	m	5.7	ft	ys

1243 Alpine Road, Suite 108
Walnut Creek, CA 94596
Phone: 925.941.0017
Fax: 925.941.0018
www.wreco.com

Alta Mesa Road at Laguna Creek
Sacramento County, California
Local Scour at Abutments - Froehlich or HIRE

100-year Flow
Calculation guideline from HEC-18 5th Edition
Input from HEC-RAS for Proposed Alternatives 2 and 3

Units = (SI or English)	English	
g = acceleration due to gravity =	32.2	ft/s^2

Left Overbank = Abutment #1 (South)

y1 = depth of flow at abutment on the overbank or in the main channe =	0.6	ft
L = length of embankment projected normal to flow =	449.6	ft
Ratio of projected embankment length to flow depth = L/y1 =	7.752E+02	
Abutment scour equation to be used =	HIRE	

Froehlich's Live Bed Abutment Scour Equation

L' = length of active flow obstructed by the embankment =	449.6	ft
Ae = flow area of the approach cross section obstructed by the embankment =	260.8	ft^2
ya = average depth of flow on the flood plain = Ae/L'	0.58	ft
Qe = flow obstructed by the abutment and approach embankment =	1453	ft^3/s
Ve = flow velocity = Qe/Ae =	5.6	ft/s
Fr = Froude Number of approach flow upstream of the abutment =	n/a	
Ɵ = abutment skew =	90	degrees
K1 = coefficient for abutment shape =	1	
K2 = coefficient for angle of embankment shape = Ɵ/90)^0.13 =	n/a	
Ys = abutment scour = ya*(2.27*k1*k2*((L'/ya)^0.43)*(Fr^0.61)+1) =	**n/a**	**ft**

HIRE Live Bed Abutment Scour Equation

V = velocity of flow at upstream face of abutment =	4.38	ft/s
Fr = Froude Number = V/((g*y1)^.5) =	1.01392909	
Ɵ = abutment skew =	90	degrees
K1 = coefficient for abutment shape =	1	
K2 = coefficient for angle of embankment shape = Ɵ/90)^0.13 =	1	
Ys = abutment scour = y1*(4*(Fr^0.33)*(K1/0.55)*K2) =	**4.24**	**ft**

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Alta Mesa Road at Laguna Creek
Sacramento County, California
Local Scour at Abutments - Froehlich or HIRE

100-year Flow
Calculation guideline from HEC-18 5th Edition
Input from HEC-RAS for Proposed Alternatives 2 and 3

Units = (SI or English)	English	
g = acceleration due to gravity =	32.2	ft/s^2

Right Overbank = Abutment #2 (North)

y1 = depth of flow at abutment on the overbank or in the main channe =	0.0	ft
L = length of embankment projected normal to flow =	547.5	ft
Ratio of projected embankment length to flow depth =	5.475E+04	
Abutment scour equation to be used =	HIRE	

Froehlich's Live Bed Abutment Scour Equation

L' = length of active flow obstructed by the embankment =		ft
Ae = flow area of the approach cross section obstructed by the embankment =	0.0	ft^2
ya = average depth of flow on the flood plain = ae/L	n/a	ft
Qe = flow obstructed by the abutment and approach embankment =	0	ft^3/s
Ve = flow velocity = Qe/Ae =		ft/s
Fr = Froude Number of approach flow upstream of the abutment =	n/a	
ϴ = abutment skew =		degrees
K1 = coefficient for abutment shape =		
K2 = coefficient for angle of embankment shape = ϴ/90)^0.13 =	n/a	
Ys = abutment scour = ya*(2.27*k1*k2*((L'/ya)^0.43)*(Fr^0.61)+1) =	**n/a**	**ft**

HIRE Live Bed Abutment Scour Equation

V = velocity of flow at upstream face of abutment =	4.47	ft/s
Fr = Froude Number = V/((g*y1)^.5) =	7.88052216	
ϴ = abutment skew =	90	degrees
K1 = coefficient for abutment shape =	1	
K2 = coefficient for angle of embankment shape = ϴ/90)^0.13 =	1	
Ys = abutment scour = y1*(4*(Fr^0.33)*(K1/0.55)*K2) =	**0.14**	**ft**

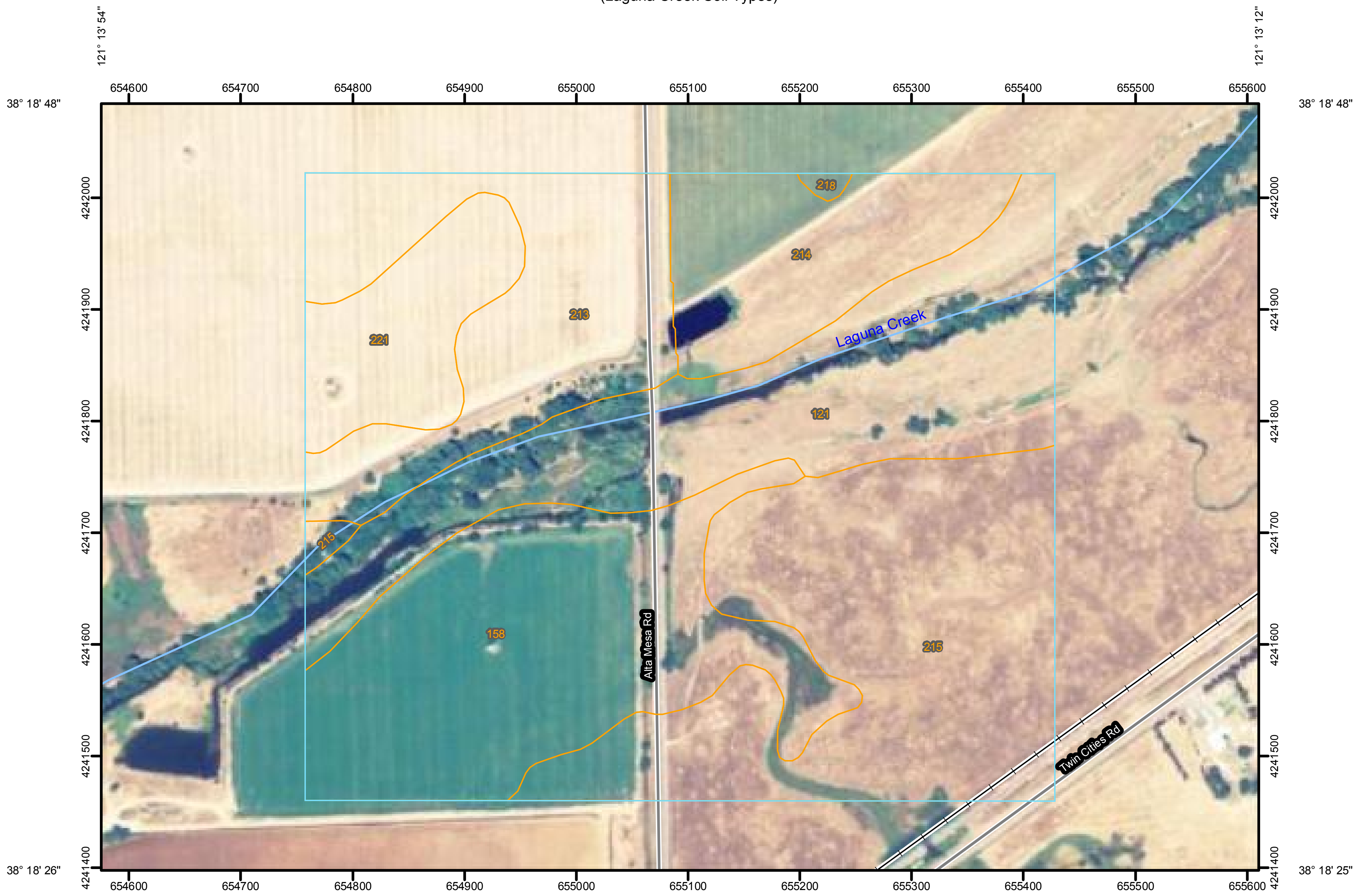
Alta Mesa Road at Laguna Creek
Sacramento County, California
Local Scour at Piers - Cohesive
100-year Flow
Calculation guideline from HEC-18 5th Edition
Input from HEC-RAS for Proposed Alternatives 2 and 3

Page 7.38, Page 204 / 340 , Section 7.12 Pier Scour In Cohesive Materials
ys = 2.2 K1 K2 a^0.65 [(2.6V1 - Vc) / sqrt (g)] ^0.7

L	39.5	ft	Pier length
a	5	ft	Pier width
L/a	7.9		If L/a is larger than 12, then use 12 as a maximum
ϴ	0	degrees	Angle of attack of flow
Group of cylinders			Pier shape
K1	1		Correction factor for pier shape
K2	1.0		Correction factor for angle of attack
V1	5.25	ft/s	Approach velocity
Vc	0.3	m/s	From Figure 4.7:
Vc	0.7	ft/s	using an erosion rate of 0.1 mm/hr
g	32.2	ft/s^2	and based on silty sand (SM)
ys	**11.16**	**ft**	

Appendix D Geotechnical Data

Soil Map—Sacramento County, California
(Laguna Creek Soil Types)



Map Scale: 1:4,910 if printed on A size (8.5" x 11") sheet.

N

Meters: 0, 50, 100, 200, 300

Feet: 0, 250, 500, 1,000, 1,500

MAP LEGEND

Area of Interest (AOI)
- Area of Interest (AOI)

Soils
- Soil Map Units

Special Point Features
- Blowout
- Borrow Pit
- Clay Spot
- Closed Depression
- Gravel Pit
- Gravelly Spot
- Landfill
- Lava Flow
- Marsh or swamp
- Mine or Quarry
- Miscellaneous Water
- Perennial Water
- Rock Outcrop
- Saline Spot
- Sandy Spot
- Severely Eroded Spot
- Sinkhole
- Slide or Slip
- Sodic Spot
- Spoil Area
- Stony Spot
- Very Stony Spot
- Wet Spot
- Other

Special Line Features
- Gully
- Short Steep Slope
- Other

Political Features
- Cities

Water Features
- Streams and Canals

Transportation
- Rails
- Interstate Highways
- US Routes
- Major Roads
- Local Roads

MAP INFORMATION

Map Scale: 1:4,910 if printed on A size (8.5" × 11") sheet.

The soil surveys that comprise your AOI were mapped at 1:24,000.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for accurate map measurements.

Source of Map: Natural Resources Conservation Service
Web Soil Survey URL: http://websoilsurvey.nrcs.usda.gov
Coordinate System: UTM Zone 10N NAD83

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Sacramento County, California
Survey Area Data: Version 11, Mar 19, 2012

Date(s) aerial images were photographed: 6/29/2005

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Map Unit Legend

Sacramento County, California (CA067)			
Map Unit Symbol	**Map Unit Name**	**Acres in AOI**	**Percent of AOI**
121	Columbia sandy loam, clayey substratum, drained, 0 to 2 percent slopes, occasionally flooded	19.4	20.9%
158	Hicksville loam, 0 to 2 percent slopes, occasionally flooded	20.4	21.9%
213	San Joaquin silt loam, leveled, 0 to 1 percent slopes	14.0	15.0%
214	San Joaquin silt loam, 0 to 3 percent slopes	9.0	9.7%
215	San Joaquin silt loam, 3 to 8 percent slopes	23.7	25.5%
218	San Joaquin-Galt complex, 0 to 3 percent slopes	0.2	0.2%
221	San Joaquin-Xerarents complex, leveled, 0 to 1 percent slopes	6.3	6.8%
Totals for Area of Interest		**93.0**	**100.0%**

GENERAL PLAN ESTIMATES

MARK THOMAS & COMPANY, INC.
Providing Engineering, Surveying and Planning Services

BRIDGE GENERAL PLAN ESTIMATE ☑ **OR PLANNING ESTIMATE** ☐

STRUCTURE Alta Mesa Rd over Laguna Creek	BR. NO. 24C-0306			
TYPE CIP Reinforced Concrete Slab (Alt 1)	DIST. 3	CO. SAC	RTE.	P.M.
LENGTH 324.0	x WIDTH 39.5	= AREA 12798	SQ FT	
QUANTITIES BY T. Pham	DATE 5/15/2013	QUANTITIES CHCKD. BY	DATE	
PRICED BY T. Pham	DATE 5/15/2013			

		CONTRACT ITEMS	UNIT	QUANTITY	PRICE	AMOUNT
1	157550	BRIDGE REMOVAL	LS	1	$75,000	$75,000
2	192003	STRUCTURE EXCAVATION (BRIDGE)	CY	81	$100	$8,100
3	193003	STRUCTURE BACKFILL (BRIDGE)	CY	36	$160	$5,760
4	490744	FURNISH PILING (CLASS 140) (ALTERNATIVE V)	LF	630	$50	$31,500
5	490745	DRIVE PILE (CLASS 140) (ALTERNATIVE V)	EA	14	$4,000	$56,000
6	490744	FURNISH PILING (15" PRECAST PILE EXTENSION)	LF	2,800	$50	$140,000
7	490745	DRIVE PILE (15" PRECAST PILE EXTENSION)	EA	56	$4,500	$252,000
8	510053	STRUCTURAL CONCRETE, BRIDGE	CY	926	$1,000	$926,000
9	510085	STRUCTURAL CONCRETE, APPROACH SLAB (TYPE EQ)	CY	22	$825	$18,150
10	520102	BAR REINFORCING STEEL (BRIDGE)	LB	171,000	$1.25	$213,750
15	XXXXXX	CALIFORNIA ST-70 BRIDGE RAIL	LF	723	$450	$325,350
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						
26						
27						

SUBTOTAL	$2,051,610
MOBILIZATION (10 %)	227,957
SUBTOTAL STRUCTURE ITEMS	$ 2,279,567
CONTINGENCIES (20 %)	455,913
BRIDGE TOTAL (214 /SQ FT)	$ 2,735,480
GRAND TOTAL	**$ 2,735,480**
FOR BUDGET PURPOSES - USE	$ 2,740,000

COMMENTS

MARK THOMAS & COMPANY, INC.
Providing Engineering, Surveying and Planning Services

BRIDGE GENERAL PLAN ESTIMATE ☑ **OR PLANNING ESTIMATE** ☐

STRUCTURE Alta Mesa Rd over Laguna Creek	BR. NO. 24C-0306			
TYPE CIP Prestressed Concrete Box Girder (Alt 2)	DIST. 3	CO. SAC	RTE.	P.M.
LENGTH 324.0 x WIDTH	39.5	= AREA 12798 SQ FT		
QUANTITIES BY T. Pham	DATE 5/20/2013	QUANTITIES CHCKD. BY		DATE
PRICED BY T. Pham	DATE 5/20/2013			

		CONTRACT ITEMS	UNIT	QUANTITY	PRICE	AMOUNT
1	157550	BRIDGE REMOVAL	LS	1	$75,000	$75,000
2	192003	STRUCTURE EXCAVATION (BRIDGE)	CY	600	$100	$60,000
3	193003	STRUCTURE BACKFILL (BRIDGE)	CY	387	$160	$61,920
4	490744	FURNISH PILING (CLASS 140) (ALTERNATIVE V)	LF	3,400	$50	$170,000
5	490745	DRIVE PILE (CLASS 140) (ALTERNATIVE V)	EA	92	$4,000	$368,000
6	510051	STRUCTURAL CONCRETE, BRIDGE FOOTING	CY	163	$575	$93,725
7	510053	STRUCTURAL CONCRETE, BRIDGE	CY	931	$900	$837,900
8	510085	STRUCTURAL CONCRETE, APPROACH SLAB (TYPE EQ)	CY	22	$825	$18,150
9	519081	JOINT SEAL (MR 1 1/2")	LF	79	$70	$5,530
10	520102	BAR REINFORCING STEEL (BRIDGE)	LB	208,023	$1.25	$260,029
15	XXXXXX	CALIFORNIA ST-70 BRIDGE RAIL	LF	723	$450	$325,350
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						
21						
22						

SUBTOTAL	$2,275,604
MOBILIZATION (10 %)	252,845
SUBTOTAL STRUCTURE ITEMS	$ 2,528,449
CONTINGENCIES (20 %)	505,690
BRIDGE TOTAL (237 /SQ FT)	$ 3,034,138
GRAND TOTAL	**$ 3,034,138**
FOR BUDGET PURPOSES - USE	$ 3,040,000

COMMENTS

MARK THOMAS & COMPANY, INC.
Providing Engineering, Surveying and Planning Services

BRIDGE GENERAL PLAN ESTIMATE ☑ **OR PLANNING ESTIMATE** ☐

STRUCTURE Alta Mesa Rd over Laguna Creek	BR. NO. 24C-0306			
TYPE Precast Prestressed Concrete Bulb-Tee Girder (Alt 3)	DIST. 3	CO. SAC	RTE.	P.M.
LENGTH 324.0	x WIDTH 39.5	= AREA 12798 SQ FT		
QUANTITIES BY T. Pham	DATE 5/20/2013	QUANTITIES CHCKD. BY		DATE
PRICED BY T. Pham	DATE 5/20/2013			

		CONTRACT ITEMS	UNIT	QUANTITY	PRICE	AMOUNT
1	157550	BRIDGE REMOVAL	LS	1	$75,000	$75,000
2	192003	STRUCTURE EXCAVATION (BRIDGE)	CY	600	$100	$60,000
3	193003	STRUCTURE BACKFILL (BRIDGE)	CY	387	$160	$61,920
4	490744	FURNISH PILING (CLASS 140) (ALTERNATIVE V)	LF	3,400	$50	$170,000
5	490745	DRIVE PILE (CLASS 140) (ALTERNATIVE V)	EA	92	$4,000	$368,000
6	510051	STRUCTURAL CONCRETE, BRIDGE FOOTING	CY	163	$575	$93,725
7	510053	STRUCTURAL CONCRETE, BRIDGE	CY	543	$900	$488,700
8	510085	STRUCTURAL CONCRETE, APPROACH SLAB (TYPE EQ)	CY	22	$825	$18,150
9	512279	FURNISH PRECAST PRESTRESSED CONCRETE BULB-TEE GIRDER (110')	EA	10	$22,000	$220,000
10	512281	FURNISH PRECAST PRESTRESSED CONCRETE BULB-TEE GIRDER (120'-130')	EA	5	$25,000	$125,000
11	512500	ERECT PRECAST PRESTRESSED CONCRETE GIRDER	EA	15	$6,000	$90,000
12	519081	JOINT SEAL (MR 1 1/2")	LF	79	$70	$5,530
13	520102	BAR REINFORCING STEEL (BRIDGE)	LB	135,040	$1.25	$168,800
15	XXXXXX	CALIFORNIA ST-70 BRIDGE RAIL	LF	723	$450	$325,350
14						
16						
17						
18						
19						
20						
21						
22						
23						
24						
25						

SUBTOTAL	$2,270,175
MOBILIZATION (10 %)	252,242
SUBTOTAL STRUCTURE ITEMS	$ 2,522,417
CONTINGENCIES (20 %)	504,483
BRIDGE TOTAL ($ 237 /SQ FT)	$ 3,026,900
GRAND TOTAL	**$3,026,900**
FOR BUDGET PURPOSES - USE	$3,030,000

COMMENTS